

Higher School of Economics

Calculus 2

Author: Andrey Mazhuga

Converted to LaTeX by Sergey Zyablyuk

Calculus 2

1 Lecture 1

Series (Prelim)

Definition 1.1. A function of the form

$$f : \mathbb{N} \rightarrow \mathbb{R}, \quad n \mapsto a_n,$$

where $\mathbb{N}$ is the set of all positive integers and $\mathbb{R}$ is the set of real numbers, is called a *sequence* (of real numbers).

Notations.

$$(a_n)_{n \in \mathbb{N}} = (a_n)_{n=1}^{\infty} = (a_n) = (a_1, a_2, \dots, a_n, \dots).$$

For example,

$$(n^2)_{n \in \mathbb{N}} = (n^2)_{n=1}^{\infty} = (n^2) = (1^2, 2^2, 3^2, \dots, n^2, \dots) = (1, 4, 9, \dots, n^2, \dots)$$

is the sequence of squares.

Definition 1.2. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the n -th *partial sum* (of $(a_n)_{n=1}^{\infty}$) is defined to be

$$S_n = \begin{cases} a_1, & n = 1, \\ S_{n-1} + a_n, & n \geq 2. \end{cases}$$

For example, if $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is a given sequence, then

$$S_1 = \frac{1}{1} = 1,$$

$$S_2 = S_1 + \frac{1}{2} = 1 + \frac{1}{2} = \frac{3}{2},$$

$$S_3 = S_2 + \frac{1}{3} = \frac{3}{2} + \frac{1}{3} = \frac{11}{6},$$

and, in general,

$$S_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}.$$

Definition 1.3. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the *series*

$$\sum_{n=1}^{\infty} a_n$$

is defined to be

$$\sum_{n=1}^{\infty} a_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} S_n,$$

where S_n is the n -th partial sum of $(a_n)_{n=1}^{\infty}$.

Notation. Instead of $\sum_{n=1}^{\infty} a_n$ one can use

$$a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots,$$

the expanded form.

Examples

1)

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \cdots + \frac{1}{n} + \cdots$$

is the *harmonic series*.

2)

$$\sum_{n=1}^{\infty} \alpha \rho^{n-1} = \alpha + \alpha \rho + \alpha \rho^2 + \cdots + \alpha \rho^{n-1} + \cdots$$

is the *geometric series*.

3) Every real number $x \in [0, 1)$ can be written as

$$x = 0.a_1 a_2 \dots a_n \dots, \quad a_n \in \{0, 1, \dots, 9\}.$$

This notation is a shorthand for the series

$$x = \sum_{n=1}^{\infty} \frac{a_n}{10^n}.$$

For instance,

$$0.2282828 \dots = \frac{2}{10} + \frac{2}{10^2} + \frac{8}{10^3} + \frac{2}{10^4} + \frac{8}{10^5} + \frac{2}{10^6} + \cdots.$$

4)

$$\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \cdots + (-1)^{n+1} + \cdots.$$

Definition 1.4. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the series $\sum_{n=1}^{\infty} a_n$ is called

- *convergent* (to S) if

$$\lim_{n \rightarrow +\infty} S_n = S \in \mathbb{R}, \quad S_n = a_1 + \cdots + a_n$$

is the n -th partial sum;

- *divergent to $\pm\infty$* if

$$\lim_{n \rightarrow +\infty} S_n = \pm\infty;$$

- *divergent* if $\lim_{n \rightarrow +\infty} S_n$ does not exist.

Examples of convergent series

1) If $|\rho| < 1$, then

$$\begin{aligned} \sum_{n=1}^{\infty} \alpha \rho^{n-1} &\stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \alpha \rho^{k-1} \\ &= \lim_{n \rightarrow +\infty} \alpha \frac{1 - \rho^n}{1 - \rho} = \alpha \frac{1 - 0}{1 - \rho} = \frac{\alpha}{1 - \rho} \in \mathbb{R}. \end{aligned}$$

2)

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n(n+1)} &\stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k(k+1)} \\ &= \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n+1}\right) = 1.\end{aligned}$$

This is a telescoping series.

3)

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}.$$

This is not trivial; we will prove it later in our course.

4)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{(-1)^{k+1}}{k} = \ln 2.$$

Examples of series divergent to $+\infty$

1)

$$\sum_{n=1}^{\infty} 1 \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n 1 = \lim_{n \rightarrow +\infty} n = +\infty.$$

2) If $\rho > 1$ and $\alpha > 0$, then

$$\sum_{n=1}^{\infty} \alpha \rho^{n-1} \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \alpha \frac{1 - \rho^n}{1 - \rho} = +\infty.$$

3)

$$\sum_{n=1}^{\infty} \frac{1}{n} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k} = +\infty.$$

We will prove this later.

Examples of divergent series

1)

$$\sum_{n=1}^{\infty} (-1)^{n+1} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n (-1)^{k+1},$$

where

$$\sum_{k=1}^n (-1)^{k+1} = \begin{cases} 1, & n \text{ is odd,} \\ 0, & n \text{ is even.} \end{cases}$$

Hence the limit does not exist.

2)

$$\sum_{n=1}^{\infty} \sin(n)$$

is divergent: the limit $\lim_{n \rightarrow +\infty} \sum_{k=1}^n \sin(k)$ does not exist. We will prove this later.

Claim 1.1. Changing a finite number of terms in a series does not affect its convergence (although it may change the sum of the series).

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the initial series and let $\sum_{n=1}^{\infty} b_n$ be obtained from it by changing a finite number of terms. Then it is clear that there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N}, \quad n \geq N \implies a_n = b_n.$$

Therefore, for every $n \geq N$,

$$A_n - B_n = A_N - B_N,$$

where

$$A_k = \sum_{i=1}^k a_i, \quad B_k = \sum_{i=1}^k b_i$$

are the k -th partial sums. Thus

$$\lim_{n \rightarrow +\infty} (A_n - B_n) = A_N - B_N \in \mathbb{R},$$

and the right-hand side does not depend on n . By the limit laws for sequences (see Calculus 1), it follows that either both limits $\lim A_n$ and $\lim B_n$ exist, or neither exists. Hence the convergence of the series is unchanged. $\square$

Claim 1.2 (Cauchy's criterion for series). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \iff \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m, k \in \mathbb{N}, m \geq N(\varepsilon) \implies \left| \sum_{n=m+1}^{m+k} a_n \right| < \varepsilon \right).$$

Proof. By [Definition 1.4](#),

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \exists \lim_{N \rightarrow +\infty} A_N, \quad A_N = \sum_{n=1}^N a_n.$$

By Cauchy's criterion for sequences (see Calculus 1), this is equivalent to

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m, k \in \mathbb{N}, m \geq N(\varepsilon) \implies |A_{m+k} - A_m| < \varepsilon.$$

Since

$$\begin{aligned} A_{m+k} - A_m &= (a_1 + \cdots + a_m + a_{m+1} + \cdots + a_{m+k}) - (a_1 + \cdots + a_m) \\ &= a_{m+1} + \cdots + a_{m+k} = \sum_{n=m+1}^{m+k} a_n, \end{aligned}$$

the claim is proved. $\square$

Claim 1.3 (The negation of the Cauchy criterion). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ diverges} \right) \iff \left(\exists \varepsilon > 0 : \forall N \in \mathbb{N} \exists m \geq N \exists k \in \mathbb{N} : \left| \sum_{n=m+1}^{m+k} a_n \right| \geq \varepsilon \right).$$

Proof. Exercise (or one can see that it follows directly from [Claim 1.2](#)). $\square$

Claim 1.4. The following statements hold:

- 1) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges;
- 2) $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof. **1)** To prove this fact we use [Claim 1.2](#). Let $\varepsilon > 0$ be arbitrary. Set

$$N(\varepsilon) = \left\lceil \frac{1}{\varepsilon} \right\rceil.$$

For every $m \geq N(\varepsilon)$ and every $k \in \mathbb{N}$,

$$\begin{aligned} \left| \sum_{n=m+1}^{m+k} \frac{1}{n^2} \right| &= \frac{1}{(m+1)^2} + \cdots + \frac{1}{(m+k)^2} \\ &\leq \left(\frac{1}{m} - \frac{1}{m+1} \right) + \cdots + \left(\frac{1}{m+k-1} - \frac{1}{m+k} \right) \\ &= \frac{1}{m} - \frac{1}{m+k} < \frac{1}{m} \leq \frac{1}{N(\varepsilon)} \leq \varepsilon, \end{aligned}$$

where we used

$$\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Therefore, by [Claim 1.2](#), $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

2) We use [Claim 1.3](#). Let $\varepsilon = \frac{1}{2}$. Let $N \in \mathbb{N}$ be arbitrary and set $m = N$, $k = N$. Then

$$\begin{aligned} \left| \sum_{n=m+1}^{m+k} \frac{1}{n} \right| &= \sum_{n=N+1}^{2N} \frac{1}{n} \\ &= \frac{1}{N+1} + \frac{1}{N+2} + \cdots + \frac{1}{N+N} \\ &\geq \underbrace{\frac{1}{2N} + \frac{1}{2N} + \cdots + \frac{1}{2N}}_{N \text{ times}} = N \cdot \frac{1}{2N} = \frac{1}{2} = \varepsilon. \end{aligned}$$

Therefore, by [Claim 1.3](#), $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. □

Claim 1.5 (Necessary condition for convergence). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \implies \left(\lim_{n \rightarrow +\infty} a_n = 0 \right).$$

Note. [Claim 1.4\(2\)](#) shows that the converse does not hold. Indeed,

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0, \quad \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.}$$

Proof. This follows directly from [Claim 1.2](#) if we set $k = 1$. □

Claim 1.6. The series

$$\sum_{n=1}^{\infty} \sin(nx)$$

diverges for all $x \notin \pi\mathbb{Z}$.

Proof. Suppose $\sum_{n=1}^{\infty} \sin(nx)$ converges. Then, due to [Claim 1.5](#),

$$\lim_{n \rightarrow +\infty} \sin(nx) = \lim_{n \rightarrow +\infty} \sin((n+1)x) = 0.$$

Since

$$\sin((n+1)x) = \sin(nx) \cos x + \cos(nx) \sin x$$

and $\sin x \neq 0$ (because $x \notin \pi\mathbb{Z}$), it follows that

$$\lim_{n \rightarrow +\infty} \cos(nx) = 0.$$

But this contradicts the identity

$$\sin^2(nx) + \cos^2(nx) = 1.$$

□

Definition 1.5. Let

$$\sum_{n=1}^{\infty} a_n = S$$

be a convergent series, and let

$$S_n = \sum_{k=1}^n a_k$$

be its n -th partial sum. Then, for every $n \in \mathbb{N}$,

$$S = S_n + r_n,$$

where the sequence $(r_n)_{n=1}^{\infty}$,

$$r_n = \sum_{k=n+1}^{\infty} a_k,$$

is called the *tail* (or the *remainder*) of the series $\sum_{n=1}^{\infty} a_n$.

Claim 1.7. Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \implies \left(\lim_{n \rightarrow +\infty} r_n = 0 \right).$$

Proof. Let $\sum_{n=1}^{\infty} a_n = S \in \mathbb{R}$. Then

$$r_n = S - S_n \quad \forall n \in \mathbb{N}.$$

Therefore,

$$\lim_{n \rightarrow +\infty} r_n = \lim_{n \rightarrow +\infty} (S - S_n) = S - \lim_{n \rightarrow +\infty} S_n = S - S = 0.$$

□

Claim 1.8 (Linear combination of series). Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two convergent series. Then, for every $\alpha, \beta \in \mathbb{R}$, the series

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n)$$

is also convergent and

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n.$$

This series is the linear combination of the series $\sum a_n$ and $\sum b_n$ with coefficients α and β .

Proof. Since $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, for some $a, b \in \mathbb{R}$ we have

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow +\infty} A_n = a, \quad A_n = \sum_{k=1}^n a_k,$$

and

$$\sum_{n=1}^{\infty} b_n = \lim_{n \rightarrow +\infty} B_n = b, \quad B_n = \sum_{k=1}^n b_k.$$

Therefore, by the limit laws for sequences (see Calculus 1),

$$\begin{aligned} \alpha a + \beta b &= \alpha \lim_{n \rightarrow +\infty} A_n + \beta \lim_{n \rightarrow +\infty} B_n \\ &= \lim_{n \rightarrow +\infty} (\alpha A_n + \beta B_n) \\ &= \lim_{n \rightarrow +\infty} \left(\sum_{k=1}^n (\alpha a_k + \beta b_k) \right). \end{aligned}$$

By [Definition 1.3](#), this is precisely

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b.$$

□

Claim 1.9 (Grouping theorem). Let

- a) $\sum_{n=1}^{\infty} a_n$ be a series;
- b) $(K_n)_{n=1}^{\infty}$ be a sequence of positive integers such that

$$K_1 = 1, \quad K_i < K_{i+1} \quad \text{for every } i \in \{1, 2, \dots\};$$

- c) $\sum_{n=1}^{\infty} b_n$, where

$$b_n = a_{K_n} + \dots + a_{K_{n+1}-1}, \quad n \in \mathbb{N},$$

be a *grouped series*.

Thus b_n is the n -th block and $K_{n+1} - K_n$ is the size of the n -th block. Then:

1)

$$\left(\sum_{n=1}^{\infty} a_n = \alpha \in \mathbb{R} \right) \implies \left(\sum_{n=1}^{\infty} b_n = \alpha \right).$$

2) If

$$\lim_{n \rightarrow +\infty} a_n = 0, \quad \exists m \in \mathbb{N} \forall n \in \mathbb{N} : K_{n+1} - K_n < m, \quad \sum_{n=1}^{\infty} b_n = b \in \mathbb{R},$$

then

$$\sum_{n=1}^{\infty} a_n = b.$$

Calculus 2

2 Lecture 2

Series with non-negative terms

Definition 2.1 (from Calculus 1). A sequence $\{a_n\}_{n=1}^{\infty}$ is called

- *increasing* if $a_{n+1} > a_n$ for all $n \in \mathbb{N}$;
- *nondecreasing* if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$;
- *decreasing* if $a_{n+1} < a_n$ for all $n \in \mathbb{N}$;
- *nonincreasing* if $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$.

Any of the above sequences is called *monotonic*. Increasing and decreasing sequences are called *strictly monotonic*.

Claim 2.1. If, for all $n \in \mathbb{N}$, $a_n > 0$ (respectively, $a_n \geq 0$, $a_n < 0$, $a_n \leq 0$), then the sequence

$$\left\{ A_n = \sum_{k=1}^n a_k \right\}_{n=1}^{\infty}$$

is increasing (respectively, nondecreasing, decreasing, nonincreasing).

Proof. Obvious. □

Claim 2.2. Let $a_n \geq 0$ for every $n \in \mathbb{N}$. Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ is convergent} \right) \iff \left(\left\{ A_n = \sum_{k=1}^n a_k \right\}_{n=1}^{\infty} \text{ is bounded} \right).$$

By definition, a sequence $\{b_n\}_{n=1}^{\infty}$ is bounded if

$$\exists M > 0 \quad \forall n \in \mathbb{N} : |b_n| < M.$$

Proof. From left to right,

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \stackrel{\text{def}}{\implies} \left(\exists \lim_{n \rightarrow +\infty} A_n \right) \implies (\{A_n\}_{n=1}^{\infty} \text{ is bounded}),$$

where the last implication is a result from Calculus 1.

Conversely, suppose $\{A_n\}_{n=1}^{\infty}$ is bounded. Since $a_k \geq 0$, the sequence $\{A_n\}_{n=1}^{\infty}$ is nondecreasing. Hence, by the corresponding result from Calculus 1, the limit $\lim_{n \rightarrow +\infty} A_n$ exists. By definition, this means that $\sum_{k=1}^{\infty} a_k$ converges. □

Claim 2.3 (Cauchy Condensation Test). Let $\{a_k\}_{k=1}^{\infty}$ be a sequence such that

- $a_k \geq 0$ for every $k \in \mathbb{N}$;
- $a_{k+1} \leq a_k$ for every $k \in \mathbb{N}$.

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges} \right).$$

The series

$$a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots + 2^k a_{2^k} + \cdots$$

is called the *condensed series*.

Proof. Since $a_{k+1} \leq a_k$, we have

$$\begin{aligned} a_2 &\leq a_2 \leq a_1, \\ 2a_4 &\leq a_3 + a_4 \leq 2a_2, \\ 4a_8 &\leq a_5 + a_6 + a_7 + a_8 \leq 4a_4, \end{aligned}$$

and, in general,

$$2^n a_{2^{n+1}} \leq a_{2^{n+1}} + \cdots + a_{2^{n+1}} \leq 2^n a_{2^n}.$$

By adding these inequalities, we obtain

$$a_2 + 2a_4 + 4a_8 + \cdots + 2^n a_{2^{n+1}} \leq \sum_{k=2}^{2^{n+1}} a_k \leq a_1 + 2a_2 + 4a_4 + \cdots + 2^n a_{2^n}. \quad (2.1)$$

Let

$$S_n = a_1 + 2a_2 + 4a_4 + \cdots + 2^n a_{2^n}$$

be the n -th partial sum of the condensed series $\sum_{k=0}^{\infty} 2^k a_{2^k}$, and let

$$A_n = a_1 + \cdots + a_n$$

be the n -th partial sum of $\sum_{k=1}^{\infty} a_k$. Then (2.1) can be rewritten as

$$\frac{1}{2}(S_{n+1} - a_1) \leq A_{2^{n+1}} - a_1 \leq S_n. \quad (2.2)$$

Since the sequences $\{A_n\}_{n=1}^{\infty}$ and $\{S_n\}_{n=1}^{\infty}$ are nondecreasing, inequality (2.2) implies

$$(\{A_n\}_{n=1}^{\infty} \text{ is bounded}) \iff (\{S_n\}_{n=1}^{\infty} \text{ is bounded}).$$

Applying Claim 2.2 to both series proves the result. $\square$

Claim 2.4 (The First Comparison Test). Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

$$0 \leq a_k \leq b_k \quad \text{for all } k \in \mathbb{N}.$$

Then the following implications hold:

1)

$$\left(\sum_{k=1}^{\infty} b_k \text{ converges} \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right) \implies \left(\sum_{k=1}^{\infty} b_k \text{ diverges} \right).$$

Proof. 1) Let

$$A_n = \sum_{k=1}^n a_k, \quad B_n = \sum_{k=1}^n b_k$$

be the n -th partial sums. Since $0 \leq a_k \leq b_k$ for every $k \in \mathbb{N}$, it is clear that

$$0 \leq A_n \leq B_n \quad (\forall n \in \mathbb{N}).$$

If $\sum_{k=1}^{\infty} b_k$ converges, then by Claim 2.2 the sequence $\{B_n\}_{n=1}^{\infty}$ is bounded. Hence $\{A_n\}_{n=1}^{\infty}$ is bounded, and Claim 2.2 gives the convergence of $\sum_{k=1}^{\infty} a_k$.

2) Assume that the statement is false. Then there are series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ such that

$$\sum_{k=1}^{\infty} a_k \text{ diverges} \quad \text{and} \quad \sum_{k=1}^{\infty} b_k \text{ converges}.$$

But, due to item 1), convergence of $\sum b_k$ implies convergence of $\sum a_k$. Thus we arrive at a contradiction. $\square$

Claim 2.5 (The Second Comparison Test). Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

- a) $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$;
- b) there exist $m > 0$ and $M > 0$ such that, for every $n \in \mathbb{N}$,

$$m \leq \frac{a_n}{b_n} \leq M.$$

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=1}^{\infty} b_k \text{ converges} \right).$$

Proof. Since $b_k > 0$, for every $k \in \mathbb{N}$ we have

$$\left(m \leq \frac{a_k}{b_k} \leq M \right) \implies (mb_k \leq a_k \leq Mb_k).$$

Therefore,

$$\sum_{k=1}^{\infty} a_k \text{ converges} \implies \sum_{k=1}^{\infty} mb_k \text{ converges} \implies \sum_{k=1}^{\infty} b_k \text{ converges},$$

where the first implication follows from [item 1\)](#) of [Claim 2.4](#) and the second from $m > 0$. Similarly,

$$\sum_{k=1}^{\infty} b_k \text{ converges} \implies \sum_{k=1}^{\infty} Mb_k \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges},$$

using [item 1\)](#) of [Claim 2.4](#) and $Mb_k \geq a_k$. □

Claim 2.6. Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

- a) $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$;
- b)

$$\lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lambda \in (0, +\infty).$$

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=1}^{\infty} b_k \text{ converges} \right).$$

Proof. The condition

$$\lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lambda \in (0, +\infty)$$

implies that

$$m \leq \frac{a_k}{b_k} \leq M$$

for some $m, M > 0$ (see Calculus 1). Therefore the statement follows from [Claim 2.5](#). □

Claim 2.7 (p -series convergence). The series

$$\sum_{k=1}^{\infty} \frac{1}{k^p}, \quad p \in \mathbb{R},$$

is called the p -series and converges if and only if $p > 1$.

Proof. If $p \leq 0$, then

$$\lim_{k \rightarrow +\infty} \frac{1}{k^p} \neq 0.$$

Therefore, due to the necessary condition for convergence (Claim 1.5), the series diverges.

If $p > 0$, then

$$\frac{1}{(k+1)^p} < \frac{1}{k^p} \quad (\forall k \in \mathbb{N}),$$

and all conditions of the Cauchy condensation test (Claim 2.3) are satisfied. Thus

$$\begin{aligned} \left(\sum_{k=1}^{\infty} \frac{1}{k^p} \text{ converges} \right) &\iff \left(\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} \text{ converges} \right) \\ &\iff \left(\sum_{k=0}^{\infty} \left(2^{1-p} \right)^k \text{ converges} \right) \\ &\iff 2^{1-p} < 1 \\ &\iff p > 1. \end{aligned}$$

□

Claim 2.8 (The Integral Test). Let

$$f : [1, +\infty) \rightarrow \mathbb{R}$$

be a positive, nonincreasing, continuous function. Then

$$\left(\sum_{k=1}^{\infty} f(k) \text{ converges} \right) \iff \left(\int_1^{+\infty} f(x) dx \text{ converges} \right).$$

Note. In general,

$$\sum_{k=1}^{\infty} f(k) \neq \int_1^{+\infty} f(x) dx.$$

Proof. For every $k \in \mathbb{N}$ and every $x \in [k, k+1]$, due to the monotonicity of f ,

$$f(k+1) \leq f(x) \leq f(k).$$

The geometric meaning of these inequalities is illustrated below.

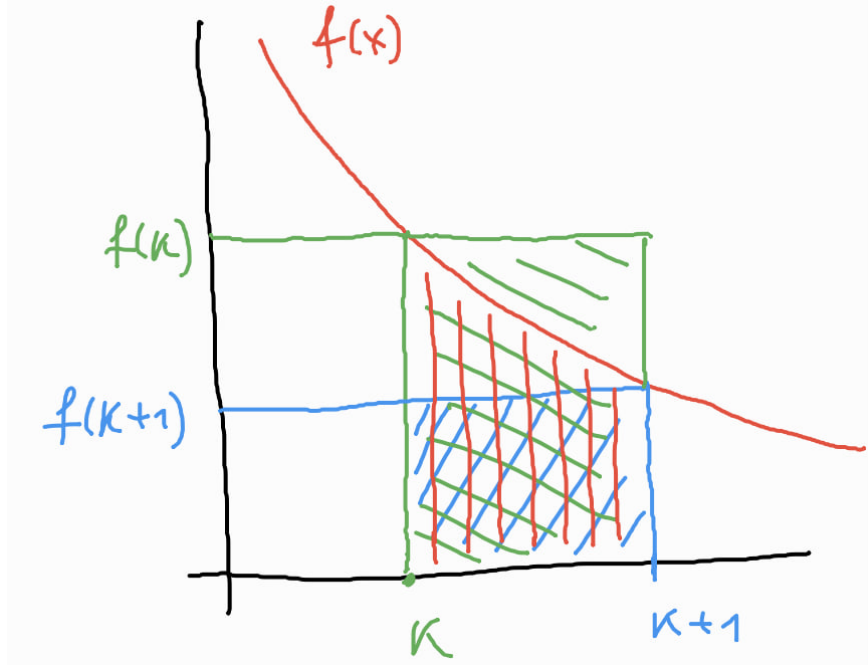


Figure 2.1

By integrating the inequalities over $[k, k+1]$, we obtain

$$f(k+1) = \int_k^{k+1} f(k+1) dx \leq \int_k^{k+1} f(x) dx \leq \int_k^{k+1} f(k) dx = f(k).$$

The integrals exist because f is continuous. By adding these inequalities from $k = 1$ to $k = n-1$, and writing

$$S_n = \sum_{k=1}^n f(k),$$

we obtain

$$S_n - f(1) = \sum_{k=1}^{n-1} f(k+1) \leq \sum_{k=1}^{n-1} \int_k^{k+1} f(x) dx = \int_1^n f(x) dx \leq \sum_{k=1}^{n-1} f(k) = S_{n-1}.$$

Hence

$$(\{S_n\}_{n=1}^{\infty} \text{ is bounded}) \iff \left(\left\{ \int_1^n f(x) dx \right\}_{n=1}^{\infty} \text{ is bounded} \right).$$

By [Claim 2.2](#), the left-hand condition is equivalent to the convergence of $\sum_{k=1}^{\infty} f(k)$. Since the sequence $\{\int_1^n f(x) dx\}$ is nondecreasing, the right-hand condition is equivalent to the convergence of $\int_1^{+\infty} f(x) dx$ (see Calculus 1). $\square$

Claim 2.9 (The Root Test). Let $a_k \geq 0$ for every $k \in \mathbb{N}$. Then:

1)

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} < 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} > 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Definition (see Calculus 1). Let $\{b_k\}_{k=1}^{\infty} \subset \mathbb{R}$ be a sequence. Then

$$\overline{\lim}_{n \rightarrow +\infty} b_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \left(\sup_{k \geq n} b_k \right)$$

is called the *upper limit* of $\{b_k\}$, and

$$\underline{\lim}_{n \rightarrow +\infty} b_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \left(\inf_{k \geq n} b_k \right)$$

is called the *lower limit* of $\{b_k\}$.

Proof. 1) Due to the definition of the upper limit,

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} < 1 \right) \implies \left(\exists q \in (0, 1) \exists N \in \mathbb{N} : \forall n \geq N, \quad 0 \leq a_n^{1/n} < q \right).$$

Equivalently, $0 \leq a_n < q^n$ for $n \geq N$. Since $q \in (0, 1)$, the geometric series $\sum_{k=0}^{\infty} q^k$ converges. By changing a finite number of terms (Claim 1.1), its tail $\sum_{k=N}^{\infty} q^k$ converges. Due to Claim 2.4 and $0 \leq a_n \leq q^n$ for $n \geq N$, the series $\sum_{k=N}^{\infty} a_k$ converges, and again by Claim 1.1, so does $\sum_{k=1}^{\infty} a_k$.

2) Due to the definition of the upper limit,

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} > 1 \right) \implies \left(\forall N \in \mathbb{N} \exists n \geq N : a_n > 1 \right).$$

Thus $\lim_{k \rightarrow +\infty} a_k \neq 0$, and Claim 1.5 implies that $\sum_{k=1}^{\infty} a_k$ diverges. □

Claim 2.10. If $a_k > 0$ for every $k \in \mathbb{N}$, then

$$\underline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} \leq \underline{\lim}_{k \rightarrow +\infty} a_k^{1/k} \leq \overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} \leq \overline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k}.$$

Proof. Without proof. □

Claim 2.11 (The Ratio Test). Let $a_k > 0$ for every $k \in \mathbb{N}$. Then:

1)

$$\left(\overline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} < 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\underline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} > 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. Follows directly from Claim 2.9 and Claim 2.10. □

Claim 2.12. The root test (Claim 2.9) is stronger than the ratio test (Claim 2.11).

Proof. Due to Claim 2.10, the root test is not weaker than the ratio test. Consider the series

$$\sum_{k=1}^{+\infty} 2^{(-1)^k - k}.$$

It ought to be clear that

$$\overline{\lim}_{k \rightarrow +\infty} \left(2^{(-1)^k - k} \right)^{1/k} = 2^{-1} < 1,$$

so by [Claim 2.9](#) the series converges. On the other hand,

$$\overline{\lim}_{k \rightarrow +\infty} \frac{2^{(-1)^{k+1}-k-1}}{2^{(-1)^k-k}} = \overline{\lim}_{k \rightarrow +\infty} 2^{(-1)^{k+1}-(-1)^k-1} = 2 > 1,$$

whereas

$$\underline{\lim}_{k \rightarrow +\infty} \frac{2^{(-1)^{k+1}-k-1}}{2^{(-1)^k-k}} = \underline{\lim}_{k \rightarrow +\infty} 2^{(-1)^{k+1}-(-1)^k-1} = 2^{-3} < 1.$$

Thus [Claim 2.11](#) cannot be applied. Since the root test works for this series and the ratio test does not, the claim is proved. $\square$

Claim 2.13 (Kummer's Test). Let $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$. Let

$$c_k = \frac{a_k}{a_{k+1}} b_k - b_{k+1} \quad \text{for every } k \in \mathbb{N}.$$

Then:

1)

$$(\exists L > 0 \exists N \in \mathbb{N} : \forall n \geq N, c_n \geq L) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\sum_{k=1}^{\infty} \frac{1}{b_k} \text{ diverges and } \exists N \in \mathbb{N} : \forall n \geq N, c_n \leq 0 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. 1) Without loss of generality, we can assume that

$$c_k \geq L \quad \forall k \in \mathbb{N}.$$

Since $a_k > 0$, we have

$$\left(\frac{a_k}{a_{k+1}} b_k - b_{k+1} \geq L \right) \iff (a_k b_k - a_{k+1} b_{k+1} \geq L a_{k+1}). \quad (*)$$

For every $n \in \mathbb{N}$, let

$$S_n = \sum_{k=1}^n a_k.$$

Then

$$\begin{aligned} S_n &= \frac{1}{L} \sum_{k=1}^n (L a_k) \\ &\leq \frac{1}{L} \sum_{k=1}^n (a_k b_k - a_{k+1} b_{k+1}) \\ &= \frac{1}{L} (a_1 b_1 - a_{n+1} b_{n+1}) \\ &\leq \frac{1}{L} a_1 b_1. \end{aligned}$$

Thus $\{S_n\}_{n=1}^{\infty}$ is bounded. Since $\{S_n\}$ is increasing, this implies that $\sum_{k=1}^{\infty} a_k$ converges.

2) Without loss of generality, we can assume that

$$c_k = \frac{a_k}{a_{k+1}} b_k - b_{k+1} \leq 0 \quad \forall k \in \mathbb{N}.$$

For every $k \in \mathbb{N}$,

$$\left(\frac{a_k}{a_{k+1}} b_k - b_{k+1} \leq 0 \right) \iff \left(\frac{a_k}{a_{k+1}} \leq \frac{b_{k+1}}{b_k} \right), \quad (**)$$

where $b_k > 0$. Multiplying the inequalities [\(**\)](#) from $k = 1$ to $k = n$, we obtain

$$\frac{a_1}{a_{n+1}} = \frac{a_1}{a_2} \frac{a_2}{a_3} \dots \frac{a_n}{a_{n+1}} \leq \frac{b_2}{b_1} \frac{b_3}{b_2} \dots \frac{b_{n+1}}{b_n} = \frac{b_{n+1}}{b_1}.$$

Thus

$$\frac{a_1}{a_{n+1}} \leq \frac{b_{n+1}}{b_1} \iff a_{n+1} \geq a_1 b_1 \frac{1}{b_{n+1}}.$$

Since

$$\sum_{k=1}^{\infty} a_1 b_1 \frac{1}{b_k}$$

diverges (it is given that $\sum 1/b_k$ diverges and $a_1 b_1$ is a nonzero constant), [item 2\)](#) of [Claim 2.4](#) implies that $\sum_{k=1}^{\infty} a_k$ diverges. $\square$

Claim 2.14 (Bertrand's Test). Let $a_k > 0$ for every $k \in \mathbb{N}$. Let

$$\lim_{k \rightarrow +\infty} \ln(k) \left(k \left(\frac{a_k}{a_{k+1}} - 1 \right) - 1 \right) = B \in [-\infty, +\infty].$$

Then:

1)

$$(B > 1) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$(B < 1) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

The cases $B = \pm\infty$ are allowed in this claim.

Proof. **1)** Let

$$b_k = k \ln k \quad (k \geq 2).$$

Then

$$\begin{aligned} c_k &= \frac{a_k}{a_{k+1}} b_k - b_{k+1} \\ &= \ln(k) \left(k \left(\frac{a_k}{a_{k+1}} - 1 \right) - 1 \right) - (k+1)(\ln(k+1) - \ln k). \end{aligned} \quad (***)$$

The first term tends to B , and

$$\lim_{k \rightarrow +\infty} (k+1)(\ln(k+1) - \ln k) = 1$$

(see Calculus 1). Hence, if $B > 1$,

$$\lim_{k \rightarrow +\infty} c_k = B - 1 > 0.$$

Therefore

$$\exists L > 0 \exists N \in \mathbb{N} : \forall n \geq N, c_n \geq L,$$

and [item 1\)](#) of [Claim 2.13](#) implies that $\sum_{k=1}^{\infty} a_k$ converges.

2) Since

$$\int_2^{+\infty} \frac{1}{x \ln x} dx = \ln \ln x \Big|_2^{+\infty}$$

diverges, [Claim 2.8](#) gives that the series

$$\sum_{k=2}^{+\infty} \frac{1}{k \ln k}$$

diverges. If $B < 1$, then by equality [\(***\)](#),

$$\lim_{k \rightarrow +\infty} c_k = B - 1 < 0,$$

so there exists N such that $c_n < 0$ for every $n \geq N$. Therefore, due to [item 2\)](#) of [Claim 2.13](#), the series $\sum_{k=1}^{\infty} a_k$ diverges. $\square$

Claim 2.15 (The Gauss Test). Let

- a) $a_k > 0$ for every $k \in \mathbb{N}$;
- b) there exist $\varepsilon > 0$ and $\lambda, \mu \in \mathbb{R}$ such that, for every $k \in \mathbb{N}$,

$$\frac{a_k}{a_{k+1}} = \lambda + \frac{\mu}{k} + \frac{\gamma_k}{k^{1+\varepsilon}},$$

where the sequence $\{\gamma_n\}_{n=1}^{\infty}$ is bounded.

Then:

1)

$$((\lambda > 1) \text{ or } (\lambda = 1 \text{ and } \mu > 1)) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$((\lambda < 1) \text{ or } (\lambda = 1 \text{ and } \mu \leq 1)) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. Follows directly from [Claim 2.14](#) (can you see why?). $\square$

Claim 2.16. Bertrand's test ([Claim 2.14](#)) is stronger than the Gauss test ([Claim 2.15](#)).

Proof. Since [Claim 2.15](#) follows from [Claim 2.14](#), Bertrand's test is not weaker than the Gauss test. One can verify by direct calculation that [Claim 2.14](#) works for

$$\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$$

and [Claim 2.15](#) does not. $\square$

Calculus 2

3 Lecture 3

Operations with series

Claim 3.1 (The Abel transformation). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences. Then, for every $n, k \in \mathbb{N}$, we have

$$\sum_{i=n+1}^{n+k} a_i b_i = A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i)$$

where

$$A_m = \sum_{j=1}^m a_j.$$

Proof. Note that $a_i = A_i - A_{i-1}$. Hence

$$\begin{aligned} \sum_{i=n+1}^{n+k} a_i b_i &= \sum_{i=n+1}^{n+k} (A_i - A_{i-1}) b_i \\ &= \sum_{i=n+1}^{n+k} A_i b_i - \sum_{i=n+1}^{n+k} A_{i-1} b_i \\ &= \sum_{i=n+1}^{n+k} A_i b_i - \sum_{i=n}^{n+k-1} A_i b_{i+1} \\ &= A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i). \end{aligned}$$

□

Claim 3.2 (The Dirichlet Test). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences of real numbers satisfying the following conditions:

a) there exists $M > 0$ such that

$$\left| \sum_{i=1}^n a_i \right| \leq M \quad \text{for every } n \in \mathbb{N};$$

b) $\{b_i\}_{i=1}^{\infty}$ is monotonic; that is, either

$$b_i \geq b_{i+1} \quad (\forall i \in \mathbb{N})$$

or

$$b_i \leq b_{i+1} \quad (\forall i \in \mathbb{N});$$

c)

$$\lim_{i \rightarrow +\infty} b_i = 0.$$

Then the series

$$\sum_{i=1}^{\infty} a_i b_i$$

converges.

Proof. Due to [Claim 1.2](#) (Cauchy's criterion),

$$\left(\sum_{i=1}^{\infty} a_i b_i \text{ converges} \right) \iff \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n, k \in \mathbb{N}, n > N(\varepsilon) \implies \left| \sum_{i=n+1}^{n+k} a_i b_i \right| < \varepsilon \right).$$

Thus, it suffices to prove the condition on the right.

Let $\varepsilon > 0$ be arbitrary. Since $\lim_{i \rightarrow +\infty} b_i = 0$, there exists $N_1(\varepsilon) \in \mathbb{N}$ such that

$$\forall n > N_1(\varepsilon), \quad |b_n| < \frac{\varepsilon}{2M}.$$

Set $N(\varepsilon) = N_1(\varepsilon)$. For every $n, k \in \mathbb{N}$ with $n > N(\varepsilon)$, the Abel transformation gives

$$\begin{aligned} \left| \sum_{i=n+1}^{n+k} a_i b_i \right| &= \left| A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i) \right| \\ &\leq |A_{n+k}| |b_{n+k}| + |A_n| |b_{n+1}| + \sum_{i=n+1}^{n+k-1} |A_i| |b_{i+1} - b_i| \\ &\leq M |b_{n+k}| + M |b_{n+1}| + M \sum_{i=n+1}^{n+k-1} |b_{i+1} - b_i|. \end{aligned} \quad (*)$$

Suppose first that $b_i \geq b_{i+1}$ for every $i \in \mathbb{N}$. Together with $\lim b_i = 0$, this implies $b_i \geq 0$ for every $i \in \mathbb{N}$. Therefore, continuing from [\(*\)](#),

$$\begin{aligned} \left| \sum_{i=n+1}^{n+k} a_i b_i \right| &\leq M b_{n+k} + M b_{n+1} + M \sum_{i=n+1}^{n+k-1} (b_i - b_{i+1}) \\ &= M (b_{n+k} + b_{n+1} + b_{n+1} - b_{n+k}) \\ &= 2M b_{n+1} < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon. \end{aligned}$$

Hence, by Cauchy's criterion, $\sum_{i=1}^{\infty} a_i b_i$ converges.

If instead $b_i \leq b_{i+1}$ for every $i \in \mathbb{N}$, then together with $\lim b_i = 0$ this implies $b_i \leq 0$ for every $i \in \mathbb{N}$. The sequence $\{-b_i\}$ is nonincreasing and tends to 0, so by the case already proved, the series $\sum a_i (-b_i)$ converges. Therefore $\sum a_i b_i$ converges as well. $\square$

Claim 3.3 (The Abel Test). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences of real numbers satisfying the following conditions:

- a) $\sum_{i=1}^{\infty} a_i$ converges;
- b) $\{b_i\}_{i=1}^{\infty}$ is monotonic;
- c) $\{b_i\}_{i=1}^{\infty}$ is bounded; that is, there exists $L > 0$ such that

$$|b_i| \leq L \quad (\forall i \in \mathbb{N}).$$

Then the series

$$\sum_{i=1}^{\infty} a_i b_i$$

converges.

Proof. Since $\{b_i\}_{i=1}^{\infty}$ is monotonic and bounded, we have (see Calculus 1)

$$\exists \lim_{i \rightarrow +\infty} b_i = b.$$

Note that:

- 1) the sequence $\{b_i - b\}_{i=1}^{\infty}$ satisfies conditions b) and c) of [Claim 3.2](#);
 2) since $\sum_{i=1}^{\infty} a_i$ converges, the sequence of partial sums

$$\left\{ \sum_{k=1}^i a_k \right\}_{i=1}^{\infty}$$

is bounded, so $\{a_i\}$ satisfies condition a) of [Claim 3.2](#).

Therefore, by [Claim 3.2](#), the series

$$\sum_{i=1}^{\infty} a_i(b_i - b)$$

converges. Since

$$\sum_{i=1}^{\infty} a_i b_i = b \sum_{i=1}^{\infty} a_i + \sum_{i=1}^{\infty} a_i(b_i - b),$$

the series $\sum_{i=1}^{\infty} a_i b_i$ converges. □

Claim 3.4 (The Leibniz Test). Let $\{b_i\}_{i=1}^{\infty}$ be a sequence of real numbers satisfying

- a) $\{b_i\}_{i=1}^{\infty}$ is nonincreasing, that is,

$$b_i \geq b_{i+1} \quad (\forall i \in \mathbb{N});$$

- b)

$$\lim_{i \rightarrow +\infty} b_i = 0.$$

Then:

- 1) the series

$$\sum_{i=1}^{\infty} (-1)^{i+1} b_i$$

converges;

- 2) for every $n \in \mathbb{N}$,

$$\left| \sum_{i=n+1}^{\infty} (-1)^{i+1} b_i \right| \leq b_{n+1}.$$

Proof. 1) This follows directly from [Claim 3.2](#) (Dirichlet's test) if we set

$$a_i = (-1)^{i+1},$$

because

$$\left| \sum_{i=1}^n a_i \right| = \left| \sum_{i=1}^n (-1)^{i+1} \right| \leq 1.$$

- 2) Let

$$S_n = \sum_{i=1}^n (-1)^{i+1} b_i \quad (n \in \mathbb{N}).$$

For every $n \in \mathbb{N}$,

$$S_{2n+2} = S_{2n} + b_{2n+1} - b_{2n+2} \geq S_{2n},$$

and

$$S_{2n+1} = S_{2n-1} - b_{2n} + b_{2n+1} \leq S_{2n-1}.$$

Therefore $\{S_{2n}\}_{n=1}^{\infty}$ is nondecreasing and $\{S_{2n-1}\}_{n=1}^{\infty}$ is nonincreasing. Due to [item 1](#)), the sequence $\{S_n\}_{n=1}^{\infty}$ is convergent, so both subsequences converge and have the same limit. Let

$$\lim_{n \rightarrow +\infty} S_n = S.$$

Hence

$$S_{2n} \leq S \leq S_{2n-1} \quad (\forall n \in \mathbb{N}).$$

Consequently,

$$0 \leq S - S_{2n} \leq S_{2n+1} - S_{2n} = b_{2n+1},$$

and

$$0 \leq S_{2n-1} - S \leq S_{2n-1} - S_{2n} = b_{2n}.$$

Thus, for every $n \in \mathbb{N}$,

$$|S - S_n| \leq b_{n+1}.$$

Since

$$S - S_n = \sum_{i=n+1}^{\infty} (-1)^{i+1} b_i,$$

the statement follows. □

Claim 3.5 (Sine and cosine sums). For every $n \in \mathbb{N}$ and $\alpha, b \in \mathbb{R}$, we have

$$\sum_{k=1}^n \sin(\alpha k + b) = \begin{cases} \frac{\sin\left(\frac{n+1}{2}\alpha + b\right) \sin\left(\frac{n}{2}\alpha\right)}{\sin\left(\frac{\alpha}{2}\right)}, & \alpha \notin 2\pi\mathbb{Z}, \\ n \sin b, & \alpha \in 2\pi\mathbb{Z}. \end{cases}$$

Proof. If $\alpha \in 2\pi\mathbb{Z}$, then $\sin(\alpha k + b) = \sin b$, and the formula is immediate.

Assume that $\alpha \notin 2\pi\mathbb{Z}$, and let

$$S_n = \sum_{k=1}^n \sin(\alpha k + b).$$

Using

$$2 \sin x \sin y = \cos(x - y) - \cos(x + y) \quad (\forall x, y \in \mathbb{R}),$$

we obtain

$$\begin{aligned} 2 \sin\left(\frac{\alpha}{2}\right) S_n &= \sum_{k=1}^n 2 \sin(\alpha k + b) \sin\left(\frac{\alpha}{2}\right) \\ &= \sum_{k=1}^n \left(\cos\left(\alpha k + b - \frac{\alpha}{2}\right) - \cos\left(\alpha k + b + \frac{\alpha}{2}\right) \right) \\ &= \cos\left(\alpha + b - \frac{\alpha}{2}\right) - \cos\left(\alpha n + b + \frac{\alpha}{2}\right). \end{aligned}$$

This is a telescoping sum. Now use

$$\cos x - \cos y = 2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{y-x}{2}\right) \quad (\forall x, y \in \mathbb{R})$$

to get

$$2 \sin\left(\frac{\alpha}{2}\right) S_n = 2 \sin\left(\frac{n+1}{2}\alpha + b\right) \sin\left(\frac{n}{2}\alpha\right).$$

It remains to divide both sides by $2 \sin(\alpha/2) \neq 0$. □

Claim 3.6. For every $n \in \mathbb{N}$, we have

1)

$$S_n = \sin 1 + \sin 2 + \cdots + \sin n = \frac{\sin\left(\frac{n+1}{2}\right) \sin\left(\frac{n}{2}\right)}{\sin\left(\frac{1}{2}\right)};$$

2)

$$C_n = \cos 1 + \cos 2 + \cdots + \cos n = \frac{\cos\left(\frac{n+1}{2}\right) \sin\left(\frac{n}{2}\right)}{\sin\left(\frac{1}{2}\right)}.$$

Proof. 1) Follows directly from [Claim 3.5](#) by setting $\alpha = 1$ and $b = 0$.

2) Follows directly from [Claim 3.5](#) by setting $\alpha = 1$ and $b = \pi/2$ and using

$$\sin\left(x + \frac{\pi}{2}\right) = \cos x \quad (\forall x \in \mathbb{R}).$$

□

Definition 3.1. A series $\sum_{k=1}^{\infty} a_k$ is called

- *absolutely convergent* if the series $\sum_{k=1}^{\infty} |a_k|$ converges;
- *conditionally convergent* if it converges but the series $\sum_{k=1}^{\infty} |a_k|$ diverges.

Thus, there are three alternatives for a series $\sum_{k=1}^{\infty} a_k$:

- the series is absolutely convergent;
- the series is conditionally convergent;
- the series is divergent.

Claim 3.7. An absolutely convergent series is convergent. That is,

$$\left(\sum_{k=1}^{\infty} |a_k| \text{ converges}\right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges}\right).$$

Proof. Since

$$0 \leq a_k + |a_k| \leq 2|a_k| \quad (\forall k \in \mathbb{N}),$$

and the series $\sum_{k=1}^{\infty} |a_k|$ converges, due to [Claim 2.4](#) (the First Comparison Test), the series

$$\sum_{k=1}^{\infty} (a_k + |a_k|)$$

converges. Since

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} (a_k + |a_k|) - \sum_{k=1}^{\infty} |a_k|,$$

due to [Claim 1.8](#), the series $\sum_{k=1}^{\infty} a_k$ converges. □

Claim 3.8 (Alternating p -series). The series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$$

1) converges absolutely if $p > 1$;

2) converges conditionally if $0 < p \leq 1$;

3) diverges if $p \leq 0$.

Proof. 1) Since

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k^p} \right| = \sum_{k=1}^{\infty} \frac{1}{k^p},$$

item 1) follows from Claim 2.7.

3) If $p \leq 0$, then

$$\lim_{k \rightarrow +\infty} \frac{(-1)^{k+1}}{k^p} \neq 0.$$

Hence item 3) follows from Claim 1.5 (the necessary condition for convergence).

2) If $p > 0$, the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$$

converges due to Claim 3.4 (the Leibniz Test). For $0 < p \leq 1$, the series of absolute values $\sum_{k=1}^{\infty} 1/k^p$ diverges by Claim 2.7. Therefore the convergence is conditional. $\square$

Definition 3.2. Let $\tau : \mathbb{N} \rightarrow \mathbb{N}$ be a *permutation* of the set of natural numbers; that is, τ is a bijection from $\mathbb{N}$ to $\mathbb{N}$. Then the series

$$\sum_{k=1}^{\infty} a_{\tau(k)}$$

is called a *rearrangement* of the series

$$\sum_{k=1}^{\infty} a_k.$$

Roughly speaking, a rearrangement changes the order of the terms of the series. For example,

$$a_2 + a_1 + a_4 + a_3 + a_6 + a_5 + \cdots + a_{2n} + a_{2n-1} + \cdots$$

is a rearrangement of $\sum_{k=1}^{\infty} a_k$.

Claim 3.9 (Cauchy's Rearrangement Theorem). Let

$$\sum_{k=1}^{\infty} a_k$$

be an absolutely convergent series. Then, for any permutation $\tau : \mathbb{N} \rightarrow \mathbb{N}$, we have

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \sum_{k=1}^{\infty} a_k.$$

Proof. Let

$$\sum_{k=1}^{\infty} a_k = \alpha.$$

We need to prove that

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \alpha.$$

By the definition of convergence, it suffices to prove that

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n > N, \quad \left| \alpha - \sum_{k=1}^n a_{\tau(k)} \right| < \varepsilon. \quad (\diamond)$$

Since the series is absolutely convergent, $\sum_{k=1}^{\infty} |a_k|$ converges. Due to [Claim 1.7](#),

$$\forall \varepsilon > 0 \exists M \in \mathbb{N} : \forall n \geq M, \quad \sum_{k=n+1}^{\infty} |a_k| < \varepsilon. \quad (\nabla)$$

Choose $N \in \mathbb{N}$ such that

$$\{1, 2, \dots, M\} \subseteq \{\tau(1), \tau(2), \dots, \tau(N)\}.$$

This can be done since $\tau : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection. Then for every $n > N$,

$$\begin{aligned} \left| \alpha - \sum_{k=1}^n a_{\tau(k)} \right| &= \left| \sum_{k=1}^{\infty} a_k - \sum_{k=1}^n a_{\tau(k)} \right| \\ &\leq \sum_{k=M+1}^{\infty} |a_k| \\ &< \varepsilon, \end{aligned}$$

where the terms $a_1, a_2, \dots, a_M$ cancel due to the choice of N . This proves [\(\diamond\)](#). $\square$

Claim 3.10. Let $\{a_k\}_{k=1}^{\infty}$ be a sequence of real numbers. Let λ_k be the k -th nonnegative term of $\{a_k\}_{k=1}^{\infty}$, and let β_k be the k -th negative term of $\{a_k\}_{k=1}^{\infty}$.

For example, if

$$\{a_k\}_{k=1}^{\infty} = (1, 0, -7, 2, -3, 1, -1, \dots),$$

then

$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = 2, \quad \lambda_4 = 1,$$

and

$$\beta_1 = -7, \quad \beta_2 = -3, \quad \beta_3 = -1.$$

If the series $\sum_{k=1}^{\infty} a_k$ converges conditionally, then:

- 1) there are infinitely many nonnegative and infinitely many negative terms in the sequence $\{a_k\}_{k=1}^{\infty}$;
- 2) both series

$$\sum_{k=1}^{\infty} \lambda_k \quad \text{and} \quad \sum_{k=1}^{\infty} \beta_k$$

are divergent.

Proof. 1) Suppose that the set of nonnegative terms of the sequence $\{a_k\}_{k=1}^{\infty}$ is finite. Then there exists $N \in \mathbb{N}$ such that

$$\forall k > N, \quad a_k < 0.$$

Hence, since $|a_k| = -a_k$ for $k > N$,

$$\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^N |a_k| + \sum_{k=N+1}^{\infty} |a_k| = M - \sum_{k=N+1}^{\infty} a_k,$$

where $M = \sum_{k=1}^N |a_k|$ is a constant. The series on the right converges because $\sum a_k$ converges. Therefore $\sum a_k$ converges absolutely, which is a contradiction. Thus, the set of nonnegative terms is infinite. In a similar way, one proves that the set of negative terms is also infinite.

- 2) Due to [item 1\)](#), we can define the series $\sum_{k=1}^{\infty} \lambda_k$ and $\sum_{k=1}^{\infty} \beta_k$.

Suppose first that both series are convergent:

$$\sum_{k=1}^{\infty} \lambda_k = \lambda \in \mathbb{R}, \quad \sum_{k=1}^{\infty} \beta_k = \beta \in \mathbb{R}.$$

Then, for every $N \in \mathbb{N}$,

$$\sum_{k=1}^N |a_k| \leq \sum_{k=1}^N \lambda_k + \sum_{k=1}^N |\beta_k| \leq \lambda + |\beta|.$$

Therefore, due to [Claim 2.2](#), the series $\sum_{k=1}^{\infty} |a_k|$ converges. This contradicts conditional convergence. Hence at least one of the series $\sum \lambda_k$ or $\sum \beta_k$ diverges.

Suppose that one of these series converges and the other diverges. Without loss of generality, assume that

$$\sum_{k=1}^{\infty} \lambda_k \text{ diverges, } \quad \sum_{k=1}^{\infty} \beta_k = \beta \in \mathbb{R}.$$

Then

$$\forall E > 0 \exists N \in \mathbb{N} : \forall n \geq N, \quad \sum_{k=1}^n \lambda_k > E - \beta.$$

Choose $M \in \mathbb{N}$ such that

$$\{\lambda_1, \dots, \lambda_N\} \subseteq \{a_1, \dots, a_M\}.$$

Then

$$\sum_{k=1}^M a_k \geq \sum_{k=1}^N \lambda_k + \beta > E - \beta + \beta = E.$$

Since $E > 0$ is arbitrary, the partial sums of $\sum a_k$ are unbounded above, so $\sum a_k$ diverges. This is a contradiction. Therefore both $\sum \lambda_k$ and $\sum \beta_k$ are divergent. $\square$

Claim 3.11 (The Riemann Rearrangement Theorem). Let

$$\sum_{k=1}^{\infty} a_k$$

be a conditionally convergent series. Then, for any $\alpha \in \mathbb{R}$, there exists a permutation $\tau : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \alpha.$$

Proof. Let λ_k be the k -th nonnegative term and β_k the k -th negative term of the sequence $\{a_k\}_{k=1}^{\infty}$. We construct the rearranged series $\sum_{k=1}^{\infty} a_{\tau(k)}$ as follows.

Let $\alpha \in \mathbb{R}$ be arbitrary.

- Let p_1 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k > \alpha.$$

Such p_1 exists because $\sum_{k=1}^{\infty} \lambda_k = +\infty$ by [item 2\)](#) of [Claim 3.10](#).

- Let q_1 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k < \alpha.$$

Such q_1 exists because $\sum_{k=1}^{\infty} \beta_k = -\infty$ by [item 2\)](#) of [Claim 3.10](#).

- Let p_2 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k + \sum_{k=p_1+1}^{p_2} \lambda_k > \alpha.$$

Again, p_2 exists because $\sum \lambda_k = +\infty$.

- Let q_2 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k + \sum_{k=p_1+1}^{p_2} \lambda_k + \sum_{k=q_1+1}^{q_2} \beta_k < \alpha.$$

Again, q_2 exists because $\sum \beta_k = -\infty$.

Continuing inductively, we obtain the sequence

$$\lambda_1, \dots, \lambda_{p_1}, \beta_1, \dots, \beta_{q_1}, \lambda_{p_1+1}, \dots, \lambda_{p_2}, \beta_{q_1+1}, \dots, \beta_{q_2}, \lambda_{p_2+1}, \dots \quad (3)$$

whose partial sums alternately pass above and below α .

The picture below illustrates the construction schematically.

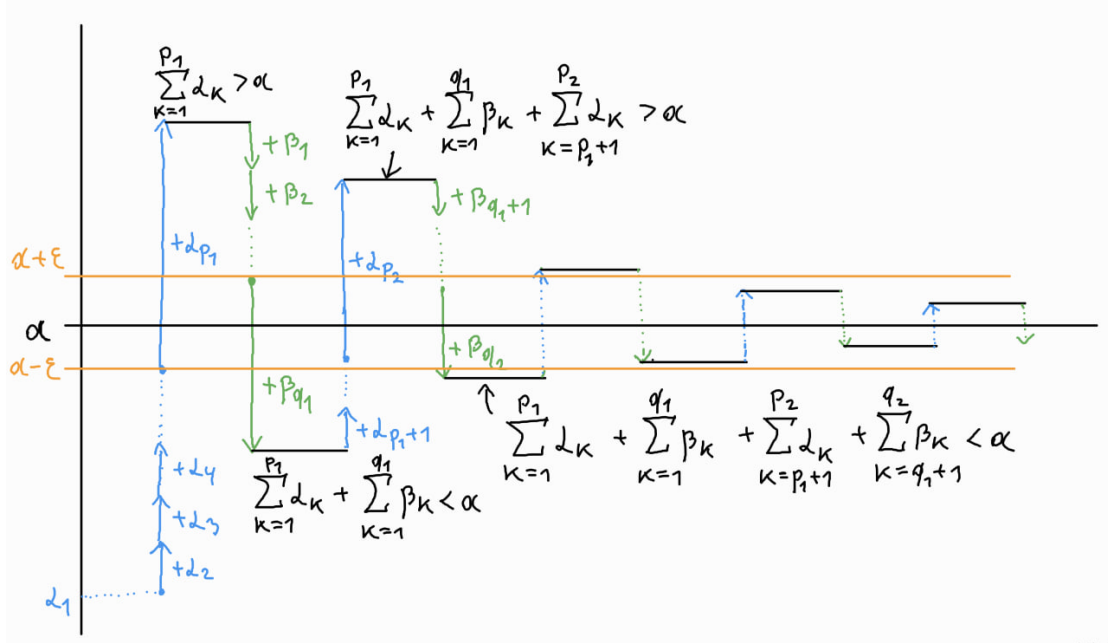


Figure 3.1

The constructed series converges to α . Indeed, note that

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \implies \left(\lim_{k \rightarrow +\infty} a_k = 0 \right) \iff \left(\lim_{k \rightarrow +\infty} \lambda_k = 0 \text{ and } \lim_{k \rightarrow +\infty} \beta_k = 0 \right).$$

Equivalently,

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall k > N, \quad \lambda_k < \varepsilon \quad \text{and} \quad |\beta_k| < \varepsilon.$$

Therefore, after all λ_k and β_k with $k \in \{1, \dots, N\}$ have been included in a partial sum of [sequence \(3\)](#)—which eventually happens because at least one λ_i and one β_i are added at each step—all further deviations of partial sums from α do not exceed ε . This follows from $\lambda_k < \varepsilon$ and $|\beta_k| < \varepsilon$ for $k > N$, together with the choice of the integers p_i and q_i . Due to [Claim 1.2](#) (Cauchy's criterion), the constructed series converges to α .

Every entry in the original sequence $\{a_k\}_{k=1}^{\infty}$ appears in [sequence \(3\)](#) exactly once. Thus [sequence \(3\)](#) is a permutation of the original sequence $\{a_k\}_{k=1}^{\infty}$. $\square$

Calculus 2

4 Lecture 4

Product of Series

Definition 4.1. Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two series. The *Cauchy product* of these two series is defined as follows:

$$\left(\sum_{k=1}^{\infty} a_k \right) \left(\sum_{k=1}^{\infty} b_k \right) \stackrel{\text{def}}{=} \sum_{k=1}^{\infty} c_k, \quad c_k = a_1 b_k + a_2 b_{k-1} + \cdots + a_k b_1.$$

For example,

$$c_1 = a_1 b_1, \quad c_2 = a_1 b_2 + a_2 b_1, \quad c_3 = a_1 b_3 + a_2 b_2 + a_3 b_1.$$

Question. Is it true that

$$\left(\sum_{k=1}^{\infty} a_k = a \in \mathbb{R} \quad \text{and} \quad \sum_{k=1}^{\infty} b_k = b \in \mathbb{R} \right) \implies \left(\sum_{k=1}^{\infty} c_k = ab \right),$$

where $c_k = a_1 b_k + \cdots + a_k b_1$? In other words, if both series converge, must their Cauchy product converge to the product of their sums?

The following example shows that the answer is **no**.

Example 4.1. Let

$$a_k = b_k = \frac{(-1)^{k+1}}{\sqrt{k}}.$$

Then the series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ are conditionally convergent, due to [Claim 3.8](#) (alternating p -series). The terms of their Cauchy product are

$$\begin{aligned} c_k &= \frac{(-1)^{1+1}}{\sqrt{1}} \frac{(-1)^{k+1}}{\sqrt{k}} + \cdots + \frac{(-1)^{i+1}}{\sqrt{i}} \frac{(-1)^{k-i+2}}{\sqrt{k-i+1}} + \cdots + \frac{(-1)^{k+1}}{\sqrt{k}} \frac{(-1)^{1+1}}{\sqrt{1}} \\ &= (-1)^{k+3} \sum_{i=1}^k \frac{1}{\sqrt{i} \sqrt{k-i+1}}. \end{aligned}$$

Hence

$$|c_k| = \sum_{i=1}^k \frac{1}{\sqrt{i} \sqrt{k-i+1}}.$$

By the arithmetic-geometric mean inequality,

$$\sqrt{ab} \leq \frac{a+b}{2} \quad (a > 0, b > 0),$$

and with $a = i$, $b = k - i + 1$,

$$\frac{1}{\sqrt{i} \sqrt{k-i+1}} \geq \frac{2}{i+k-i+1} = \frac{2}{k+1}.$$

Therefore

$$|c_k| \geq \sum_{i=1}^k \frac{2}{k+1} = \frac{2k}{k+1} \geq 1.$$

Since $\lim_{k \rightarrow \infty} c_k \neq 0$, by [Claim 1.5](#) (the necessary condition for convergence), the series

$$\sum_{k=1}^{\infty} c_k$$

diverges.

The following two claims, which we give without proof, provide sufficient conditions for the implication above to hold.

Claim 4.1 (The Mertens Theorem). Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two convergent series with sums a and b , respectively. If at least one of them converges absolutely, then their Cauchy product ([Definition 4.1](#)) converges to ab .

Thus, the implication in the question is true if at least one of the series is absolutely convergent.

Claim 4.2 (The Abel Theorem on the Cauchy product). Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be two convergent series with sums a and b , respectively. If their Cauchy product

$$\sum_{k=1}^{\infty} c_k$$

(see [Definition 4.1](#)) is a convergent series, then

$$\sum_{k=1}^{\infty} c_k = ab.$$

Definition 4.2. Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two series. Let

$$\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2, \quad i \mapsto (m_i, n_i),$$

be a bijection from $\mathbb{N}$ to $\mathbb{N}^2$. We define the $\tilde{\tau}$ -product of the series $\sum a_k$ and $\sum b_k$ by

$$\left(\sum_{k=1}^{\infty} a_k \right) \dot{\underset{\tau}{\cdot}} \left(\sum_{k=1}^{\infty} b_k \right) = \sum_{i=1}^{\infty} a_{m_i} b_{n_i}.$$

Note 1. Since there are infinitely many bijections of the form $\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2$, there are infinitely many distinct definitions of the product of two series.

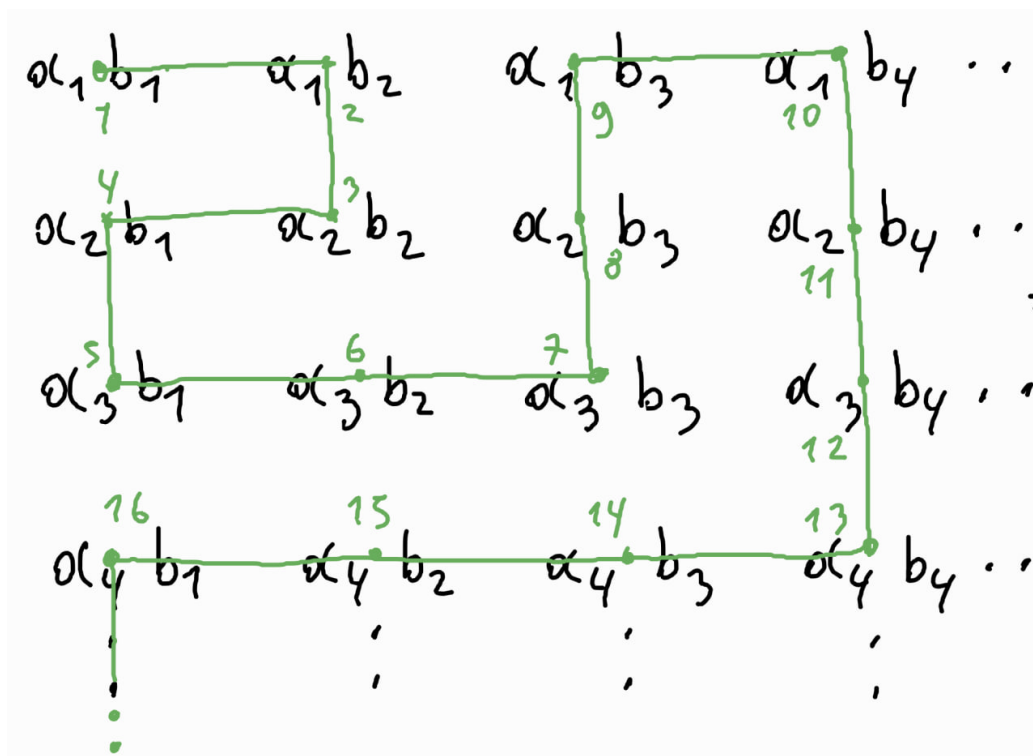
Note 2. The idea behind [Definition 4.2](#) is the following. If we need to multiply

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k,$$

we need to expand the brackets

$$(a_1 + a_2 + \cdots + a_k + \cdots)(b_1 + b_2 + \cdots + b_k + \cdots).$$

For example:
 “the serpent” order (= bijection $\tilde{\tau}$):


$$\left(\sum_{i=1}^{\infty} a_i\right) \cdot_{\tau} \left(\sum_{i=1}^{\infty} b_i\right) = a_1 b_1 + a_1 b_2 + a_2 b_2 + a_2 b_1 + a_3 b_1 + a_3 b_2 + a_3 b_3 + a_2 b_3 + a_1 b_3 + a_1 b_4 + a_2 b_4 + \cdots$$

28

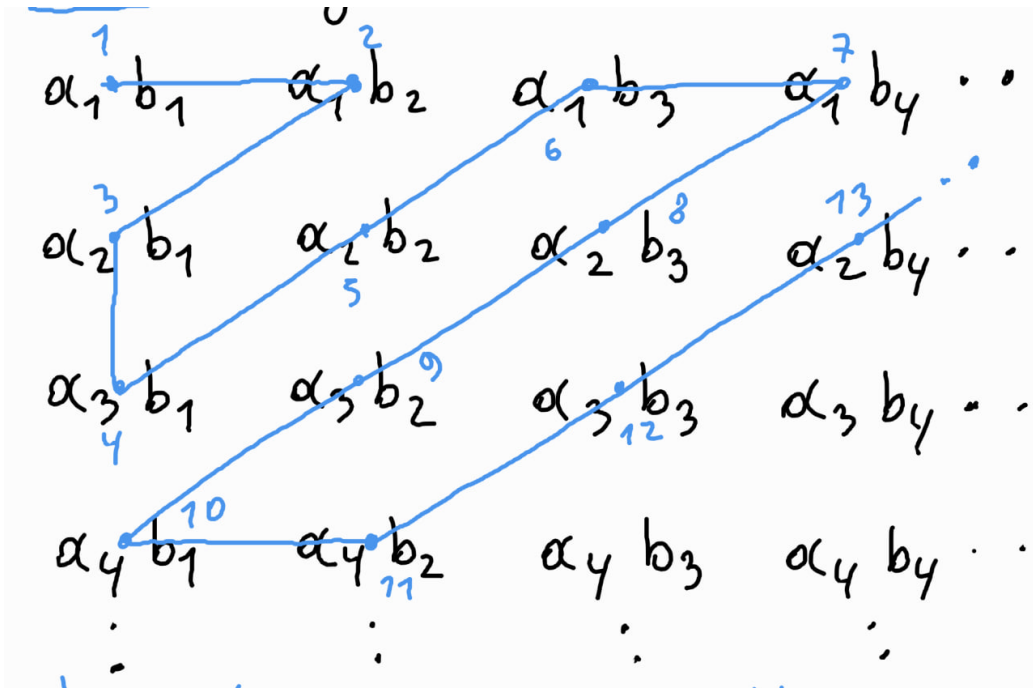


Figure 4.2

$$\left(\sum_{i=1}^{\infty} a_i \right) \cdot_{\tilde{\tau}} \left(\sum_{i=1}^{\infty} b_i \right) = a_1 b_1 + (a_1 b_2 + a_2 b_1) + (a_3 b_1 + a_2 b_2 + a_1 b_3) + (a_1 b_4 + a_2 b_3 + a_3 b_2 + a_4 b_1) + \cdots,$$

where terms c_i are the same as in the Cauchy product ([Definition 4.1](#)).

The next theorem shows that if both series are absolutely convergent, then the choice of the order does not matter.

Claim 4.3 (The Abel Theorem on Series Product). Let

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=1}^{\infty} b_k$$

be two absolutely convergent series with sums a and b , respectively. Then, for every bijection

$$\tilde{\tau} : \mathbb{N} \rightarrow \mathbb{N}^2, \quad i \mapsto (m_i, n_i),$$

we have

$$\left(\sum_{k=1}^{\infty} a_k \right) \cdot_{\tilde{\tau}} \left(\sum_{k=1}^{\infty} b_k \right) = \sum_{i=1}^{\infty} a_{m_i} b_{n_i} = ab.$$

Proof. For every $N \in \mathbb{N}$, let

$$K = \max\{m_1, \dots, m_N, n_1, \dots, n_N\}.$$

Then

$$\sum_{i=1}^N |a_{m_i} b_{n_i}| \leq \left(\sum_{i=1}^K |a_i| \right) \left(\sum_{i=1}^K |b_i| \right).$$

The inequality holds because all terms are nonnegative and every term from the left-hand sum is present in the product on the right. Hence

$$\sum_{i=1}^N |a_{m_i} b_{n_i}| \leq d\beta,$$

where

$$\sum_{i=1}^{\infty} |a_i| = d, \quad \sum_{i=1}^{\infty} |b_i| = \beta.$$

Since the two series are absolutely convergent, $d, \beta \in \mathbb{R}$, and the sequences of partial sums

$$\left\{ \sum_{i=1}^n |a_i| \right\}_{n=1}^{\infty}, \quad \left\{ \sum_{i=1}^n |b_i| \right\}_{n=1}^{\infty}$$

are nondecreasing; in particular their partial sums are bounded by d and β , respectively.

Thus the sequence of partial sums

$$\left\{ \sum_{i=1}^N |a_{m_i} b_{n_i}| \right\}_{N=1}^{\infty}$$

is bounded by $d\beta$. Since this sequence is nondecreasing, [Claim 2.2](#) implies that

$$\sum_{i=1}^{\infty} |a_{m_i} b_{n_i}|$$

is convergent. By [Definition 3.1](#), the series

$$\sum_{i=1}^{\infty} a_{m_i} b_{n_i}$$

is absolutely convergent.

Therefore, by [Claim 3.9](#) (Cauchy's Rearrangement Theorem), no permutation $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ affects its sum. We may rearrange the series so that the order is diagonal:

$$a_1 b_1 + (a_1 b_2 + a_2 b_1) + (a_1 b_3 + a_2 b_2 + a_3 b_1) + \cdots + (a_1 b_k + a_2 b_{k-1} + \cdots + a_k b_1) + \cdots .$$

Grouping the terms gives

$$c_1 + c_2 + c_3 + \cdots + c_k + \cdots .$$

By [Definition 4.1](#), this is the Cauchy product of $\sum a_i$ and $\sum b_i$. Since these series are absolutely convergent, [Claim 4.1](#) (Mertens' theorem) gives

$$\sum_{k=1}^{\infty} c_k = ab.$$

Therefore

$$\sum_{i=1}^{\infty} a_{m_i} b_{n_i} = ab.$$

□

The Stirling Formula

To prove the formula, we will need the following fact.

Claim 4.4. Let

$$I_n = \int_0^{\pi/2} \sin^n(x) dx, \quad n \in \mathbb{N} \cup \{0\}.$$

Then:

- 1) the sequence $\{I_n\}_{n=1}^{\infty}$ is nonincreasing;

2) for every $n \in \mathbb{N}$,

$$I_{2n} = \frac{\pi}{2} \cdot \frac{(2n-1)!!}{(2n)!!}, \quad I_{2n+1} = \frac{(2n)!!}{(2n+1)!!}.$$

Proof. 1) For every $x \in [0, \pi/2]$,

$$0 \leq \sin x \leq 1.$$

Hence, for every $n \in \mathbb{N} \cup \{0\}$,

$$0 \leq \sin^{n+1} x \leq \sin^n x \leq 1,$$

and therefore

$$I_{n+1} = \int_0^{\pi/2} \sin^{n+1} x \, dx \leq \int_0^{\pi/2} \sin^n x \, dx = I_n.$$

Thus $\{I_n\}$ is nonincreasing.

2) We prove the equalities by induction on $n \in \mathbb{N} \cup \{0\}$.

For $n = 0$,

$$I_0 = \int_0^{\pi/2} \sin^0 x \, dx = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2} = \frac{\pi}{2} \cdot \frac{(2 \cdot 0 - 1)!!}{(2 \cdot 0)!!}.$$

For $n = 1$,

$$I_1 = \int_0^{\pi/2} \sin x \, dx = -\cos x \Big|_0^{\pi/2} = 1 = \frac{(2 \cdot 0)!!}{(2 \cdot 0 + 1)!!}.$$

For the induction step, note that

$$\begin{aligned} I_{n+2} &= \int_0^{\pi/2} \sin^{n+1} x \sin x \, dx \\ &= -\cos x \sin^{n+1} x \Big|_0^{\pi/2} + \int_0^{\pi/2} (n+1) \sin^n x \cos^2 x \, dx \\ &= (n+1) \int_0^{\pi/2} \sin^n x (1 - \sin^2 x) \, dx \\ &= (n+1) I_n - (n+1) I_{n+2}. \end{aligned}$$

Thus

$$I_{n+2} = \frac{n+1}{n+2} I_n \quad (n \in \mathbb{N} \cup \{0\}).$$

Therefore

$$I_{2n+2} = \frac{2n+1}{2n+2} I_{2n} = \frac{2n+1}{2n+2} \cdot \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} = \frac{\pi}{2} \frac{(2n+1)!!}{(2n+2)!!},$$

and

$$I_{2n+3} = \frac{2n+2}{2n+3} I_{2n+1} = \frac{2n+2}{2n+3} \cdot \frac{(2n)!!}{(2n+1)!!} = \frac{(2n+2)!!}{(2n+3)!!}.$$

□

Claim 4.5 (The Wallis Formula). Let

$$P_n = \prod_{k=1}^n \frac{(2k)^2}{(2k)^2 - 1} \quad (n \in \mathbb{N}).$$

Then

$$\lim_{n \rightarrow \infty} P_n = \frac{\pi}{2}.$$

Proof. For every $n \in \mathbb{N}$,

$$\frac{P_n}{P_{n-1}} = \frac{(2n)^2}{(2n)^2 - 1} > 1,$$

so the sequence $\{P_n\}_{n=1}^{\infty}$ is increasing.

Also,

$$\begin{aligned} P_n &= \prod_{k=1}^n \frac{(2k)^2}{(2k-1)(2k+1)} \\ &= \frac{(2 \cdot 4 \cdot 6 \cdots 2n)^2}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots (2n-1)(2n+1)} \\ &= \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1). \end{aligned} \tag{**}$$

By item 1) of Claim 4.4,

$$I_{2n+1} \leq I_{2n} \leq I_{2n-1}.$$

Using item 2) of Claim 4.4,

$$\frac{(2n)!!}{(2n+1)!!} \leq \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} \leq \frac{(2n-2)!!}{(2n-1)!!}.$$

Multiplying by $\frac{(2n)!!}{(2n-1)!!}$ gives

$$\left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1) \leq \frac{\pi}{2} \leq \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 \frac{(2n+1)^2}{2n}.$$

By (**),

$$P_n \leq \frac{\pi}{2} \leq P_n \frac{2n+1}{2n}.$$

Hence

$$\lim_{n \rightarrow \infty} P_n = \frac{\pi}{2}.$$

□

Claim 4.6 (Almost the Stirling Formula). There exists a limit

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{n} (n/e)^n} = A > 0.$$

That is, we prove that the limit exists and its value is positive.

Proof. Let

$$a_n = \frac{n!}{\sqrt{n} (n/e)^n} = \frac{n! e^n}{n^{n+1/2}} \quad (n \in \mathbb{N}).$$

We need to prove that $\lim_{n \rightarrow \infty} a_n = A > 0$.

Note that

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{(n+1)! e^{n+1}}{(n+1)^{n+1+1/2}} \cdot \frac{n^{n+1/2}}{n! e^n} \\ &= e \left(\frac{n}{n+1} \right)^{n+1/2} = e \left(1 + \frac{1}{n} \right)^{-n-1/2}. \end{aligned}$$

Hence

$$\begin{aligned}
\ln\left(\frac{a_{n+1}}{a_n}\right) &= 1 - \left(n + \frac{1}{2}\right) \ln\left(1 + \frac{1}{n}\right) \\
&= 1 - \left(n + \frac{1}{2}\right) \left(\frac{1}{n} - \frac{1}{2} \frac{1}{n^2} + \frac{1}{3} \frac{1}{n^3} + O\left(\frac{1}{n^4}\right)\right) \\
&= 1 - \left(1 - \frac{1}{2} \frac{1}{n} + \frac{1}{3} \frac{1}{n^2} + O\left(\frac{1}{n^3}\right)\right) - \left(\frac{1}{2} \frac{1}{n} - \frac{1}{4} \frac{1}{n^2} + \frac{1}{6} \frac{1}{n^3} + O\left(\frac{1}{n^4}\right)\right) \\
&= \left(-\frac{1}{3} + \frac{1}{4}\right) \frac{1}{n^2} + O\left(\frac{1}{n^3}\right) \\
&= \frac{1}{n^2} \left(-\frac{1}{12} + O\left(\frac{1}{n}\right)\right). \tag{1}
\end{aligned}$$

Let

$$P_n = \prod_{k=1}^n \frac{a_{k+1}}{a_k} = \frac{a_2}{a_1} \frac{a_3}{a_2} \dots \frac{a_{n+1}}{a_n} = \frac{a_{n+1}}{a_1}. \tag{3}$$

Then, by (1),

$$\ln(P_n) = \sum_{k=1}^n \ln\left(\frac{a_{k+1}}{a_k}\right) = \sum_{k=1}^n \frac{1}{k^2} \left(-\frac{1}{12} + O\left(\frac{1}{k}\right)\right). \tag{2}$$

Since the factor in parentheses is bounded for all sufficiently large k , the series converges absolutely by [Claim 2.4](#) and [Claim 2.7](#):

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \left(-\frac{1}{12} + O\left(\frac{1}{k}\right)\right)$$

converges, say to $b \in \mathbb{R}$. Therefore, from (2),

$$\lim_{n \rightarrow \infty} \ln(P_n) = b.$$

Thus

$$\lim_{n \rightarrow \infty} P_n = e^b \in \mathbb{R}. \tag{4}$$

Finally,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n+1} = a_1 \lim_{n \rightarrow \infty} P_n = a_1 e^b.$$

Since $a_1 = e$, we obtain

$$A = e^{b+1} > 0.$$

□

Claim 4.7 (The Stirling Formula).

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} (n/e)^n} = 1,$$

equivalently,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad n \rightarrow +\infty.$$

Note.

$$\lim_{n \rightarrow \infty} \left(n! - \sqrt{2\pi n} \left(\frac{n}{e} \right)^n \right) = +\infty.$$

Proof. Due to [Claim 4.5](#) (the Wallis formula) and equality [\(**\)](#), we have

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 (2n+1) = \frac{\pi}{2}.$$

Thus

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \sqrt{2n+1} \right) = \sqrt{\frac{\pi}{2}}. \quad (5)$$

On the other hand,

$$\frac{(2n)!!}{(2n+1)!!} = \frac{2 \cdot 4 \cdot 6 \cdots 2n}{1 \cdot 3 \cdot 5 \cdots (2n+1)} = \frac{2^{2n} (n!)^2}{(2n+1)!}.$$

Hence

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^{2n} (n!)^2}{(2n+1)!} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^{2n} (A\sqrt{n} (n/e)^n)^2}{A\sqrt{2n+1} ((2n+1)/e)^{2n+1}} \sqrt{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} \frac{2^{2n} e A n^{2n+1}}{(2n+1)^{2n+1}} \\ &= \lim_{n \rightarrow \infty} \left(e A 2^{2n} \left(\frac{2n+1}{n} \right)^{-2n-1} \right) \\ &= \lim_{n \rightarrow \infty} \left(e A \cdot \frac{1}{2} \left(1 + \frac{1}{2n} \right)^{-2n} \left(1 + \frac{1}{2n} \right)^{-1} \right) \\ &= \frac{1}{2} A. \end{aligned}$$

Therefore, by [\(5\)](#),

$$\sqrt{\frac{\pi}{2}} = \frac{1}{2} A,$$

so

$$A = \sqrt{2\pi}.$$

By [Claim 4.6](#),

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi} \sqrt{n} (n/e)^n} = 1,$$

which is the Stirling formula. □

Claim 4.8 (The Robbins Formula). For every $n \in \mathbb{N}$, we have

$$\sqrt{2\pi n} \left(\frac{n}{e} \right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n} \left(\frac{n}{e} \right)^n e^{\frac{1}{12n}}.$$

Without proof.

Calculus 2

5 Lecture 5

Functional Sequences and Series

Definition 5.1. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the sequence $\{f_n(x)\}_{n=1}^{\infty}$ is called a *functional sequence*;
2. the series $\sum_{n=1}^{\infty} f_n(x)$ is called a *functional series*.

Definition 5.2. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the sequence $\{f_n(x)\}_{n=1}^{\infty}$ *converges pointwise* to a function $f : E \rightarrow \mathbb{R}$ (notation: $f_n \xrightarrow{E} f$) on the set E if

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \text{for every } x \in E.$$

That is,

$$(f_n \xrightarrow{E} f) \iff (\forall x \in E \ \forall \varepsilon > 0 \ \exists N(\varepsilon, x) \in \mathbb{N} \ \forall n \geq N(\varepsilon, x) \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (1)$$

2. the functional series $\sum_{k=1}^{\infty} f_k(x)$ *converges pointwise* to a function $f : E \rightarrow \mathbb{R}$ (notation: $\sum_{k=1}^{\infty} f_k \xrightarrow{E} f$) on the set E if the sequence of partial sums

$$\left\{ S_n(x) = \sum_{k=1}^n f_k(x) \right\}_{n=1}^{\infty}$$

converges pointwise to f .

Claim 5.1 (Properties of Finite Sums). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

Then:

1. if every u_k is continuous on E , then f_n is also continuous on E ;
2. if every u_k is differentiable on E , then f_n is also differentiable on E and

$$f'_n(x) = \sum_{k=1}^n u'_k(x), \quad \forall x \in E;$$

3. if every u_k is Riemann integrable on $[a, b] \subseteq E$, then f_n is also Riemann integrable on $[a, b]$ and

$$\int_a^b f_n(x) dx = \sum_{k=1}^n \int_a^b u_k(x) dx.$$

Proof. See Calculus-1. □

Can we extend Properties 1), 2), and 3) from [Claim 5.1](#) to the series

$$f(x) = \sum_{k=1}^{\infty} u_k(x)?$$

In general, **NO** (see the next Claim), but **YES** under some assumptions.

The main goal of this and the next lectures is to establish such assumptions.

Claim 5.2. Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

Then:

1. If every u_k is continuous on E and $f_n \xrightarrow{E} f$, then f may not be continuous on E .
2. If every u_k is differentiable on E and $f_n \xrightarrow{E} f$, then f may not be differentiable on E .
3. If every u_k is Riemann integrable on $[a, b] \subseteq E$ and $f_n \xrightarrow{E} f$, then f may not be Riemann integrable on $[a, b] \subseteq E$.

Note that, [Claim 5.2](#) shows that the assumption $f_n \xrightarrow{E} f$ is not strong enough, if we want to extend the properties (i.e. continuity, differentiability and Riemann integrability) to

$$f(x) = \sum_{k=1}^{\infty} u_k(x),$$

we need to replace the assumption $f_n \xrightarrow{E} f$ with a stronger one. We will prove the corresponding claims in the next lecture.

Proof. 1) and 2). Let $E = \mathbb{R}$,

$$u_k(x) = \frac{\sin(kx)}{k}, \quad \forall k \in \mathbb{N}.$$

Then

$$\sum_{k=1}^{\infty} \frac{\sin(kx)}{k} \xrightarrow{\mathbb{R}} f(x),$$

that is, the series converges pointwise on $\mathbb{R}$ (due to [Claim 3.2](#) [Dirichlet's Test]).

It can be shown that (we will be able to do it later in our course) that the function f is of the form:

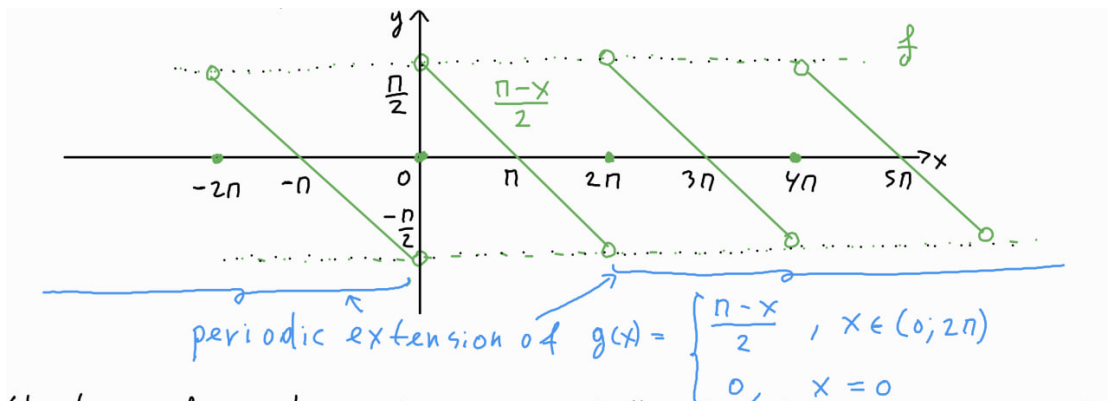


Figure 5.1

It is the periodic extension of

$$g(x) = \begin{cases} \frac{\pi - x}{2}, & x \in (0, 2\pi), \\ 0, & x = 0. \end{cases}$$

That is, f is discontinuous at the points $x = 2\pi m$, $m \in \mathbb{Z}$.

Since, for every $k \in \mathbb{N}$,

$$u_k(x) = \frac{\sin(kx)}{k}$$

is infinitely differentiable on $\mathbb{R}$, [Items 1\)](#) and [2\)](#) are proved.

Also note that the equality

$$f'(x) = \sum_{k=1}^{\infty} \left(\frac{\sin(kx)}{k} \right)' = \sum_{k=1}^{\infty} \cos(kx)$$

has no sense (since the latter series diverges for every $x \in \mathbb{R}$).

3) Let $E = [0, 1]$, $\tilde{\tau} : \mathbb{N} \rightarrow [0, 1] \cap \mathbb{Q}$ be a bijection (you should know that such a bijection exists; why?) and

$$u_k(x) = \begin{cases} 1, & x = \tilde{\tau}(k), \\ 0, & x \in [0, 1] \setminus \{\tilde{\tau}(k)\}, \end{cases} \quad \forall k \in \mathbb{N}.$$

Then,

$$\sum_{k=1}^{\infty} u_k(x) \xrightarrow{E} f(x) = \begin{cases} 1, & x \in [0, 1] \cap \mathbb{Q}, \\ 0, & x \in [0, 1] \text{ and } x \notin \mathbb{Q}. \end{cases}$$

That is, f is the Dirichlet function on $[0, 1]$.

It ought to be clear that f is not Riemann-integrable on $E = [0, 1]$ (since f is nowhere continuous on $[0, 1]$). Since every u_k has finitely many points of discontinuities (actually just one point: $x = \tilde{\tau}(k)$) and it is bounded on $[0, 1]$, it is Riemann-integrable on $[0, 1]$. Thus, the item is proved. $\square$

Definition 5.3. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$.

Then:

1. the functional sequence $\{f_n(x)\}_{n=1}^{\infty}$ is called *uniformly convergent* to a function $f(x)$ on a set E (notation $f_n \xrightarrow{E} f$) if

$$\lim_{n \rightarrow \infty} \sup_{x \in E} |f_n(x) - f(x)| = 0.$$

That is,

$$(f_n \xrightarrow{E} f) \iff (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (2)$$

2. the functional series $\sum_{k=1}^{\infty} f_k(x)$ is called *uniformly convergent* to a function $f(x)$ on a set E (notation $\sum_{k=1}^{\infty} f_k(x) \xrightarrow{E} f(x)$) if the sequence of its partial sums

$$\left\{ S_n(x) = \sum_{k=1}^n f_k(x) \right\}_{n=1}^{\infty}$$

converges uniformly to $f(x)$ on E .

Definition 2) means that, in order to converge uniformly on E , the functions $f_n(x)$ ought to be inside the ε -neighbourhood of $f(x)$ for all $n \geq N(\varepsilon)$ and for all $x \in E$.

Claim 5.3. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$(f_n \xrightarrow{E} f) \implies (f_n \xrightarrow{E} f).$$

That is, if the sequence $\{f_n(x)\}_{n=1}^\infty$ converges uniformly to $f(x)$ on E , then it converges pointwise to $f(x)$ on E .

Note that the converse does not hold true in general.

Proof. ($\implies$) The statement follows directly from [Def 5.2](#) and [Def 5.3](#):

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ & \Updownarrow \quad \text{Def 5.3} \\ & (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \varepsilon) \\ & \Downarrow \quad \text{since we can use } N(\varepsilon, x) = N(\varepsilon) \\ & (\forall \varepsilon > 0 \forall x \in E \exists N(\varepsilon, x) \in \mathbb{N} : \forall n \geq N(\varepsilon, x) \rightarrow |f_n(x) - f(x)| < \varepsilon) \\ & \Updownarrow \quad \text{Def 5.2} \\ & (f_n \xrightarrow{E} f). \end{aligned}$$

For the converse, let

$$f_n(x) = \begin{cases} 1, & x \in [n, n+1], \\ 0, & x \in \mathbb{R} \setminus [n, n+1], \end{cases} \quad f(x) = 0 \quad \forall x \in \mathbb{R}, \quad E = \mathbb{R}.$$

Then $\forall x \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} f_n(x) = 0 = f(x).$$

That is, due to [Def 5.2](#), $f_n \xrightarrow{\mathbb{R}} f$. Since

$$\sup_{x \in \mathbb{R}} |f_n(x) - f(x)| = 1,$$

we have

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| = 1 \neq 0.$$

Therefore, due to [Def 5.3](#), $f_n \not\xrightarrow{\mathbb{R}} f$. □

Claim 5.4 (The Cauchy Criterion for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ & \Updownarrow \\ & (\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon). \end{aligned} \quad (3)$$

Equivalently,

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon).$$

Proof. ($\Downarrow$) Note that

$$(f_n \xrightarrow{E} f)$$

by [Formula 2](#) implies

$$(\forall \varepsilon > 0 \exists N_1(\varepsilon) : \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{2}).$$

Let $N(\varepsilon) = N_1(\varepsilon)$ in [Formula \(3\)](#). Then, for $\forall k \geq N(\varepsilon), \forall m \geq N(\varepsilon), \forall x \in E$, we have

$$\begin{aligned} |f_m(x) - f_k(x)| &= |f_m(x) - f(x) + f(x) - f_k(x)| \\ &\leq |f_m(x) - f(x)| + |f_k(x) - f(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus, we proved ($\Downarrow$).

($\Uparrow$) Note that

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon)$$

implies

$$(\forall x \in E \forall \varepsilon > 0 \exists N(\varepsilon, x) \in \mathbb{N} : \forall k \geq N(\varepsilon, x) \forall m \geq N(\varepsilon, x) \rightarrow |f_m(x) - f_k(x)| < \varepsilon).$$

For a fixed $x \in E$, this is the Cauchy convergence criterion for the sequence $\{f_n(x)\}_{n=1}^{\infty}$ (see Calculus-1). Therefore,

$$\forall x \in E \rightarrow \exists \lim_{n \rightarrow \infty} f_n(x).$$

Therefore, we can define $f : E \rightarrow \mathbb{R}$ as

$$f(x) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} f_n(x), \quad \forall x \in E.$$

Thus, we proved that $f_n \xrightarrow{E} f$.

Now, if in [Formula \(3\)](#) we take the limit $k \rightarrow \infty$, we will receive [Formula 2](#) from [Def 5.3](#):

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f(x)| \leq \varepsilon.$$

Here we used that $\forall x \in E \exists \lim_{k \rightarrow \infty} f_k(x) = f(x)$. Therefore, due to [Def 5.3](#), $f_n \xrightarrow{E} f$. $\square$

Claim 5.5 (The Negation of the Cauchy Criterion for the UC). Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $n \in \mathbb{N}$. Then

$$(f_n \not\xrightarrow{E} f)$$

$$\Uparrow$$

$$(\exists \varepsilon > 0 \forall N \in \mathbb{N} \exists k \geq N \exists m \geq N \exists x \in E \rightarrow |f_m(x) - f_k(x)| \geq \varepsilon). \quad (4)$$

Example. Study the uniform convergence of the sequence

$$\left\{ f_n(x) = \frac{\ln(nx)}{\sqrt{nx}} \right\}_{n=1}^{\infty}$$

on the set $E = (0, 1)$.

Solution. Since, for every $x \in (0, 1)$, we have

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{\ln(nx)}{\sqrt{nx}} = 0 \quad // \text{see Calculus-1} //$$

due to [Def 5.2](#), the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges pointwise to the function $f \equiv 0$ on $E = (0, 1)$.

Now, using [Claim 5.5](#), we will prove that $f_n \not\stackrel{E}{\rightrightarrows} f \equiv 0$. In [Formula \(4\)](#), let

$$\varepsilon = \frac{1}{4}, \quad k = N, \quad m = 2N, \quad x = \frac{1}{N} \in E = (0, 1).$$

Then, we have:

$$\begin{aligned} |f_m(x) - f_k(x)| &= \left| f_{2N}\left(\frac{1}{N}\right) - f_N\left(\frac{1}{N}\right) \right| \\ &= \frac{\ln(2N \cdot \frac{1}{N})}{\sqrt{2N \cdot \frac{1}{N}}} - \frac{\ln(N \cdot \frac{1}{N})}{\sqrt{N \cdot \frac{1}{N}}} \\ &= \frac{\ln 2}{\sqrt{2}} - \frac{\ln 1}{\sqrt{1}} = \frac{\ln 2}{\sqrt{2}} > \frac{1}{4} = \varepsilon. \end{aligned}$$

Thus, due to [Claim 5.5](#), $f_n \not\stackrel{E}{\rightrightarrows} f \equiv 0$.

Claim 5.6 (Necessary Condition for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad n \in \mathbb{N}.$$

Then

$$(f_n \stackrel{E}{\rightrightarrows} f) \implies (u_k \stackrel{E}{\rightrightarrows} 0).$$

That is, if the series $\sum_{k=1}^{\infty} u_k(x)$ converges uniformly on E , then the sequence $\{u_k(x)\}_{k=1}^{\infty}$ converges uniformly to the zero function.

Compare [Claim 5.6](#) with [Claim 1.5](#).

Proof. Note that

$$(f_n \stackrel{E}{\rightrightarrows} f)$$

due to [Claim 5.4](#), for $k = m - 1$, implies

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_{m-1}(x)| < \varepsilon).$$

Since

$$f_m(x) - f_{m-1}(x) = \sum_{k=1}^m u_k(x) - \sum_{k=1}^{m-1} u_k(x) = u_m(x),$$

this is equivalent to

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |u_m(x) - 0| < \varepsilon),$$

which, due to [Def 5.3](#) and [Formula \(2\)](#), is

$$u_m \stackrel{E}{\rightrightarrows} 0.$$

□

Note. The necessary condition ([Claim 5.6](#)) is not sufficient. It cannot even guarantee the pointwise convergence. Let $E = (0, 1)$, and

$$u_k(x) = \frac{1}{k+x}.$$

Then $u_k(x) \xrightarrow{E} 0$, but the series

$$\sum_{k=1}^{\infty} \frac{1}{k+x}$$

is divergent for every $x \in E = (0, 1)$.

Claim 5.7 (The Weierstrass M-test). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad n \in \mathbb{N}.$$

Suppose that there is a sequence of real numbers $\{M_k\}_{k=1}^{\infty}$ satisfying the conditions:

a) $\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k$;

b) $\sum_{k=1}^{\infty} M_k$ converges.

Then, the series

$$\sum_{k=1}^{\infty} u_k(x)$$

converges uniformly on E . That is, $f_n \xrightarrow{E} f$.

Proof. First, note that

$$\left| \sum_{k=m+1}^{m+p} M_k \right| < \varepsilon.$$

From [Item a](#)),

$$(\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k) \implies (\forall k \in \mathbb{N} \rightarrow M_k \geq 0).$$

Therefore,

$$\sum_{k=m+1}^{m+p} M_k < \varepsilon.$$

Using again $\forall k \in \mathbb{N} \forall x \in E \rightarrow |u_k(x)| \leq M_k$, we obtain

$$\forall x \in E \rightarrow \sum_{k=m+1}^{m+p} |u_k(x)| < \varepsilon,$$

and by

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|, \quad \forall a_1, a_2, \dots, a_n \in \mathbb{R},$$

we get

$$\forall x \in E \rightarrow \left| \sum_{k=m+1}^{m+p} u_k(x) \right| < \varepsilon \iff \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon. \quad (**)$$

Due to [Claim 1.2](#), we have

$$\left(\sum_{k=1}^{\infty} M_k \text{ converges} \right)$$

(Item b)), hence

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \rightarrow \left| \sum_{k=m+1}^{m+p} M_k \right| < \varepsilon.$$

Due to Formula (**),

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m \geq N(\varepsilon) \forall p \in \mathbb{N} \forall x \in E \rightarrow |f_{m+p}(x) - f_m(x)| < \varepsilon,$$

which is equivalent to Formula (3) from Claim 5.4. Therefore, due to Claim 5.4, $f_n \xrightarrow{E} f$. $\square$

Claim 5.8 (The Dirichlet Test for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $a_k : E \rightarrow \mathbb{R}$, and $b_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$. Suppose that:

a) $\exists M > 0 : \forall n \in \mathbb{N} \forall x \in E \rightarrow \left| \sum_{k=1}^n a_k(x) \right| < M;$

that is, the sequence

$$\left\{ A_n = \sum_{k=1}^n a_k(x) \right\}_{n=1}^{\infty}$$

is uniformly bounded on E ;

b) either

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \geq b_{k+1}(x),$$

or

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \leq b_{k+1}(x);$$

c) $b_k \xrightarrow{E} 0$.

Then, the series

$$\sum_{k=1}^{\infty} a_k(x) b_k(x)$$

converges uniformly on E .

Claim 5.9 (The Abel Test for the Uniform Convergence). Let $E \subseteq \mathbb{R}$, $a_k : E \rightarrow \mathbb{R}$, and $b_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$. Suppose that:

a) the series $\sum_{k=1}^{\infty} a_k(x)$ converges uniformly on E ;

b) either

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \geq b_{k+1}(x),$$

or

$$\forall k \in \mathbb{N} \forall x \in E \rightarrow b_k(x) \leq b_{k+1}(x);$$

c) $\exists M > 0 \forall k \in \mathbb{N} \forall x \in E \rightarrow |b_k(x)| \leq M;$

that is, the sequence $\{b_k(x)\}_{k=1}^{\infty}$ is uniformly bounded on E .

Then, the series

$$\sum_{k=1}^{\infty} a_k(x) b_k(x)$$

converges uniformly on E .

Claim 5.10. Let

$$S_n(x) = \sum_{k=1}^n \sin(kx) \quad \text{and} \quad C_n(x) = \sum_{k=1}^n \cos(kx), \quad \forall n \in \mathbb{N}.$$

Then:

1. $\forall \delta \in (0, \pi) \ \forall x \in [\delta, 2\pi - \delta] \ \forall n \in \mathbb{N} \rightarrow |S_n(x)| \leq \frac{1}{\sin(\delta/2)};$
2. $\forall \delta \in (0, \pi) \ \forall x \in [\delta, 2\pi - \delta] \ \forall n \in \mathbb{N} \rightarrow |C_n(x)| \leq \frac{1}{\sin(\delta/2)}.$

Proof. 1) Follows directly from [Claim 3.5](#) for $a = x$ and $b = 0$.

2) Follows directly from [Claim 3.5](#) for $a = x$ and $b = \frac{\pi}{2}$.

□

Calculus 2

6 Lecture 6

Properties of Limit Functions

Definition 6.1. Let $E \subseteq \mathbb{R}$. Then, $\lambda \in \mathbb{R}$ is called a *limit point* (or *accumulation point*) of a set E if

$$\forall \varepsilon > 0 \quad \exists x \in E \setminus \{\lambda\} \quad \rightarrow \quad |\lambda - x| < \varepsilon.$$

Claim 6.1 (Theorem on Changing the Order of Two Limits). Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) λ is a *finite limit point* of E (finite means that $\lambda \neq \pm\infty$);
- b) $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow \lambda} f_n(x) = \alpha_n$;
- c) $f_n \xrightarrow{E} f$ (see [Def 5.3](#)),

then:

1. $\exists \lim_{n \rightarrow \infty} \alpha_n$, that is, the sequence $\{\alpha_n\}_{n=1}^{\infty}$ converges;
- 2.

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \lambda} f_n(x) \right) = \lim_{x \rightarrow \lambda} \left(\lim_{n \rightarrow \infty} f_n(x) \right). \quad (1)$$

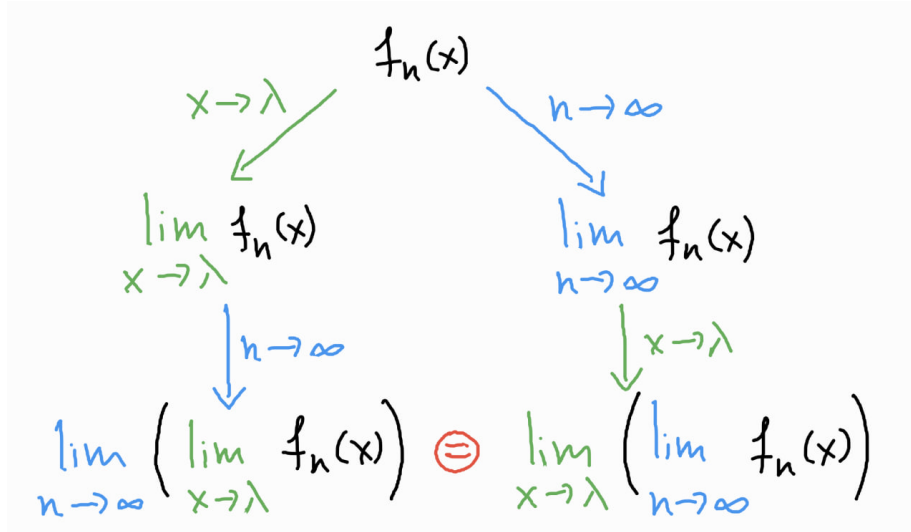


Figure 6.1

The equality in [Formula \(1\)](#) is the statement of [Item 2](#) of [Claim 6.1](#) and the reason for the name of the claim.

Proof. 1) Since, due to [Item b](#)), $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow \lambda} f_n(x) = \alpha_n$, we have

$$(\forall x \rightarrow \lambda \quad |f_m(x) - f_k(x)| < \varepsilon) \implies (|\alpha_m - \alpha_k| \leq \varepsilon). \quad (*)$$

Now note that:

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ \Updownarrow & \text{ due to } \text{Claim 5.4} \end{aligned}$$

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon)$$

$\Downarrow$ due to [Formula \(*\)](#)

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \rightarrow |\alpha_m - \alpha_k| \leq \varepsilon)$$

$\Updownarrow$ due to the Cauchy criterion for convergence of numerical sequences (see Calculus-1)

$$\left(\exists \lim_{n \rightarrow \infty} \alpha_n \right).$$

2) Let

$$\lim_{n \rightarrow \infty} \alpha_n = \alpha \quad (\text{the limit exists due to [Item 1](#)}).$$

Then, in [Formula \(1\)](#):

the left hand side is

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \lambda} f_n(x) \right) = \lim_{n \rightarrow \infty} \alpha_n = \alpha,$$

the right hand side is

$$\lim_{x \rightarrow \lambda} \left(\lim_{n \rightarrow \infty} f_n(x) \right) = \lim_{x \rightarrow \lambda} f(x),$$

since, due to [Item c](#)) and [Claim 5.3](#), we have $f_n \xrightarrow{E} f$.

Thus, we need to prove that

$$\exists \lim_{x \rightarrow \lambda} f(x) = \alpha.$$

That is, due to a definition of the limit (see Calculus-1), we need to prove that

$$\forall \varepsilon > 0 \exists \delta > 0 : \forall x \in (\alpha - \delta, \alpha + \delta) \cap E \rightarrow |f(x) - \alpha| < \varepsilon. \quad (**)$$

Note that, for some $\varepsilon > 0$ we have:

i)

$$(f_n \xrightarrow{E} f) \implies \left(\exists N_1(\varepsilon) : \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{3} \right). \quad (1a)$$

ii)

$$\left(\lim_{n \rightarrow \infty} \alpha_n = \alpha \right) \implies \left(\exists N_2 : \forall n \geq N_2 \rightarrow |\alpha_n - \alpha| < \frac{\varepsilon}{3} \right). \quad (2)$$

Let $N(\varepsilon) = \max\{N_1(\varepsilon), N_2\}$. Then

$$\left(\lim_{x \rightarrow \lambda} f_N(x) = \alpha_N \right)$$

(for simplicity of notation we write N instead of $N(\varepsilon)$), hence

$$\left(\exists \delta > 0 : \forall x \in (\alpha - \delta, \alpha + \delta) \cap E \rightarrow |f_N(x) - \alpha_N| < \frac{\varepsilon}{3} \right). \quad (3)$$

Now we are ready to verify (**). Indeed, for every $x \in (\alpha - \delta, \alpha + \delta)$ we have

$$\begin{aligned} |f(x) - \alpha| &= |f(x) - f_N(x) + f_N(x) - \alpha_N + \alpha_N - \alpha| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - \alpha_N| + |\alpha_N - \alpha| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned}$$

due to (1a), (3), and (2), respectively.

Thus, we verified (**) and proved [Item 2](#). □

Claim 6.2 (Continuity of a Limit Function). Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}, \forall n \in \mathbb{N}$. If:

a) for every $n \in \mathbb{N}$, f_n is a continuous function on E , that is, $\forall n \in \mathbb{N} \rightarrow f_n \in C(E)$;

b) $f_n \xrightarrow{E} f$ (see Def 5.3),

then, the function f is continuous on E . That is, $f \in C(E)$.

Compare Claim 6.2 with Item 1) of Claim 5.1 and Claim 5.2.

Proof. If $\lambda \in E$ is an *isolated point* of E (that is, if λ is an element of E and there exists a neighborhood of λ which does not contain any other elements of E), then, f is continuous at λ (see Calculus-1).

If $\lambda \in E$ is a *limit point* of E , then from Calculus-1, it is known that

$$(f \text{ is continuous at } \lambda \in E) \iff \left(\lim_{x \rightarrow \lambda} f(x) = f(\lambda) \right). \quad (4)$$

Therefore, it remains to prove that for every limit point $\lambda \in E$ we have $\lim_{x \rightarrow \lambda} f(x) = f(\lambda)$.

Let $\lambda \in E$ be a limit point of E . Then, due to Item a) and Expression (4), we have

$$\forall n \in \mathbb{N} \rightarrow \lim_{x \rightarrow \lambda} f_n(x) = f_n(\lambda) = \alpha_n.$$

Therefore, all conditions of Claim 6.1 are satisfied and we have

$$\lim_{x \rightarrow \lambda} f(x) = \lim_{x \rightarrow \lambda} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow \lambda} f_n(x) = \lim_{n \rightarrow \infty} f_n(\lambda) = f(\lambda),$$

since $(f_n \xrightarrow{E} f) \Rightarrow (f_n \xrightarrow{E} f)$, due to Item 2) of Claim 6.1, since $\forall n \in \mathbb{N}$ f_n is continuous on E , and since $(f_n \xrightarrow{E} f) \Rightarrow (f_n \xrightarrow{E} f)$.

That is, the claim is proved. $\square$

To prove the next claim, we will need the following fact.

Claim 6.3 (The Darboux Criterion for Riemann Integrability). Let $I = [a, b] \subseteq \mathbb{R}$ and $f : I \rightarrow \mathbb{R}$. Then

$$(f \text{ is Riemann integrable on } I = [a, b], \text{ notation } f \in \mathcal{R}([a, b]))$$

$$\Updownarrow$$

$$\left(\forall \varepsilon > 0 \exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f) |I_i| < \varepsilon \right). \quad (5)$$

where: T is a *partition* of $I = [a, b]$ (see Calculus-1); I_i is a *subinterval* of I from the partition T ; $|I_i|$ is the *length* of a subinterval I_i (if $I_i = [x_{i-1}, x_i]$, then $|I_i| = x_i - x_{i-1}$);

$$|T| = \max_{I_i \in T} |I_i|$$

is the *mesh* of the partition T ; and

$$\omega_{I_i}(f) = \sup_{x \in I_i} f(x) - \inf_{x \in I_i} f(x)$$

is the *oscillation* of f on an interval I_i .

Proof. See Calculus-1. $\square$

Claim 6.4 (Riemann Integrability of a Limit Function). Let $I = [a, b] \subset \mathbb{R}$ and $f_n : I \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

a) for every $n \in \mathbb{N}$, f_n is Riemann integrable on $I = [a, b]$, that is, $\forall n \in \mathbb{N} \rightarrow f_n \in \mathcal{R}([a, b])$;

b) $f_n \xrightarrow{I} f$ (see [Def 5.3](#)),

then, the function f is Riemann integrable on I , that is, $f \in \mathcal{R}([a, b])$.

Compare [Claim 6.4](#) with [Item 3](#)) of [Claim 5.1](#) and [Claim 5.2](#).

Proof. Due to [Claim 6.3](#) and [Formula \(5\)](#) it suffices to show that

$$\forall \varepsilon > 0 \exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f) |I_i| < \varepsilon. \quad (**)$$

For some $\varepsilon > 0$, note that:

i)

$$(f_n \xrightarrow{I} f) \implies \left(\exists N(\varepsilon) \forall x \in I \rightarrow |f(x) - f_N(x)| < \frac{\varepsilon}{4|I|} \right). \quad (6)$$

For simplicity of notation we write N instead of $N(\varepsilon)$, and $|I| = b - a$.

If we know that

$$\forall x \in I \rightarrow |f(x) - f_N(x)| < \frac{\varepsilon}{4|I|},$$

then

$$\forall x \in I \rightarrow f_N(x) - \frac{\varepsilon}{4|I|} < f(x) < f_N(x) + \frac{\varepsilon}{4|I|}.$$

Then, for every subset $J \subseteq I$, we have

$$\inf_{x \in J} f_N(x) - \frac{\varepsilon}{4|I|} \leq \inf_{x \in J} f(x) \quad (7a)$$

and

$$\sup_{x \in J} f(x) \leq \sup_{x \in J} f_N(x) + \frac{\varepsilon}{4|I|}. \quad (7b)$$

Therefore, due to [\(6\)](#), for every subset $J \subseteq I$, we have

$$\begin{aligned} \omega_J(f) &= \sup_{x \in J} f(x) - \inf_{x \in J} f(x) \\ &\leq \sup_{x \in J} f_N(x) + \frac{\varepsilon}{4|I|} - \left(\inf_{x \in J} f_N(x) - \frac{\varepsilon}{4|I|} \right) \\ &= \omega_J(f_N) + \frac{\varepsilon}{2|I|}. \end{aligned} \quad (8)$$

ii) For f_N from [Formula \(6\)](#), due to [Claim 6.3](#), we have

$$(f_N \in \mathcal{R}(I)) \implies \left(\exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f_N) |I_i| < \frac{\varepsilon}{2} \right). \quad (9)$$

This holds true due to [Item a\)](#). Here, for a given $\varepsilon > 0$ and $N(\varepsilon)$, we define $d(\varepsilon)$ which we are going to use to verify [\(**\)](#).

We will show now that $d(\varepsilon)$ from [Formula \(9\)](#) satisfies [\(**\)](#). For every partition T of I such that $|T| < d(\varepsilon)$, we have

$$\begin{aligned} \sum_{I_i \in T} \omega_{I_i}(f) |I_i| &\leq \sum_{I_i \in T} \left(\omega_{I_i}(f_N) + \frac{\varepsilon}{2|I|} \right) |I_i| \\ &= \sum_{I_i \in T} \omega_{I_i}(f_N) |I_i| + \frac{\varepsilon}{2|I|} \sum_{I_i \in T} |I_i| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2|I|} |I| = \varepsilon. \end{aligned}$$

The first inequality holds due to [Inequality \(8\)](#), which holds true for every subset J of I ; the second term uses $\sum_{I_i \in T} |I_i| = |I|$, since we sum up the lengths of all subintervals from the partition T .

Thus, we verified [\(**\)](#) and proved the claim. $\square$

Claim 6.5 (Integration of a Uniformly Convergent Sequence). Let $I = [a, b] \subset \mathbb{R}$ and $f_n : I \rightarrow \mathbb{R}$. If:

- a) for every $n \in \mathbb{N}$, f_n is Riemann integrable on I , that is, $\forall n \in \mathbb{N} \rightarrow f_n \in \mathcal{R}(I)$;
- b) $f_n \xrightarrow{I} f$,

then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

Proof. Due to [Claim 6.4](#), the limit function

$$f(x) = \lim_{n \rightarrow \infty} f_n(x), \quad x \in I,$$

is Riemann integrable on I . Therefore, exists the integral:

$$\int_a^b f(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

Due to [Item b\)](#) and [Def 5.3](#), we have:

$$(f_n \xrightarrow{I} f) \implies \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in I \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{b-a} \right). \quad (10)$$

Therefore, for $\forall \varepsilon > 0$ and $\forall n \geq N(\varepsilon)$ we have

$$\begin{aligned} \left| \int_a^b f_n(x) dx - \int_a^b f(x) dx \right| &= \left| \int_a^b (f_n(x) - f(x)) dx \right| \\ &\leq \int_a^b |f_n(x) - f(x)| dx \\ &\leq \int_a^b \frac{\varepsilon}{b-a} dx = \frac{\varepsilon}{b-a} (b-a) = \varepsilon. \end{aligned}$$

The first equality uses linearity of the Riemann integral, and the inequality

$$\left| \int_a^b g(x) dx \right| \leq \int_a^b |g(x)| dx, \quad \forall g \in \mathcal{R}([a, b]),$$

see Calculus-1.

Thus, due to a definition of the limit of a numerical sequence (see Calculus-1), we proved that

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

$\square$

Note that if all the functions f_n are differentiable on an interval $J = (a, b)$ and converge uniformly to a limit function f on J , the limit function f need not be differentiable.

Example. Let $J = (0, 2\pi)$ and

$$f_n(x) = \sum_{k=1}^n \frac{\sin(k^2 x)}{k^2}, \quad \forall n \in \mathbb{N}.$$

Then, due to [Claim 5.7](#) [The Weierstrass M-test], we have

$$f_n \xrightarrow{J} f, \quad \text{where } f(x) = \sum_{k=1}^{\infty} \frac{\sin(k^2 x)}{k^2}$$

is called the Riemann function.

It is clear that the functions f_n are differentiable on J . It can be shown (though it is rather difficult and is therefore not included in our course) that the Riemann function f is differentiable almost nowhere on J .

That is, a direct analogue of [Claim 6.2](#) and [Claim 6.4](#) does not hold true for differentiation. Our next goal is to establish conditions under which the limit function of a sequence of differentiable functions is a differentiable function.

Claim 6.6. Let $E \subseteq \mathbb{R}$, $A \subseteq E$, $B \subseteq E$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) $f_n \xrightarrow{E} f$ (see [Def 5.2](#));
- b) $f_n \xrightarrow{A} f$ and $f_n \xrightarrow{B} f$ (see [Def 5.3](#)),

then

$$f_n \xrightarrow{A \cup B} f.$$

Proof. Follows directly from [Def 5.3](#):

$$(f_n \xrightarrow{A} f) \iff (\forall \varepsilon > 0 \exists N_1(\varepsilon) \forall n \geq N_1(\varepsilon) \forall x \in A \rightarrow |f_n(x) - f(x)| < \varepsilon), \quad (11a)$$

$$(f_n \xrightarrow{B} f) \iff (\forall \varepsilon > 0 \exists N_2(\varepsilon) \forall n \geq N_2(\varepsilon) \forall x \in B \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (11b)$$

Let $N(\varepsilon) = \max\{N_1, N_2\}$. Then, from Formulas (11a)–(11b), we have:

$$\forall \varepsilon > 0 \exists N(\varepsilon) \forall n \geq N(\varepsilon) \forall x \in A \cup B \rightarrow |f_n(x) - f(x)| < \varepsilon.$$

Therefore, due to [Def 5.3](#), we have $f_n \xrightarrow{A \cup B} f$. □

Claim 6.7. Let $E \subseteq \mathbb{R}$, $g : E \rightarrow \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) $f_n \xrightarrow{E} f$;
- b) g is bounded on E (that is, $\exists M > 0 \forall x \in E \rightarrow |g(x)| \leq M$),

then

$$g \cdot f_n \xrightarrow{E} g \cdot f.$$

That is, when multiplying a uniformly convergent sequence by a bounded function, we obtain a uniformly convergent sequence.

Proof. Due to [Item b\)](#), we have $\exists M > 0 \forall x \in E \rightarrow |g(x)| \leq M$. Then, due to [Def 5.3](#), we have:

$$(f_n \xrightarrow{E} f) \implies \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{M} \right). \quad (12)$$

Since, for every $x \in E$, we have

$$|g(x)f_n(x) - g(x)f(x)| = |g(x)| |f_n(x) - f(x)| \leq M |f_n(x) - f(x)|,$$

[Formula \(12\)](#) implies that

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |g(x)f_n(x) - g(x)f(x)| < \varepsilon,$$

which, due to [Def 5.3](#), means that $g f_n \xrightarrow{E} g f$. □

Claim 6.8. Let $E \subseteq \mathbb{R}$, $f_n : E \rightarrow \mathbb{R}$, and $g_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If $f_n \xrightarrow{E} f$ and $g_n \xrightarrow{E} g$, then, $f_n + g_n \xrightarrow{E} f + g$.

Proof. Follows directly from [Def 5.3](#):

$$(f_n \xrightarrow{E} f) \implies \left(\forall \varepsilon > 0 \exists N_1(\varepsilon) \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{2} \right), \quad (12a)$$

$$(g_n \xrightarrow{E} g) \implies \left(\forall \varepsilon > 0 \exists N_2(\varepsilon) \forall n \geq N_2(\varepsilon) \forall x \in E \rightarrow |g_n(x) - g(x)| < \frac{\varepsilon}{2} \right). \quad (12b)$$

Let $N(\varepsilon) = \max\{N_1(\varepsilon), N_2(\varepsilon)\}$. Then, from Formulas [\(12a\)–\(12b\)](#), $\forall \varepsilon > 0$, $\forall n \geq N(\varepsilon)$, $\forall x \in E$ we have

$$|f_n(x) + g_n(x) - f(x) - g(x)| \leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which, due to [Def 5.3](#), implies that $f_n + g_n \xrightarrow{E} f + g$. □

Claim 6.9. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If E is a *finite set*, then

$$(f_n \xrightarrow{E} f) \iff (f_n \xrightarrow{E} f).$$

Proof. $(\implies)$ Let $E = \{\alpha_1, \alpha_2, \dots, \alpha_K\}$. Then, due to [Def 5.2](#), for every $\alpha_k \in E$, we have

$$(f_n \xrightarrow{E} f) \implies (\forall \varepsilon > 0 \exists N(\varepsilon, \alpha_k) \in \mathbb{N} \forall n \geq N(\varepsilon, \alpha_k) \rightarrow |f_n(\alpha_k) - f(\alpha_k)| < \varepsilon). \quad (13)$$

Let

$$N(\varepsilon) = \max\{N(\varepsilon, \alpha_1), N(\varepsilon, \alpha_2), \dots, N(\varepsilon, \alpha_K)\}.$$

Then, from [Formula \(13\)](#), $\forall \varepsilon > 0$, $\forall n \geq N(\varepsilon)$, $\forall x \in E$, we have

$$|f_n(x) - f(x)| < \varepsilon,$$

which, due to [Def 5.3](#), implies that $f_n \xrightarrow{E} f$.

$(\impliedby)$ The claim holds true for every set E (not necessarily finite) due to [Claim 5.3](#). □

Claim 6.10 (Differentiability of a Limit Function). Let $J = (a, b) \subset \mathbb{R}$ and $f_n : J \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) for every $n \in \mathbb{N}$, f_n is differentiable on $J = (a, b)$, that is, $\forall n \in \mathbb{N} \forall x \in (a, b) \rightarrow \exists f'_n(x) \stackrel{\text{def}}{=} \lim_{\delta \rightarrow 0} \frac{f_n(x + \delta) - f_n(x)}{\delta}$;
- b) there is $\lambda \in J = (a, b)$ such that $\exists \lim_{n \rightarrow \infty} f_n(\lambda)$;
- c) the sequence $\{f'_n(x)\}_{n=1}^{\infty}$ converges uniformly on $J = (a, b)$, that is, $f'_n \xrightarrow{J} g$ for some function $g : J \rightarrow \mathbb{R}$,

then:

- 1. $f_n \xrightarrow{J} f$;
- 2. the limit function f is differentiable on J and

$$\forall x \in J \rightarrow f'(x) = \lim_{n \rightarrow \infty} f'_n(x).$$

On the diagram, the green shows what is given and the red shows what need to be proved.

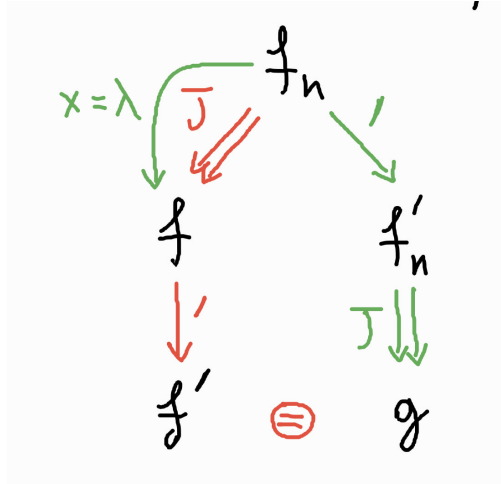


Figure 6.2

Proof. 1) Note that, due to [Item c\)](#) and [Claim 5.4](#), for every $\varepsilon > 0$ there is $N(\varepsilon) \in \mathbb{N}$ such that

$$\forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in J \rightarrow |f'_k(x) - f'_m(x)| < \varepsilon. \quad (14)$$

For any $y \in J$, let

$$J_y = J \setminus \{y\} = (a, y) \cup (y, b).$$

Consider the functions

$$h_n : J_y \rightarrow \mathbb{R}, \quad h_n(x) = \frac{f_n(x) - f_n(y)}{x - y}, \quad \forall n \in \mathbb{N}, \forall x \in J_y. \quad (***)$$

Let us prove that the sequence

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(y)}{x - y} \right\}_{n=1}^{\infty}$$

converges uniformly on J_y .

Due to [Claim 5.4](#), it suffices to show that

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in J_y \rightarrow |h_k(x) - h_m(x)| < \varepsilon. \quad (15)$$

Let $N(\varepsilon) \in \mathbb{N}$ be defined as in [\(14\)](#). Then, $\forall k \geq N(\varepsilon), \forall m \geq N(\varepsilon), \forall x \in J_y$ we have

$$\begin{aligned} |h_k(x) - h_m(x)| &= \left| \frac{f_k(x) - f_k(y)}{x - y} - \frac{f_m(x) - f_m(y)}{x - y} \right| \\ &= \left| \frac{(f_k(x) - f_m(x)) - (f_k(y) - f_m(y))}{x - y} \right|. \end{aligned} \quad (16)$$

Note that, due to [Item a\)](#), the function $f_k - f_m$ is continuous on the closed interval

$$[\min\{x, y\}, \max\{x, y\}]$$

and differentiable on the open interval

$$(\min\{x, y\}, \max\{x, y\}).$$

Therefore, due to the Mean Value Theorem (see Calculus-1), we have

$$(f_k(x) - f_m(x)) - (f_k(y) - f_m(y)) = (f'_k(z) - f'_m(z))(x - y),$$

for some $z \in (\min\{x, y\}, \max\{x, y\})$. Thus, due to [Equality \(16\)](#), we have

$$|h_k(x) - h_m(x)| = \left| \frac{(f'_k(z) - f'_m(z))(x - y)}{x - y} \right| = |f'_k(z) - f'_m(z)| < \varepsilon,$$

for some $z \in (\min\{x, y\}, \max\{x, y\})$, since [\(14\)](#) holds true for every $x \in J$ and

$$z \in (\min\{x, y\}, \max\{x, y\}) \subset J.$$

That is, we verified [\(15\)](#). Therefore the sequence

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(y)}{x - y} \right\}_{n=1}^{\infty} \text{ converges uniformly on } J_y = J \setminus \{y\}. \quad (17)$$

Let in [Formula \(17\)](#) $y = \lambda$ (where $\lambda \in J$ is from [Item b\)](#)). Note that

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(\lambda)}{x - \lambda} \right\}_{n=1}^{\infty} \text{ conv. uniformly on } J_{\lambda}.$$

Due to [Claim 6.7](#), since the function $g(x) = x - \lambda$ is bounded on J_{λ} ,

$$\left\{ g(x)h_n(x) = \frac{f_n(x) - f_n(\lambda)}{x - \lambda}(x - \lambda) = f_n(x) - f_n(\lambda) \right\}_{n=1}^{\infty}$$

converges uniformly on J_{λ} .

Due to [Claim 6.8](#), since

$$f_n(x) = (f_n(x) - f_n(\lambda)) + f_n(\lambda)$$

and the sequence $\{f_n(\lambda)\}_{n=1}^{\infty}$ converges uniformly on J_{λ} (since it converges due to [Item b\)](#) and does not depend on x), it follows that

$$\{f_n(x)\}_{n=1}^{\infty} \text{ conv. uniformly on } J_{\lambda}.$$

Due to [Item b\)](#), the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges pointwise on the one element set $\{\lambda\}$. Due to [Claim 6.9](#), this implies that the sequence converges uniformly on the set $\{\lambda\}$. Since the sequence converges uniformly on the sets J_{λ} and $\{\lambda\}$, due to [Claim 6.6](#) it converges uniformly on the set $J \cup \{\lambda\} = J$. Thus,

$$\{f_n(x)\}_{n=1}^{\infty} \text{ conv. uniformly on } J.$$

2) Due to [Item 1\)](#), we know that the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges uniformly on J . Let

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad (\forall x \in J). \quad (18)$$

That is, f is the limit function of the sequence $\{f_n(x)\}_{n=1}^{\infty}$ on J .

Then, for every $n \in \mathbb{N}$, consider the function

$$g_n : J_y \rightarrow \mathbb{R}, \quad g_n(x) = \frac{f_n(x) - f_n(y)}{x - y}, \quad \forall x \in J_y = J \setminus \{y\}. \quad (19)$$

Note that:

a) y is a finite limit point of J_y ;

b) $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow y} g_n(x) = \lim_{x \rightarrow y} \frac{f_n(x) - f_n(y)}{x - y} = f'_n(y)$ (holds true due to [Item a\)](#));

c)

$$g_n(x) \xrightarrow{J_y} \frac{f(x) - f(y)}{x - y}.$$

Due to [Formula \(17\)](#), we know that the sequence $\{g_n(x)\}_{n=1}^{\infty}$ converges uniformly on J_y . The limit function can be found from [Formulas \(18\) and \(19\)](#):

$$\lim_{n \rightarrow \infty} g_n(x) = \lim_{n \rightarrow \infty} \frac{f_n(x) - f_n(y)}{x - y} = \frac{f(x) - f(y)}{x - y}. \quad (20)$$

Therefore, all conditions of [Claim 6.1](#) are satisfied and due to [Item 2\)](#) of [Claim 6.1](#), we have:

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow y} g_n(x) \right) = \lim_{x \rightarrow y} \left(\lim_{n \rightarrow \infty} g_n(x) \right). \quad (21)$$

The left hand side is $\lim_{n \rightarrow \infty} f'_n(y)$ due to [\(19\)](#) and [Item a\)](#), and the right hand side is

$$\lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y}$$

due to [Formula \(20\)](#).

Due to [Item c\)](#) and [Claim 5.3](#), the limit $\lim_{n \rightarrow \infty} f'_n(y)$ exists for every $y \in J$. Therefore, due to [Equalities \(21\)](#), we have

$$\exists f'(y) \stackrel{\text{def}}{=} \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} = \lim_{n \rightarrow \infty} f'_n(y), \quad (\forall y \in J).$$

□

Claim 6.11 (Continuity of a Series). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, u_k is a continuous function on E ;
- b) $f_n \xrightarrow{E} f$,

then, the function f is continuous on E .

Proof. Since a finite sum of continuous functions is a continuous function (see Calculus-1),

$$f_n(x) = \sum_{k=1}^n u_k(x)$$

is continuous on E . Thus, the sequence $\{f_n\}_{n=1}^{\infty}$ satisfies all conditions of [Claim 6.2](#). Therefore, due to [Claim 6.2](#), the function f is continuous on E . □

Claim 6.12 (Term-by-term Integration of a Uniformly Conv. Series). Let $I = [a, b] \subset \mathbb{R}$, $u_k : I \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, u_k is Riemann integrable on I ;
- b) $f_n \xrightarrow{I} f$,

then

$$\sum_{k=1}^{\infty} \int_a^b u_k(x) dx = \int_a^b \left(\sum_{k=1}^{\infty} u_k(x) \right) dx = \int_a^b f(x) dx.$$

Proof. Since a finite sum of Riemann integrable functions is a Riemann integrable function (see Calculus-1), the function

$$f_n(x) = \sum_{k=1}^n u_k(x)$$

is Riemann integrable on I . Thus, the sequence $\{f_n(x)\}_{n=1}^\infty$ satisfies all conditions of [Claim 6.5](#). Therefore, we have:

$$\begin{aligned} \sum_{k=1}^\infty \int_a^b u_k(x) dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_a^b u_k(x) dx \\ &= \lim_{n \rightarrow \infty} \int_a^b \sum_{k=1}^n u_k(x) dx \\ &= \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx \\ &= \int_a^b \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx \\ &= \int_a^b \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n u_k(x) \right) dx \\ &= \int_a^b f(x) dx, \end{aligned}$$

where: 1) due to [Def 1.3](#); 2) since the sum is finite; 3) and 5) since $f_n(x) = \sum_{k=1}^n u_k(x)$; 4) due to [Claim 6.5](#); 6) since $f_n \xrightarrow{I} f$ due to [Item b\)](#) and [Claim 5.3](#). $\square$

Claim 6.13 (Term-by-term Differentiation of Series). Let $J = (a, b) \subset \mathbb{R}$, $u_k : J \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, $u_k(x)$ is differentiable on J ;
- b) there is $\lambda \in J$ such that the series $\sum_{k=1}^\infty u_k(\lambda)$ converges;
- c) the sequence $\{f'_n(x)\}_{n=1}^\infty$ converges uniformly on J ,

then:

- 1. $f_n \xrightarrow{J} f$;
- 2. the limit function f is differentiable on J and

$$\sum_{k=1}^\infty u'_k(x) = \left(\sum_{k=1}^\infty u_k(x) \right)' = f'(x), \quad (\forall x \in J).$$

Proof. Follows directly from [Claim 6.10](#). $\square$

Claim 6.14 (Stone-Weierstrass Theorem). Let $f : [a, b]$ be a continuous on $[a, b]$ function. Then, for every $\varepsilon > 0$ there is a polynomial function

$$p : [a, b] \rightarrow \mathbb{R}$$

such that

$$\forall x \in [a, b] \rightarrow |f(x) - p(x)| < \varepsilon.$$

Proof. Without proof, this is not a simple theorem.

□

Calculus 2

7 Lecture 7

Power Series

Definition 7.1. A functional series of the form

$$\sum_{k=0}^{\infty} a_k(x - x_0)^k = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_k(x - x_0)^k + \cdots \quad (1)$$

is called a *power series centered at x_0* .

$x_0 \in \mathbb{R}$ is called the *center* of the series.

$a_k \in \mathbb{R}$, $k \in \mathbb{N} \cup \{0\}$ are the *coefficients* of the series.

Since by the linear substitution $t = x - x_0$, the power series (1) can be rewritten as a power series centered at 0:

$$\sum_{k=0}^{\infty} a_k t^k = a_0 + a_1 t + a_2 t^2 + \cdots + a_k t^k + \cdots, \quad (2)$$

without loss of generality, we can study only power series of the form (2).

Definition 7.2. Let

$$D = \left\{ x \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\} \subseteq \mathbb{R}.$$

That is, D is the *set of all points of convergence* of $\sum_{k=0}^{\infty} a_k x^k$.

Then,

$$R = \sup\{|x| \mid x \in D\}$$

is called the *radius of convergence* of the power series $\sum_{k=0}^{\infty} a_k x^k$. The set D is called the *interval of convergence* of the power series $\sum_{k=0}^{\infty} a_k x^k$.

Note that, since $0 \in D$, we have $D \neq \emptyset$, which implies that $R = \sup\{|x| \mid x \in D\}$ exists.

Claim 7.1. Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series and let R be its radius of convergence. Then:

1. $\forall x \in \mathbb{R} : |x| < R$, the series $\sum_{k=0}^{\infty} a_k x^k$ converges *absolutely*;
2. $\forall x \in \mathbb{R} : |x| > R$, the series $\sum_{k=0}^{\infty} a_k x^k$ *diverges*.

Proof. 1) Due to [Def 3.1](#), it suffices to show that the series

$$\sum_{k=0}^{\infty} |a_k x^k|$$

converges.

Since

$$|x| < R = \sup\{|y| \mid y \in D\}, \quad D = \left\{ y \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k y^k \text{ converges} \right\},$$

by the definition of the supremum, there exists $y \in D$ such that $|x| < |y|$. Since $y \in D$, the series $\sum_{k=0}^{\infty} a_k y^k$ converges, therefore, due to [Claim 1.5](#), we have

$$\lim_{k \rightarrow \infty} a_k y^k = 0,$$

which implies that

$$\exists M > 0 \forall k \in \mathbb{N} \cup \{0\} \rightarrow |a_k y^k| < M.$$

Therefore, for every $k \in \mathbb{N} \cup \{0\}$ we have

$$|a_k x^k| = |a_k y^k| \left| \frac{x}{y} \right|^k < M \left| \frac{x}{y} \right|^k = M q^k,$$

where

$$0 \leq |x| < |y| \implies q = \left| \frac{x}{y} \right| \in [0, 1).$$

The latter implies that the series

$$\sum_{k=0}^{\infty} M q^k$$

converges. Therefore, due to [Claim 5.7](#) [The Weierstrass M-test], the series

$$\sum_{k=0}^{\infty} |a_k x^k|$$

converges.

2) Assume that $|x| > R$ and $\sum_{k=0}^{\infty} a_k x^k$ converges. The latter implies (see [Def 7.2](#)) that $x \in D$. Therefore, since

$$R = \sup\{|y| \mid y \in D\},$$

we have $|x| \leq R$, which contradicts $|x| > R$. Thus, our assumption is false and the series $\sum_{k=0}^{\infty} a_k x^k$ diverges. $\square$

Claim 7.2 (The Cauchy–Hadamard Formula). For any power series $\sum_{k=0}^{\infty} a_k x^k$, its radius of convergence R is given by

$$\frac{1}{R} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \lim_{k \rightarrow \infty} \sup_{n \geq k} |a_n|^{1/n}. \quad (*)$$

Proof. Recall that, by [Def 7.2](#), we have

$$D = \left\{ x \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\} \quad \text{and} \quad R = \sup\{|x| \mid x \in D\}.$$

It is clear that $0 \in D$. Let $x \neq 0$, then we have

$$L = \overline{\lim}_{k \rightarrow \infty} |a_k x^k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} |x| = |x| \tau, \quad (1)$$

where

$$\tau = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}.$$

[Equality \(1\)](#) holds true since x does not depend on k and $|x| > 0$ (see [Claim 7.4](#)).

Then, due to [Claim 2.9](#) [The Root Test], the series $\sum_{k=0}^{\infty} |a_k x^k|$ converges if $L = |x| \tau < 1$ and diverges if $L = |x| \tau > 1$.

- If $\tau = 0$, then $|x| \tau = 0 < 1$, $\forall x \in \mathbb{R}$, thus $D = \mathbb{R}$ and $R = +\infty$; that is, [Formula \(*\)](#) holds true if $\tau = 0$.
- If $\tau = +\infty$, then $|x| \tau = +\infty > 1$, $\forall x \in \mathbb{R} \setminus \{0\}$, thus $D = \{0\}$ and $R = 0$; that is, [Formula \(*\)](#) holds true if $\tau = +\infty$.
- If $\tau \in (0, +\infty)$, then

$$|x| \tau < 1 \iff |x| < \frac{1}{\tau},$$

hence

$$\left\{ x \in \mathbb{R} \mid |x| < \frac{1}{\tau} \right\} \subseteq D,$$

which implies that $R \geq \frac{1}{\tau}$.

Let us assume that $R > \frac{1}{\tau}$. Then, due to [Def 7.2](#), there is $x \in D$ such that

$$R > |x| > \frac{1}{\tau}.$$

Inequality $|x| > \frac{1}{\tau}$ implies that the series $\sum_{k=0}^{\infty} |a_k x^k|$ diverges (since $L = |x|\tau > 1$), and inequality $R > |x|$ implies that the series $\sum_{k=0}^{\infty} a_k x^k$ converges (due to [Claim 7.1](#)). Thus, the assumption is false.

Therefore, $R = \frac{1}{\tau}$ and [Formula \(*\)](#) holds true if $\tau \in (0, +\infty)$. $\square$

Note that [Claim 7.1](#) tells nothing about the behavior of the series $\sum_{k=0}^{\infty} a_k x^k$ at the boundary points $x = R$ and $x = -R$, if $R \in (0, +\infty)$.

In fact, all logically admissible possibilities can occur:

- the series can be absolutely convergent at $x = R$ and $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} x^k, \quad R = 1, \quad D = [-1, 1];$$

- the series can be divergent at $x = R$ and $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k} x^{2k}, \quad R = 1, \quad D = (-1, 1);$$

- the series can be divergent at $x = R$ and conditionally convergent at $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k} x^k, \quad R = 1, \quad D = [-1, 1);$$

- the series can be conditionally convergent at $x = R$ and divergent at $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} x^k, \quad R = 1, \quad D = (-1, 1].$$

Claim 7.3 (On Uniform Convergence of Power Series). Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with the radius of convergence $R > 0$. Then:

1. the series converges *uniformly* on every interval

$$I = [a, b] \subset (-R, R);$$

2. if the series *converges* at $x = R$, then, it converges *uniformly* on every interval

$$I = [a, R] \subset (-R, R];$$

3. if the series *diverges* at $x = R$, then, it converges *non-uniformly* on every interval

$$I = [a, R) \subset (-R, R).$$

Proof. 1) Let

$$m = \max\{|a|, |b|\}.$$

It is clear that $0 \leq m < R$. Note that:

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in I = [a, b],$

$$|a_k x^k| \leq |a_k| m^k =: M_k;$$

- since $|m| < R$, due to [Item 1\)](#) of [Claim 7.1](#), the series

$$\sum_{k=0}^{\infty} M_k = \sum_{k=0}^{\infty} |a_k| m^k$$

converges.

Thus, due to [Claim 5.7](#) [The Weierstrass M-test], the series $\sum_{k=0}^{\infty} a_k x^k$ converges uniformly on $[a, b] \subset (-R, R)$.

2) Due to [Item 1\)](#) and [Claim 6.6](#), it suffices to prove the claim for $I = [0, R]$.

Let

$$f_k(x) = a_k R^k \quad \text{and} \quad g_k(x) = \left(\frac{x}{R}\right)^k, \quad x \in [0, R], \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Note that:

- the series

$$\sum_{k=0}^{\infty} f_k(x) = \sum_{k=0}^{\infty} a_k R^k$$

converges uniformly on $[0, R]$.

Indeed, the series converges since this is one of the conditions of the claim; since the series $\sum_{k=0}^{\infty} a_k R^k$ does not effectively depend on x , we can say, due to [Def 5.3](#), that it converges uniformly on $[0, R]$.

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in [0, R],$

$$\left(\frac{x}{R}\right)^k \geq \left(\frac{x}{R}\right)^{k+1},$$

i.e. $g_k(x) \geq g_{k+1}(x);$

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in [0, R],$

$$\left|\frac{x}{R}\right| < 1,$$

i.e. $g_k(x) \downarrow 0.$

Thus, due to [Claim 5.9](#) [The Abel Test for the UC], the series

$$\sum_{k=0}^{\infty} f_k(x) g_k(x) = \sum_{k=0}^{\infty} a_k R^k \left(\frac{x}{R}\right)^k = \sum_{k=0}^{\infty} a_k x^k$$

converges uniformly on $[0, R]$.

3) The pointwise convergence of $\sum_{k=0}^{\infty} a_k x^k$ on $[a, R)$ follows from [Item 1\)](#) of [Claim 7.1](#). Let us assume that the series

$$\sum_{k=0}^{\infty} a_k x^k = \lim_{n \rightarrow \infty} f_n(x), \quad f_n(x) = \sum_{k=0}^n a_k x^k,$$

converges uniformly on $[a, R)$. Note that:

- $\lambda = R$ is a finite limit point of the set $[a, R)$;

- $\forall n \in \mathbb{N}$, we have

$$\lim_{x \rightarrow R} f_n(x) = \sum_{k=0}^n a_k R^k \in \mathbb{R}$$

(since n is finite).

Thus, due to [Item 1](#)) of [Claim 6.1](#) [On Changing the Order of Two Limits], the limit

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n a_k R^k = \sum_{k=0}^{\infty} a_k R^k$$

exists. Since this contradicts the conditions of the claim, our assumption is false and the series $\sum_{k=0}^{\infty} a_k x^k$ converges non-uniformly on $[a, R) \subset (-R, R)$. $\square$

Claim 7.4. Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two numerical sequences. If

$$\exists \lim_{k \rightarrow \infty} a_k = \alpha > 0,$$

then

$$\overline{\lim}_{k \rightarrow \infty} (a_k b_k) = \alpha \cdot \overline{\lim}_{k \rightarrow \infty} b_k.$$

Proof. See Calculus-1. $\square$

Claim 7.5 (On Derivative of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the power series. Then:

1. the radius of convergence of the series

$$\sum_{k=1}^{\infty} k a_k x^{k-1}$$

is equal to R ;

2. for every $x \in (-R, R)$, we have

$$S'(x) = \left(\sum_{k=0}^{\infty} a_k x^k \right)' = \sum_{k=1}^{\infty} k a_k x^{k-1}. \quad (3)$$

Proof. 1) Note that:

- the series

$$\sum_{k=1}^{\infty} k a_k x^{k-1} \quad \text{and} \quad \sum_{k=1}^{\infty} k a_k x^k = x \sum_{k=1}^{\infty} k a_k x^{k-1}$$

have the same interval of convergence, therefore, due to [Def 7.2](#), they have the same radius of convergence;

- since

$$\lim_{k \rightarrow \infty} k^{1/k} = 1 > 0,$$

due to [Claim 7.4](#), we have the equality

$$\begin{aligned} \overline{\lim}_{k \rightarrow \infty} |k a_k|^{1/k} &= \overline{\lim}_{k \rightarrow \infty} \left(k^{1/k} |a_k|^{1/k} \right) \\ &= 1 \cdot \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}. \end{aligned}$$

which, due to [Claim 7.2](#) [The Cauchy–Hadamard Formula], implies that the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence.

Therefore, the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence and [Item 1](#)) is proved.

2) Let $x_0 \in (-R, R)$ be arbitrary. Then it ought to be clear that there is an open interval $J = (-r, r)$ such that $x_0 \in J$ and $0 < r < R$.

Let

$$u_k(x) = a_k x^k, \quad \forall k \in \mathbb{N} \cup \{0\},$$

and

$$f_n(x) = \sum_{k=0}^n u_k(x), \quad \forall n \in \mathbb{N} \cup \{0\}.$$

Then note that:

- a) $\forall k \in \mathbb{N} \cup \{0\}$, $u_k(x) = a_k x^k$ is differentiable on $J = (-r, r)$;
- b) the series $\sum_{k=0}^{\infty} u_k(x)$ converges at $x = 0 \in J$;
- c) the series

$$\sum_{k=0}^{\infty} u'_k(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

converges uniformly on J .

Indeed, due to [Item 1](#)), the radius of convergence of the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ is equal to $R > 0$. Therefore, due to [Item 1](#)) of [Claim 7.3](#), it converges uniformly on the interval $[-r, r] \subset (-R, R)$, which implies the uniform convergence of the series on $(-r, r)$.

That is, all conditions of [Claim 6.13](#) [Term-by-term Diff. of Series] are satisfied and [Equality \(3\)](#) holds true $\forall x \in (-r, r)$ due to [Item 2](#)) of [Claim 6.13](#). Then, since $x_0 \in (-R, R)$ is arbitrary, [Equality \(3\)](#) holds true $\forall x \in (-R, R)$ and [Item 2](#)) is proved. $\square$

Note that the series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=1}^{\infty} k a_k x^{k-1}$ have the same radius of convergence, but the series may have different intervals of convergence (see [Def 7.2](#)).

For example, if $a_0 = 0$, $a_k = \frac{1}{k^2}$, $\forall k \in \mathbb{N}$, we have

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=1}^{\infty} \frac{1}{k^2} x^k \quad \text{with } R = 1 \text{ and } D = [-1, 1],$$

and

$$\sum_{k=1}^{\infty} k a_k x^{k-1} = \sum_{k=1}^{\infty} \frac{1}{k} x^{k-1} \quad \text{with } R = 1 \text{ and } D = [-1, 1).$$

Claim 7.6 (Analyticity of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the series. Then $S \in C^\infty(-R, R)$ and, for $\forall n \in \mathbb{N}$, $\forall x \in (-R, R)$, we have

$$S^{(n)}(x) = \sum_{k=n}^{\infty} k(k-1) \cdots (k-n+1) a_k x^{k-n},$$

where $C^\infty(-R, R)$ is the set of all infinitely differentiable functions on $(-R, R)$ and $S^{(n)}(x)$ is the n -th derivative of $S(x)$.

Proof. Follows directly from [Claim 7.5](#) via mathematical induction on $n \in \mathbb{N}$. $\square$

Claim 7.7 (On Antiderivative of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the series. Then:

1. the radius of convergence of the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$$

is equal to R ;

2. for every $x \in (-R, R)$, we have

$$\int_0^x S(t) dt = \int_0^x \left(\sum_{k=0}^{\infty} a_k t^k \right) dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = \sum_{k=0}^{\infty} \left(\int_0^x a_k t^k dt \right). \quad (4)$$

Proof. 1) Note that:

- the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = x \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$$

and the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$$

have the same interval of convergence, therefore, due to [Def 7.2](#), they have the same radius of convergence;

- since

$$\lim_{k \rightarrow \infty} \left(\frac{1}{k+1} \right)^{1/k} = 1 > 0,$$

due to [Claim 7.4](#), we have the equality

$$\begin{aligned} \overline{\lim}_{k \rightarrow \infty} \left| \frac{1}{k+1} a_k \right|^{1/k} &= \overline{\lim}_{k \rightarrow \infty} \left(\left(\frac{1}{k+1} \right)^{1/k} |a_k|^{1/k} \right) \\ &= 1 \cdot \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}. \end{aligned}$$

which, due to [Claim 7.2](#) [The Cauchy–Hadamard Formula], implies that the series $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence.

Therefore, the series $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence and [Item 1\)](#) is proved.

- 2) Let $x \in (-R, R)$, $I = [0, x] \subset (-R, R)$,

$$u_k(t) = a_k t^k, \quad \forall k \in \mathbb{N} \cup \{0\},$$

and

$$f_n(t) = \sum_{k=0}^n u_k(t).$$

Then note that:

- a) $\forall k \in \mathbb{N} \cup \{0\}$, $u_k(t) = a_k t^k$ is Riemann integrable on $I = [0, x]$;

b) the series $\sum_{k=0}^{\infty} a_k t^k$ converges uniformly on $I = [0, x]$.

The latter holds since $I = [0, x] \subset (-R, R)$ and due to [Item 1](#)) of [Claim 7.3](#). Thus, all conditions of [Claim 6.12](#) [Term-by-term Int. of Series] are satisfied and, for every $x \in (-R, R)$, we have

$$\int_0^x S(t) dt = \int_0^x \left(\sum_{k=0}^{\infty} a_k t^k \right) dt \stackrel{1}{=} \sum_{k=0}^{\infty} \left(\int_0^x a_k t^k dt \right) \stackrel{2}{=} \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}.$$

Here, 1) is due to [Claim 6.12](#) and 2) follows from

$$\int_0^x a_k t^k dt = \frac{a_k}{k+1} t^{k+1} \Big|_0^x = \frac{a_k}{k+1} x^{k+1}.$$

Thus, [Equality \(4\)](#) is proved. $\square$

Note that [Item 1](#)) of [Claim 7.7](#) tells us that the series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$ have the same radius of convergence, but the series may have different intervals of convergence (see [Def 7.2](#)).

For example, if $a_0 = 0$, $a_k = \frac{1}{k}$, $\forall k \in \mathbb{N}$, we have

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=1}^{\infty} \frac{1}{k} x^k \quad \text{with } R = 1 \text{ and } D = [-1, 1),$$

and

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} x^{k+1} \quad \text{with } R = 1 \text{ and } D = [-1, 1].$$

Claim 7.8 (On Equality of Power Series). Let

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad \text{on } (-R_1, R_1)$$

and

$$g(x) = \sum_{k=0}^{\infty} b_k x^k \quad \text{on } (-R_2, R_2),$$

where $R_1 > 0$ and $R_2 > 0$ are radii of convergence of the power series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} b_k x^k$ respectively. If there is a numerical sequence $\{x_n\}_{n=1}^{\infty}$ such that:

a) $x_n \in (-r, r) \setminus \{0\}$, $\forall n \in \mathbb{N}$, where

$$r = \min\{R_1, R_2\} > 0;$$

b) $\lim_{n \rightarrow \infty} x_n = 0$;

c) $f(x_n) = g(x_n)$ for every $n \in \mathbb{N}$,

then $a_k = b_k$, $\forall k \in \mathbb{N} \cup \{0\}$.

Proof. Let

$$c_k = a_k - b_k, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Then, for every $x \in (-r, r)$, we have

$$h(x) := \sum_{k=0}^{\infty} c_k x^k = \sum_{k=0}^{\infty} (a_k - b_k) x^k \stackrel{1}{=} \sum_{k=0}^{\infty} a_k x^k - \sum_{k=0}^{\infty} b_k x^k = f(x) - g(x), \quad (**)$$

where [Equality 1](#)) holds since both series converge due to [Item 1](#)) of [Claim 7.1](#).

From [Formula \(**\)](#) it follows that:

- the series

$$h(x) = \sum_{k=0}^{\infty} c_k x^k$$

has the radius of convergence $R \geq r > 0$;

•

$$h(x_n) = 0, \quad \forall n \in \mathbb{N}; \tag{5}$$

- it suffices to prove that

$$c_k := a_k - b_k = 0, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Let us assume that there is $k \in \mathbb{N} \cup \{0\}$ such that $c_k \neq 0$. Then

$$m = \min\{k \in \mathbb{N} \cup \{0\} \mid c_k \neq 0\}$$

is well-defined and, for every $x \in (-r, r)$, we have

$$\begin{aligned} h(x) &= \sum_{k=0}^{\infty} c_k x^k = \sum_{k=0}^{m-1} c_k x^k + \sum_{k=m}^{\infty} c_k x^k \\ &= 0 + \sum_{k=m}^{\infty} c_k x^k = x^m \sum_{k=0}^{\infty} c_{k+m} x^k. \end{aligned}$$

Here, the first finite sum is 0 since $c_k = 0$ for every $k \in \{0, 1, \dots, m-1\}$.

It ought to be clear that the series

$$\widehat{h}(x) := \sum_{k=0}^{\infty} c_{k+m} x^k$$

and

$$h(x) = x^m \sum_{k=0}^{\infty} c_{m+k} x^k = x^m \widehat{h}(x)$$

have the same interval of convergence, therefore, due to [Def 7.2](#), the series have the same radius of convergence $R \geq r > 0$.

Since, due to [Item a\)](#), $x_n \neq 0$ for every $n \in \mathbb{N}$, the equalities

$$h(x) = x^m \widehat{h}(x), \quad \forall x \in (-r, r),$$

and [Formula \(5\)](#) imply that

$$\widehat{h}(x_n) = 0, \quad \forall n \in \mathbb{N}.$$

Now note that

$$c_m \stackrel{1}{=} \sum_{k=0}^{\infty} c_{k+m} 0^k \stackrel{2}{=} \widehat{h}(0) \stackrel{3}{=} \widehat{h}\left(\lim_{n \rightarrow \infty} x_n\right) \stackrel{4}{=} \lim_{n \rightarrow \infty} \widehat{h}(x_n) \stackrel{5}{=} \lim_{n \rightarrow \infty} 0 = 0.$$

Indeed:

1. since $0^0 := 1$ and $0^k = 0$, $\forall k \in \mathbb{N}$;
2. by definition, $\widehat{h}(x) = \sum_{k=0}^{\infty} c_{m+k} x^k$, $\forall x \in (-r, r)$, $r > 0$;
3. $\lim_{n \rightarrow \infty} x_n = 0$ due to [Item b\)](#);

4. due to [Item 2](#) of [Claim 7.5](#), $\hat{h}$ is continuous on $(-r, r)$; since $\lim_{n \rightarrow \infty} x_n = 0 \in (-r, r)$, this implies that

$$\lim_{n \rightarrow \infty} \hat{h}(x_n) = \hat{h}\left(\lim_{n \rightarrow \infty} x_n\right);$$

5. due to [Equality \(5\)](#), we have $\hat{h}(x_n) = 0, \forall n \in \mathbb{N}$.

That is, we arrived at a contradiction (with $c_m \neq 0$). Therefore, the assumption is false and $c_k = 0, \forall k \in \mathbb{N} \cup \{0\}$. $\square$

Now we can easily prove [Claim 4.2](#) [The Abel Theorem on the Cauchy product].

Claim 7.9 (The Abel Theorem on the Cauchy product). Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be two convergent series with sums a and b respectively. If their Cauchy product $\sum_{k=0}^{\infty} c_k$, where

$$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0, \quad \forall k \in \mathbb{N} \cup \{0\},$$

is a convergent series with the sum c , then $c = a \cdot b$.

Proof. Let us consider the series:

$$f(x) := \sum_{k=0}^{\infty} a_k x^k, \quad g(x) := \sum_{k=0}^{\infty} b_k x^k, \quad h(x) := \sum_{k=0}^{\infty} c_k x^k.$$

Let R_1, R_2 , and R_3 be the radii of convergence of $f(x), g(x)$, and $h(x)$ respectively. Since

$$f(1) = \sum_{k=0}^{\infty} a_k 1^k = a, \quad g(1) = \sum_{k=0}^{\infty} b_k 1^k = b, \quad h(1) = \sum_{k=0}^{\infty} c_k 1^k = c,$$

due to [Def 7.2](#), we have $R_1 \geq 1, R_2 \geq 1$, and $R_3 \geq 1$. Therefore, due to [Item 1](#)) (if $R_i > 1$) or [Item 2](#)) (if $R_i = 1$) of [Claim 7.3](#), the series $f(x), g(x)$, and $h(x)$ converge uniformly on the interval $[0, 1]$.

Therefore, due to [Claim 6.11](#) [Continuity of a Series], the functions $f(x), g(x)$, and $h(x)$ are continuous on $[0, 1]$ and we have

$$\lim_{x \rightarrow 1-0} f(x) \stackrel{1}{=} f(1) = a, \quad \lim_{x \rightarrow 1-0} g(x) \stackrel{1}{=} g(1) = b, \quad \lim_{x \rightarrow 1-0} h(x) \stackrel{1}{=} h(1) = c. \quad (6)$$

[Equalities 1](#)) hold since the functions are continuous at $x = 1$.

Since $R_1 \geq 1$ and $R_2 \geq 1$, due to [Item 1](#)) of [Claim 7.1](#), the series $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$ converge absolutely for every $x \in [0, 1)$.

Note that, for every $k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, we have

$$c_k x^k = (a_0 x^0)(b_k x^k) + (a_1 x^1)(b_{k-1} x^{k-1}) + \cdots + (a_k x^k)(b_0 x^0).$$

That is, due to [Def 4.1](#),

$$h(x) = \sum_{k=0}^{\infty} c_k x^k$$

is the Cauchy product of the series

$$\sum_{k=0}^{\infty} a_k x^k = f(x) \quad \text{and} \quad \sum_{k=0}^{\infty} b_k x^k = g(x).$$

Therefore, due to [Claim 4.1](#) [The Mertens Theorem], we have

$$h(x) = f(x) \cdot g(x), \quad \text{for every } x \in [0, 1). \quad (7)$$

To finish the proof, note that:

$$c \stackrel{1}{=} \lim_{x \rightarrow 1-0} h(x) \stackrel{2}{=} \lim_{x \rightarrow 1-0} (f(x)g(x)) \stackrel{3}{=} \left(\lim_{x \rightarrow 1-0} f(x) \right) \left(\lim_{x \rightarrow 1-0} g(x) \right) \stackrel{4}{=} ab.$$

Here, 1) and 4) are due to [Equalities \(6\)](#), 2) is due to [Equality \(7\)](#), and 3) is due to the [Equality \(7\)](#) for every $x < 1$ and we can assume that $x > 0$. $\square$

Claim 7.10. Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with the radius of convergence $R > 0$. Then

$$\forall p \in (0, R) \quad \exists C > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad |a_k| \leq \frac{C}{p^k}.$$

Proof. Since $p \in (0, R)$, due to [Item 1\)](#) of [Claim 7.1](#), the series

$$\sum_{k=0}^{\infty} |a_k| p^k$$

converges. Therefore, due to [Claim 1.5](#),

$$\lim_{k \rightarrow \infty} |a_k| p^k = 0.$$

It is well-known (see Calculus-1) that every convergent sequence is bounded. Thus we have

$$\exists C > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad |a_k| p^k < C.$$

Since $p > 0$, the claim is proved. $\square$

Calculus 2

8 Lecture 8

Analytic Functions

Definition 8.1. Let $n \in \mathbb{N} \cup \{0\}$ and let $I \subseteq \mathbb{R}$ be an open interval. Then, a function $f : I \rightarrow \mathbb{R}$ is said to be

- a) *n-differentiable* on a set $E \subseteq I$ if the derivatives

$$f^0 = f, f', f'', \dots, f^{(n)}$$

exist at every point of E ;

- b) of *differentiable class* $C^n(E)$ if the derivatives

$$f^0 = f, f', f'', \dots, f^{(n)}$$

exist and are continuous at every point of E ;

- c) *smooth* (or *infinitely differentiable*) on E , denoted $f \in C^\infty(E)$, if it has derivatives of all orders at every point of E .

It ought to be clear that (see Calculus-1)

$$C^n(E) \subset C^{n-1}(E), \quad C^\infty(E) = \bigcap_{n=0}^{\infty} C^n(E).$$

Definition 8.2. Let $I \subseteq \mathbb{R}$ be an open interval and let a function $f : I \rightarrow \mathbb{R}$ be smooth at a point $\lambda \in I$ (i.e. $f \in C^\infty(\{\lambda\})$). Then, the *Taylor series* of the function $f(x)$ at the point λ (denoted $T_{f,\lambda}(x)$) is the power series of the form

$$T_{f,\lambda}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k = f(\lambda) + f'(\lambda)(x - \lambda) + \frac{f''(\lambda)}{2!} (x - \lambda)^2 + \dots \quad (1)$$

Coefficients $\frac{f^{(k)}(\lambda)}{k!}$ are called the *Taylor coefficients*.

Definition 8.3. Let $I \subseteq \mathbb{R}$ be an open interval. Then a function $f : I \rightarrow \mathbb{R}$ is said to be *analytic* at a point $\lambda \in I$, denoted $f \in C^\omega(\{\lambda\})$, if there is a sequence $\{a_k\}_{k=0}^{\infty} \subset \mathbb{R}$ and $\delta > 0$ such that

$$f(x) = \sum_{k=0}^{\infty} a_k (x - \lambda)^k, \quad \forall x \in (\lambda - \delta, \lambda + \delta).$$

That is, a function f is analytic at a point $\lambda \in I$ if it can be given by a convergent power series centered at λ in some neighbourhood of λ .

A function f is *analytic on an open interval* $J \subseteq I$ if it is analytic at every point $\lambda \in J$.

Claim 8.1 (Any Analytic Function is Smooth). Let $I \subseteq \mathbb{R}$ be an open interval. If a function $f : I \rightarrow \mathbb{R}$ is analytic at a point $\lambda \in I$ (i.e. $f \in C^\omega(\{\lambda\})$), then:

1. f is smooth at λ (i.e. $f \in C^\infty(\{\lambda\})$);
- 2.

$$f(x) = T_{f,\lambda}(x), \quad \forall x \in (\lambda - \delta, \lambda + \delta),$$

for some $\delta > 0$, where $T_{f,\lambda}(x)$ is the Taylor series of the function $f(x)$ at the point λ (see [Def 8.2](#)).

Proof. 1) Due to [Def 8.3](#), we have

$$f(x) = \sum_{n=0}^{\infty} a_n(x - \lambda)^n, \quad \forall x \in (\lambda - \delta, \lambda + \delta), \quad (1)$$

for some $\delta > 0$. Therefore, due to [Def 7.2](#), the series $\sum_{n=0}^{\infty} a_n(x - \lambda)^n$ has a positive radius of convergence $R \geq \delta > 0$. Thus, due to [Claim 7.6](#), [Equality \(1\)](#) implies that, for every $k \in \mathbb{N} \cup \{0\}$ and every $x \in (\lambda - \delta, \lambda + \delta)$, we have

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n (x - \lambda)^{n-k}.$$

If we substitute $x = \lambda$, then

$$f^{(k)}(\lambda) = k! a_k, \quad a_k = \frac{f^{(k)}(\lambda)}{k!}, \quad \forall k \in \mathbb{N} \cup \{0\}. \quad (2)$$

Thus, due to [Def 8.1.c](#)), f is smooth on $E = \{\lambda\}$.

2) Note that, $\forall x \in (\lambda - \delta, \lambda + \delta)$, we have

$$f(x) \stackrel{1}{=} \sum_{k=0}^{\infty} a_k (x - \lambda)^k \stackrel{2}{=} \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k \stackrel{3}{=} T_{f, \lambda}(x).$$

Here 1 holds due to [Equality \(1\)](#), 2 holds due to [Equality \(2\)](#), and 3 holds due to [Def 8.2](#) and since $f \in C^\infty(\{\lambda\})$ due to [Item 1](#). $\square$

Claim 8.2 (Example of non-analytic smooth function). Let $\tilde{\tau} : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$\tilde{\tau}(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Then:

1.

$$\tilde{\tau}^{(n)}(x) = \begin{cases} \frac{P_n(x)}{x^{2n}} \tilde{\tau}(x), & x > 0, \\ 0, & x \leq 0, \end{cases} \quad \forall n \in \mathbb{N}, \quad (3)$$

where $P_n(x)$ is a polynomial of degree $n - 1$ given recursively by $P_1(x) = 1$ and

$$P_{n+1}(x) = x^2 P'_n(x) - (2nx - 1)P_n(x), \quad \forall n \geq 1;$$

2. $\tilde{\tau} \in C^\infty(\mathbb{R})$ and $\tilde{\tau}^{(n)}(0) = 0, \forall n \in \mathbb{N} \cup \{0\}$;

3. $\tilde{\tau}$ is not analytic at $\lambda = 0$.

Proof. 1) and 2) One can prove by math. induction on $n \in \mathbb{N}$. We leave it as an exercise for the reader.

3) Let us assume that $\tilde{\tau}(x)$ is analytic at $\lambda = 0$. Then, for some $\delta > 0$ and $\forall x \in (0 - \delta, 0 + \delta) = (-\delta, \delta)$ we have

$$\tilde{\tau}(x) = T_{\tilde{\tau}, 0}(x) = \sum_{k=0}^{\infty} \frac{\tilde{\tau}^{(k)}(0)}{k!} (x - 0)^k = \sum_{k=0}^{\infty} 0 \cdot x^k = 0. \quad (4)$$

The first equality holds due to [Item 2\)](#) of [Claim 8.1](#) (since we assumed that $\tilde{\tau} \in C^\omega(\{0\})$), the second due to [Def 8.2](#) and since $\tilde{\tau} \in C^\infty(\{0\}) \subset C^\infty(\mathbb{R})$, and the third since $\tilde{\tau}^{(k)}(0) = 0, \forall k \in \mathbb{N} \cup \{0\}$.

[Equality \(4\)](#) contradicts the definition of $\tilde{\tau}$ (indeed, $\forall x > 0$ we have $\tilde{\tau}(x) = e^{-1/x} > 0$). Thus, our assumption is false and $\tilde{\tau}(x)$ is not analytic at $\lambda = 0$. $\square$

Later we will need the following well-known fact.

Claim 8.3 (The Taylor Formula). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$ be an $(n+1)$ -differentiable on I function (see [Def 8.1.a](#)), and let $\lambda \in I$. Then, for each $x \in I$, there is a point β located between λ and x such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k + R_n(x), \quad (5)$$

where the remainder term $R_n(x)$ can be written as

$$\bullet \quad R_n(x) = \frac{f^{(n+1)}(\beta)}{(n+1)!} (x - \lambda)^{n+1} \quad [\text{the Lagrange form of } R_n(x)]$$

or

$$\bullet \quad R_n(x) = \frac{f^{(n+1)}(\beta)}{n!} (x - \beta)^n (x - \lambda) \quad [\text{the Cauchy form of } R_n(x)].$$

Proof. See Calculus-1. □

Claim 8.4 (Criterion for Analyticity). Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$ be a smooth function on I (i.e. $f \in C^\infty(I)$). Then, the following two conditions are equivalent:

1. f is analytic at $\lambda \in I$ (i.e. $f \in C^\omega(\{\lambda\})$);
2. there exist constants $L > d > 0$ and $C > 0$ such that

$$(\lambda - d, \lambda + d) \subset I \quad \text{and} \quad |f^{(k)}(x)| \leq C \cdot \frac{k!}{L^k}$$

for all $k \in \mathbb{N} \cup \{0\}$ and all $x \in (\lambda - d, \lambda + d)$.

Proof. 2) $\Rightarrow$ 1) Due to [Def 8.3](#), it suffices to prove that

$$\exists \delta > 0 \quad \forall x \in (\lambda - \delta, \lambda + \delta) \quad \rightarrow \quad f(x) = T_{f,\lambda}(x), \quad (6)$$

where $T_{f,\lambda}(x)$ is the Taylor series of f at λ (see [Def 8.2](#)).

Since $f \in C^\infty(I)$, conditions of [Claim 8.3](#) are satisfied for $\forall n \in \mathbb{N}$ and, due to [Formula \(5\)](#), we have

$$f(x) = T_{f,\lambda}^n(x) + R_n(x), \quad \forall n \in \mathbb{N} \cup \{0\}, \quad \forall x \in I, \quad (7)$$

where

$$T_{f,\lambda}^n(x) = \sum_{k=0}^n \frac{f^{(k)}(\lambda)}{k!} (x - \lambda)^k$$

is the n -th partial sum of $T_{f,\lambda}(x)$ and

$$R_n(x) = \frac{f^{(n+1)}(\beta)}{(n+1)!} (x - \lambda)^{n+1}$$

for some β between λ and x .

Since, due to [Equality \(7\)](#), for every $x \in I$ we have

$$\lim_{n \rightarrow \infty} R_n(x) = \lim_{n \rightarrow \infty} (f(x) - T_{f,\lambda}^n(x)) = f(x) - \lim_{n \rightarrow \infty} T_{f,\lambda}^n(x),$$

[Condition \(6\)](#) can be written as

$$\exists \delta > 0 \quad \forall x \in (\lambda - \delta, \lambda + \delta) \quad \rightarrow \quad \lim_{n \rightarrow \infty} R_n(x) = 0. \quad (8)$$

Let $\delta = d$, where $d > 0$ is from [Item 2](#)), then, $\forall x \in (\lambda - d, \lambda + d)$, we have

$$|R_n(x)| \leq C \cdot \frac{(n+1)!}{L^{n+1}} \cdot \frac{1}{(n+1)!} \cdot d^{n+1} = C \left(\frac{d}{L}\right)^{n+1}. \quad (9)$$

Indeed, the first inequality holds due to the definition of $R_n(x)$, since β is between λ and x , hence $\beta \in (\lambda - d, \lambda + d)$, and since $|x - \lambda| < d$. Since $L > d > 0$, we have

$$\lim_{n \rightarrow \infty} C \left(\frac{d}{L}\right)^{n+1} = 0.$$

Therefore, [Inequality \(9\)](#) implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for every $x \in (\lambda - d, \lambda + d)$. Thus [Condition \(8\)](#) (and [\(6\)](#), since they are equivalent) is proved.

1) $\Rightarrow$ 2) First note that if we differentiate $n \in \mathbb{N} \cup \{0\}$ times term-by-term the well-known equality

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad \forall x \in (-1, 1),$$

due to [Claim 7.6](#), we will receive

$$\sum_{k=n}^{\infty} k^n x^{k-n} = \frac{n!}{(1-x)^{n+1}}, \quad \forall x \in (-1, 1), \quad (10)$$

where

$$k^n = \begin{cases} k(k-1) \cdots (k-n+1), & n \in \mathbb{N}, \\ 1, & n = 0. \end{cases}$$

Due to [Item 2](#)) of [Claim 8.1](#), $f \in C^\omega(\{\lambda\})$ implies that

$$f(x) = T_{f,\lambda}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda)}{k!} (x-\lambda)^k, \quad \forall x \in (\lambda - \delta, \lambda + \delta) \subset I \quad (11)$$

for some $\delta > 0$. Note that, due to [Def 7.2](#) and [Equality \(11\)](#), the radius of convergence of the series $T_{f,\lambda}(x)$ is at least $\delta > 0$. Let $p = \frac{3}{4}\delta > 0$, then, due to [Claim 7.9](#),

$$\exists \widehat{C} > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad \left| \frac{f^{(k)}(\lambda)}{k!} \right| \leq \frac{\widehat{C}}{p^k}. \quad (12)$$

Let $d = \frac{1}{4}\delta > 0$, then

$$(\lambda - d, \lambda + d) \subset (\lambda - \delta, \lambda + \delta) \subset I.$$

If we differentiate n times term-by-term [Equality \(11\)](#), due to [Claim 7.6](#), we will receive

$$f^{(n)}(x) = \sum_{k=n}^{\infty} k^n \frac{f^{(k)}(\lambda)}{k!} (x-\lambda)^{k-n}, \quad \forall x \in (\lambda - \delta, \lambda + \delta), \quad (13)$$

where coefficients k^n are defined as in [\(10\)](#).

Let $y \in (\lambda - d, \lambda + d)$.

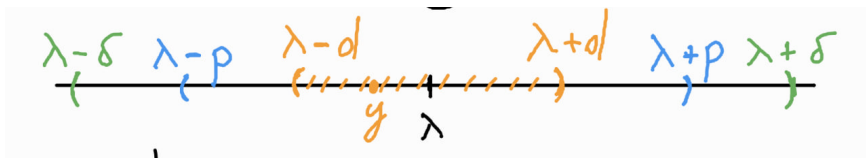


Figure 8.1

Then, for $\forall n \in \mathbb{N} \cup \{0\}$, we have

$$\begin{aligned}
|f^{(n)}(y)| &\leq \sum_{k=n}^{\infty} k^n \left| \frac{f^{(k)}(\lambda)}{k!} \right| |y - \lambda|^{k-n} \\
&\leq \sum_{k=n}^{\infty} k^n \frac{\widehat{C}}{p^k} |y - \lambda|^{k-n} \\
&= \frac{\widehat{C}}{p^n} \sum_{k=n}^{\infty} k^n \left(\frac{|y - \lambda|}{p} \right)^{k-n} \\
&= \frac{\widehat{C}}{p^n} \cdot \frac{n!}{\left(1 - \frac{|y - \lambda|}{p}\right)^{n+1}} \\
&= \frac{p\widehat{C}n!}{(p - |y - \lambda|)^{n+1}} \\
&\leq \frac{p\widehat{C}n!}{(\frac{1}{2}\delta)^{n+1}} = \frac{2p\widehat{C}}{\delta} \cdot \frac{n!}{(\frac{1}{2}\delta)^n} = C \cdot \frac{n!}{L^n},
\end{aligned}$$

where

$$C = \frac{2p\widehat{C}}{\delta} = \frac{2 \cdot \frac{3}{4}\delta\widehat{C}}{\delta} = \frac{3}{2}\widehat{C} > 0, \quad L = \frac{1}{2}\delta > 0.$$

Here we used [Formula \(10\)](#), $\frac{|y - \lambda|}{p} < \frac{d}{p} = \frac{1}{3} < 1$, and, since $p = \frac{3}{4}\delta$ and $|y - \lambda| < d = \frac{1}{4}\delta$, we have $p - |y - \lambda| > \frac{1}{2}\delta$. Since $L = \frac{1}{2}\delta > \frac{1}{4}\delta = d > 0$, the statement is proved. $\square$

Claim 8.5. 1. the functions e^x , $\sin x$, and $\cos x$ are analytic on $\mathbb{R}$;

2. $\ln x$ is analytic on $(0, +\infty)$.

Proof. 2) Due to [Claim 8.4](#), it suffices to show that for every $\lambda \in (0, +\infty)$ there are constants $L > d > 0$ and $C > 0$ such that

$$|((\ln x)^{(k)})| \leq C \cdot \frac{k!}{L^k}, \quad \forall x \in (\lambda - d, \lambda + d) \subset (0, +\infty).$$

Let $d = \frac{1}{4}\lambda$ and $L = \frac{1}{2}\lambda$. Then

$$(\lambda - d, \lambda + d) = \left(\frac{3}{4}\lambda, \frac{5}{4}\lambda\right) \subset (0, +\infty),$$

and, for every $k \in \mathbb{N}$ and every $x \in (\frac{3}{4}\lambda, \frac{5}{4}\lambda)$, we have

$$|((\ln x)^{(k)})| = \frac{(k-1)!}{x^k} \leq (k-1)! \left(\frac{4}{3\lambda}\right)^k \leq \frac{k!}{L^k}.$$

For $k = 0$, the function $\ln x$ is bounded on the closed interval $[\frac{3}{4}\lambda, \frac{5}{4}\lambda]$. Hence, choosing

$$C = \max \left\{ 1, \max_{x \in [\frac{3}{4}\lambda, \frac{5}{4}\lambda]} |\ln x| \right\},$$

we obtain

$$|((\ln x)^{(k)})| \leq C \cdot \frac{k!}{L^k} \quad \forall k \in \mathbb{N} \cup \{0\}, \quad \forall x \in (\lambda - d, \lambda + d),$$

where $L = \frac{1}{2}\lambda > d = \frac{1}{4}\lambda > 0$.

1) We leave it as exercise (similar to [Item 2](#)). $\square$

We will use the following claim to construct an important example of a smooth non-analytic function.

Claim 8.6 (The Borel Lemma). For every sequence $\{\alpha_n\}_{n=0}^\infty$ there exists a smooth function f on $\mathbb{R}$ such that

$$f^{(n)}(0) = \alpha_n, \quad \forall n \in \mathbb{N} \cup \{0\}.$$

Proof. For $\forall x \in \mathbb{R}$ let us consider the following functions

$$\tilde{\tau}(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad g(x) := \frac{\tilde{\tau}(x)}{\tilde{\tau}(x) + \tilde{\tau}(1-x)}, \quad h(x) := g(2+x)g(2-x).$$

$\tilde{\tau} \in C^\infty(\mathbb{R})$ due to Item 2 of [Claim 8.2](#).

$g \in C^\infty(\mathbb{R})$ since $\tilde{\tau}(x) + \tilde{\tau}(1-x) > 0, \forall x \in \mathbb{R}$, and $\tilde{\tau} \in C^\infty(\mathbb{R})$.

$h \in C^\infty(\mathbb{R})$ since $g \in C^\infty(\mathbb{R})$.

It ought to be clear that the function $h(x)$ equals 1 on the closed interval $[-1, 1]$ and vanishes outside the open interval $(-2, 2)$. In particular,

$$h^{(n)}(0) = 0, \quad \forall n \in \mathbb{N}.$$

For $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, let

$$u_k(x) := x^k h(x).$$

Then it is clear that, $\forall k \in \mathbb{N} \cup \{0\}$, we have

- $u_k(x) = x^k, \forall x \in [-1, 1]$, and $u_k(x) = 0, \forall x \in \mathbb{R} \setminus (-2, 2)$;
- $u_k(x) \in C^\infty(\mathbb{R})$;
-

$$u_k^{(n)}(0) = \begin{cases} n!, & n = k, \\ 0, & n \neq k, \end{cases} \quad \forall n \in \mathbb{N} \cup \{0\}, \quad (14)$$

where $u_k^{(n)}(0)$ is the n -th derivative of u_k at $x = 0$.

$\forall n \in \mathbb{N} \cup \{0\}$ the function $u_k^{(n)}(x)$ is bounded on $\mathbb{R}$, that is

$$\forall k, n \in \mathbb{N} \cup \{0\} \quad \exists L_{k,n} > 0 \quad \forall x \in \mathbb{R} \quad \rightarrow \quad |u_k^{(n)}(x)| \leq L_{k,n}. \quad (15)$$

Indeed, $\forall n \in \mathbb{N} \cup \{0\}$ we have $u_k^{(n)}(x) = 0, \forall x \in \mathbb{R} \setminus (-2, 2)$, thus it suffices to prove that $u_k^{(n)}(x)$ is bounded on $[-2, 2]$. Since $u_k^{(n)}$ is continuous on $[-2, 2]$, the claim follows from the fact that a continuous on a closed interval function attains a maximum and a minimum on the interval (see Calculus-1).

Let

$$M_k = \max\{1, |\alpha_k|, L_{k,0}, L_{k,1}, \dots, L_{k,k}\}, \quad \forall k \in \mathbb{N} \cup \{0\}, \quad (16)$$

where α_k is the k -th element of the sequence $\{\alpha_k\}_{k=0}^\infty$ and constants $L_{k,i}$ are defined in [\(15\)](#).

For $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, let

$$f_k(x) = \frac{\alpha_k}{(M_k)^k k!} u_k(M_k x).$$

Then we have

$$f_k^{(n)}(x) = \frac{\alpha_k}{(M_k)^{k-n} k!} u_k^{(n)}(M_k x), \quad \forall k, n \in \mathbb{N} \cup \{0\}, \quad \forall x \in \mathbb{R}, \quad (17)$$

and, due to [Equality \(14\)](#),

$$f_k^{(n)}(0) = \begin{cases} \alpha_n, & n = k, \\ 0, & n \neq k, \end{cases} \quad \forall k, n \in \mathbb{N} \cup \{0\}.$$

Note that, $\forall n \in \mathbb{N} \cup \{0\}$, $\forall x \in \mathbb{R}$, we have

$$\begin{aligned} \left| \sum_{k=0}^{\infty} f_k^{(n)}(x) \right| &\leq \sum_{k=0}^{\infty} |f_k^{(n)}(x)| \leq \sum_{k=0}^{\infty} \frac{|\alpha_k|}{M_k^{k-n} k!} L_{k,n} \\ &= \sum_{k=0}^{n+1} \frac{|\alpha_k|}{M_k^{k-n} k!} L_{k,n} + \sum_{k=n+2}^{\infty} \frac{1}{k!} \cdot \frac{|\alpha_k|}{M_k} \cdot \frac{L_{k,n}}{M_k} \cdot \frac{1}{M_k^{k-n-2}} \\ &\leq C_n + \sum_{k=n+2}^{\infty} \frac{1}{k!}, \end{aligned} \quad (18)$$

where $C_n \in \mathbb{R}$. Here we used $|\alpha_k|/M_k \in [0, 1]$, $L_{k,n}/M_k \in [0, 1]$ for $k > n$, and $1/M_k^{k-n-2} \in (0, 1]$ for $k \geq n+2$, due to [\(16\)](#).

Since the series $\sum_{k=n+2}^{\infty} \frac{1}{k!}$ converges (due to [Claim 2.11](#)), due to [Inequality \(18\)](#) and [Claim 5.7](#) [The Weierstrass M-test], $\forall n \in \mathbb{N} \cup \{0\}$ the functional series $\sum_{k=0}^{\infty} f_k^{(n)}(x)$ converges uniformly on $\mathbb{R}$. Therefore, due to [Claim 6.13](#), we have

$$f^{(n)}(x) = \sum_{k=0}^{\infty} f_k^{(n)}(x), \quad \forall n \in \mathbb{N}, \forall x \in \mathbb{R}, \quad (19)$$

where we defined

$$f(x) := \sum_{k=0}^{\infty} f_k(x), \quad \forall x \in \mathbb{R}.$$

It remains to note that:

- $f \in C^\infty(\mathbb{R})$ due to [\(19\)](#);
-

$$f^{(n)}(0) = \sum_{k=0}^{\infty} f_k^{(n)}(0) = \alpha_n, \quad \forall n \in \mathbb{N} \cup \{0\}.$$

□

Claim 8.7 (Pathological Examples of Smooth Functions). There exists a smooth function f on $\mathbb{R}$ such that

1. its Taylor series $T_{f,0}(x)$ has zero radius of convergence;
2. it is not analytic at any point.

Proof. 1) Let $a_0 = 1$, $a_k = k! k^k$, $\forall k \in \mathbb{N}$. Then, due to [Claim 8.6](#), there is $f \in C^\infty(\mathbb{R})$ such that

$$f^{(k)}(0) = a_k, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Then, due to [Def 8.2](#), we have

$$T_{f,0}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (x-0)^k = \sum_{k=0}^{\infty} k^k x^k.$$

It is clear that the radius of convergence of the series $\sum_{k=0}^{\infty} k^k x^k$ is equal to zero (see [Claim 7.2](#)).

2) Let

$$f(x) = \sum_{k=1}^{\infty} \cos(2^k x) e^{-\sqrt{2^k}}, \quad \forall x \in \mathbb{R}.$$

Then it can be shown that $f \in C^\infty(\mathbb{R})$ (this is not difficult and we leave it as an exercise) and f is nowhere analytic on $\mathbb{R}$ (the proof is difficult, so we shall merely state the fact). $\square$

Calculus 2

9 Lecture 9

n -Dimensional Riemann Integral: Part I

Definition 9.1. The set

$$I = \{\bar{x} \in \mathbb{R}^n \mid a_i \leq x_i \leq b_i, i = 1, 2, \dots, n\} = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n], \quad (1)$$

where $\bar{x} = (x_1, \dots, x_n)$, $\bar{a} = (a_1, \dots, a_n)$, $\bar{b} = (b_1, \dots, b_n) \in \mathbb{R}^n$, is called a *nondegenerative n -dimensional closed interval*.

Definition 9.2. Let an interval I be defined by [Equality \(1\)](#). Then, the value

$$|I| \stackrel{\text{def}}{=} \prod_{i=1}^n (b_i - a_i)$$

is called the *measure* (or *n -measure* or *n -volume*) of the interval I .

Note that, since I is a nondegenerative interval (that is, $a_i < b_i$, $i = 1, 2, \dots, n$), we have $|I| > 0$.

Definition 9.3. A *partition* P of a one-dimensional nondegenerative interval $I = [a, b] \subset \mathbb{R}$ is a finite system of points $x_0, \dots, x_n$ of the interval such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

The intervals $[x_{i-1}, x_i]$, $i = 1, 2, \dots, n$, are called the intervals of the partition P . Note that

$$I = \bigcup_{i=1}^n [x_{i-1}, x_i].$$

Definition 9.4. Let an interval I be defined by [Equality \(1\)](#). Then, partitions of the coordinate intervals $[a_i, b_i]$, $i = 1, \dots, n$, induce a partition of the interval I into finer intervals obtained as the direct products of the intervals of the partitions of the coordinate intervals. More precisely:

Let $I = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ be defined by [Equality \(1\)](#). Then, partitions of the coordinate intervals $[a_i, b_i]$

$$\begin{aligned} P_1 : \quad & a_1 = x_{10} < x_{11} < x_{12} < \cdots < x_{1t_1} = b_1, \\ P_2 : \quad & a_2 = x_{20} < x_{21} < x_{22} < \cdots < x_{2t_2} = b_2, \\ & \vdots \\ P_n : \quad & a_n = x_{n0} < x_{n1} < x_{n2} < \cdots < x_{nt_n} = b_n, \end{aligned}$$

induce a partition of I into $t_1 t_2 \cdots t_n$ many intervals

$$I_{i_1 i_2 \dots i_n} = [x_{1(i_1-1)}, x_{1i_1}] \times [x_{2(i_2-1)}, x_{2i_2}] \times \cdots \times [x_{n(i_n-1)}, x_{ni_n}] \subset \mathbb{R}^n.$$

The representation of the interval I as the union

$$I = \bigcup_{i_1=1}^{t_1} \bigcup_{i_2=1}^{t_2} \cdots \bigcup_{i_n=1}^{t_n} I_{i_1 i_2 \dots i_n}$$

of finer intervals $I_{i_1 i_2 \dots i_n}$ just described will be called a *partition of the (n -dimensional) interval I* , and will be denoted by P .

Note that, all intervals $I_{i_1 i_2 \dots i_n}$ are nondegenerate by construction.

For example: Let $I = [a_1, b_1] \times [a_2, b_2]$ be a 2-dimensional interval. Then partitions of the coordinate intervals

$$P_1 : a_1 = x_{10} < x_{11} < x_{12} < x_{13} = b_1,$$

(partition of $[a_1, b_1]$ into $t_1 = 3$ many intervals $[x_{1(i_1-1)}, x_{1i_1}]$), and

$$P_2 : a_2 = x_{20} < x_{21} < x_{22} = b_2,$$

(partition of $[a_2, b_2]$ into $t_2 = 2$ many intervals $[x_{2(i_2-1)}, x_{2i_2}]$), induce a partition of I into $t_1 t_2 = 3 \cdot 2 = 6$ many intervals:

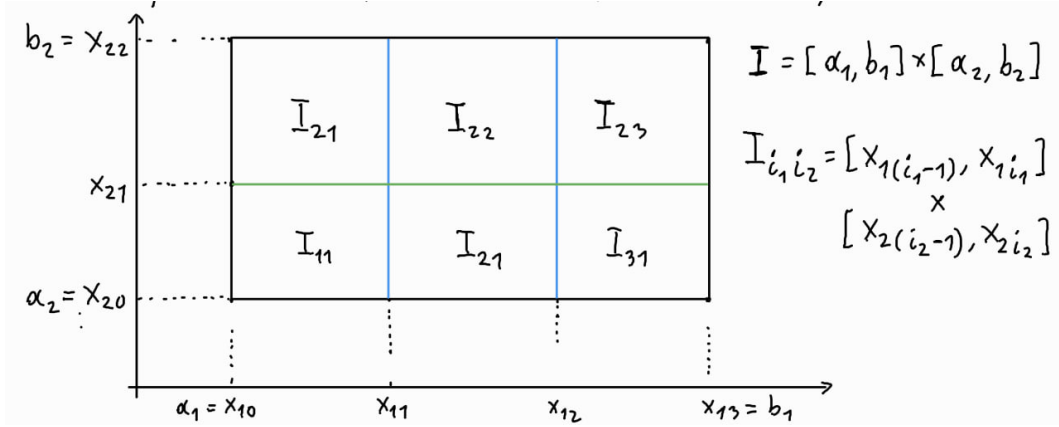


Figure 9.1

Definition 9.5. Let P be a partition of an interval I into finer intervals $I_{i_1 i_2 \dots i_n}$ (see Def 9.4). If in every interval $I_{i_1 i_2 \dots i_n}$ we fix a point $\zeta_{i_1 i_2 \dots i_n} \in I_{i_1 i_2 \dots i_n}$, then we say that we have a *tagged partition* (or a *partition with distinguished points*). The set of all $t_1 t_2 \dots t_n$ many points $\zeta_{i_1 i_2 \dots i_n}$ we will denote by the single letter ζ , and the tagged partition of I we will denote by (P, ζ) .

Definition 9.6. Let X be a non-empty subset of $\mathbb{R}^n$. Then, the *diameter* of X we will define as

$$d(X) = \sup_{\bar{a}, \bar{b} \in X} \|\bar{a} - \bar{b}\|,$$

where

$$\|\bar{a} - \bar{b}\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

for $\bar{a} = (a_1, a_2, \dots, a_n)$, $\bar{b} = (b_1, b_2, \dots, b_n)$.

Note that, if an interval I is defined by Equality (1), then it ought to be clear that

$$d(I) = \|\bar{a} - \bar{b}\| > 0.$$

Definition 9.7. Let P be a partition of an interval I (see Def 9.4). Then, the maximum among the diameters of the intervals $I_{i_1 i_2 \dots i_n}$ of the partition P is called the *mesh* of the partition P and denoted by $\lambda(P)$. That is,

$$\lambda(P) = \max_{\substack{i_1 \in \{1, \dots, t_1\} \\ \vdots \\ i_n \in \{1, \dots, t_n\}}} d(I_{i_1 i_2 \dots i_n}).$$

Definition 9.8. Let

- I be an interval defined by Equality (1);
- $(P, \bar{\zeta})$ be a tagged partition of I (see Def 9.5);
- $|I_{i_1 i_2 \dots i_n}|$ be the measure of an interval $I_{i_1 i_2 \dots i_n}$ from partition $(P, \bar{\zeta})$ (see Def 9.2);

- $f : I \rightarrow \mathbb{R}$ be a function.

Then, the sum

$$\sigma(f, P, \bar{\zeta}) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} f(\zeta_{i_1 i_2 \dots i_n}) |I_{i_1 i_2 \dots i_n}|$$

is called the *Riemann sum* (of the function f corresponding to the tagged partition $(P, \bar{\zeta})$ of the interval I).

Definition 9.9. Let I be an interval defined by [Equality \(1\)](#) and let $f : I \rightarrow \mathbb{R}$ be a function. We say that the *(multiple) Riemann integral* of the function f over the interval I (denoted by $\int_I f(\bar{x}) d\bar{x}$) exists and equals s if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for any tagged partition $(P, \bar{\zeta})$ of the interval I whose mesh (i.e. $\lambda(P)$) is less than δ , we have

$$|\sigma(f, P, \bar{\zeta}) - s| < \varepsilon,$$

where $\sigma(f, P, \bar{\zeta})$ is the Riemann sum (see [Def 9.8](#)). That is,

$$\left(\int_I f(\bar{x}) d\bar{x} = s \right) \iff \left(\lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \bar{\zeta}) = s \right), \quad (2)$$

or, equivalently,

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall (P, \bar{\zeta}) \left[\lambda(P) < \delta \implies |\sigma(f, P, \bar{\zeta}) - s| < \varepsilon \right].$$

If limit (2) exists, we say that the function $f : I \rightarrow \mathbb{R}$ is *Riemann integrable over the interval I* .

The set of all Riemann integrable over the interval $I \subset \mathbb{R}^n$ functions we shall denote by $\mathcal{R}(I)$.

Equivalent but more expanded notation of the multiple integral is

$$\int_I f(\bar{x}) d\bar{x} = \int_I f(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_I \cdots \int_I f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

If $n = 2$, then the integral is called the *double integral* and usually denoted by

$$\iint_I f(x, y) dx dy.$$

If $n = 3$, then the integral is called the *triple integral* and usually denoted by

$$\iiint_I f(x, y, z) dx dy dz.$$

Note that our process of constructing the multiple Riemann integral over an interval $I \subset \mathbb{R}^n$ is a verbatim repetition of the procedure of constructing the Riemann integral over a one-dimensional interval from Calculus-1.

Criteria for Riemann Integrability

Definition 9.10. A set $E \subset \mathbb{R}^n$ has *(n -dimensional) Lebesgue measure zero* if, for every $\varepsilon > 0$, there exist at most countable system $\{I_i\}$ of n -dimensional intervals I_i (see [Def 9.1](#)) such that

$$E \subseteq \bigcup_i I_i \quad \text{and} \quad \sum_i |I_i| \leq \varepsilon,$$

where $|I_i|$ is the measure of I_i (see [Def 9.2](#)).

If $E \subset \mathbb{R}^n$ is a set of Lebesgue measure zero, we denote this by $\mu(E) = 0$.

Note that, since the series $\sum |I_i|$ converges absolutely, by [Claim 3.9](#), the exact order of summation does not affect the sum, so that [Def 9.10](#) is unambiguous.

Claim 9.1 (Examples of sets of Lebesgue measure zero). 1. The union of a finite or countable number of sets of Lebesgue measure zero is a set of Lebesgue measure zero.

2. Any finite or countable set is a set of Lebesgue measure zero.

3. A subset of a set of Lebesgue measure zero is itself of Lebesgue measure zero.

4. There are examples of uncountable 1-dimensional sets of Lebesgue measure zero.

Proof. 1) Let $E = \bigcup_n E_n$ be at most countable union of sets $E_n \subset \mathbb{R}^n$ such that $\mu(E_n) = 0$ (see [Def 9.10](#)). Then, given $\varepsilon > 0$, for each E_n , by [Def 9.10](#), there exists at most countable system $\{I_{nk}\}$ of intervals $I_{nk} \subset \mathbb{R}^n$ such that

$$E_n \subseteq \bigcup_k I_{nk} \quad \text{and} \quad \sum_k |I_{nk}| < \frac{\varepsilon}{2^n}.$$

Since the union of at most countably many sets is itself at most countable, the system $\bigcup_n \{I_{nk}\}$ is at most countable. Since, for every n , $E_n \subseteq \bigcup_k I_{nk}$, we have

$$E = \bigcup_n E_n \subseteq \bigcup_n \bigcup_k I_{nk} = \bigcup I_{nk}.$$

Note that, by the definition of the union of sets, the order in which we unite the sets I_{nk} in $\bigcup I_{nk}$ is of no importance. Since for every n , $\sum_k |I_{nk}| < \varepsilon/2^n$, it should be clear that any finite sum of $|I_{nk}|$ is bounded by the value

$$\frac{\varepsilon}{2} + \frac{\varepsilon}{2^2} + \cdots = \varepsilon.$$

Therefore, we have

$$\sum_{n,k} |I_{nk}| \leq \varepsilon.$$

Note that, if the sum is countable, then the series $\sum_{n,k} |I_{nk}|$ converges absolutely (by [Claim 2.2](#)), and the value of $\sum_{n,k} |I_{nk}|$ does not depend on the order of summation (due to [Claim 3.9](#)). Thus, by [Def 9.10](#), $E = \bigcup_n E_n$ is a set of Lebesgue measure zero.

2) It should be clear that if $E = \{\bar{a}\} \subset \mathbb{R}^n$ is a one-point set, then, for every $\varepsilon > 0$, there exists a nondegenerate interval $I \subset \mathbb{R}^n$ such that $E = \{\bar{a}\} \subset I$ and $|I| < \varepsilon$. Indeed, if $\bar{a} = (a_1, a_2, \dots, a_n)$, then you can consider

$$I = [a_1 - \delta, a_1 + \delta] \times \cdots \times [a_n - \delta, a_n + \delta] \subset \mathbb{R}^n.$$

Then, by [Def 9.2](#), $|I| = (2\delta)^n$ and it is enough to set $\delta < \frac{1}{2} \sqrt[n]{\varepsilon}$. Since any finite (countable) set is a union of finitely (countably) many one-point sets, now [Item 2](#)) follows directly from [Item 1](#)).

3) The statement follows directly from [Def 9.10](#).

4) Such examples are relatively difficult to construct, thus we state this without proof. $\square$

Claim 9.2 (The Lebesgue Criterion for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerate closed interval (see [Def 9.1](#));
- $f : I \rightarrow \mathbb{R}$ be a function on I ;
- $T(f) = \{\bar{x} \in I \mid f \text{ is discontinuous at } \bar{x}\}$ be the set of all points of discontinuity of the function f on the interval I .

Then, f is Riemann integrable on I (i.e. $f \in \mathcal{R}(I)$) if and only if f is bounded on I (i.e. $f \in \mathcal{B}(I)$) and the set $T(f)$ is of Lebesgue measure zero (i.e. $\mu(T(f)) = 0$). Thus,

$$(f \in \mathcal{R}(I)) \iff (f \in \mathcal{B}(I) \text{ and } \mu(T(f)) = 0).$$

Proof. Since the proof of Lebesgue's Criterion is very difficult (especially the $\Leftarrow$ direction), we state it without a proof. $\square$

Note that if some property holds at all points of a set X except possibly the points of a set of Lebesgue measure zero, then we say that the property holds *almost everywhere* on X . Thus, we can rephrase the Lebesgue Criterion like this: f is Riemann integrable on I iff f is bounded on I and f is almost everywhere continuous on I .

Let us consider another useful criterion for Riemann integrability.

Definition 9.11. Let

- I be an interval defined by Equality (1);
- P be a partition of the interval I defined as in Def 9.4;
- $f : I \rightarrow \mathbb{R}$ be a function on I ;
-

$$m_{i_1 i_2 \dots i_n} \stackrel{\text{def}}{=} \inf_{\bar{x} \in I_{i_1 i_2 \dots i_n}} f(\bar{x}), \quad M_{i_1 i_2 \dots i_n} \stackrel{\text{def}}{=} \sup_{\bar{x} \in I_{i_1 i_2 \dots i_n}} f(\bar{x}).$$

Then, the quantities

$$\underline{S}(f, P) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} m_{i_1 i_2 \dots i_n} |I_{i_1 i_2 \dots i_n}|$$

and

$$\overline{S}(f, P) \stackrel{\text{def}}{=} \sum_{i_1=1}^{t_1} \sum_{i_2=1}^{t_2} \cdots \sum_{i_n=1}^{t_n} M_{i_1 i_2 \dots i_n} |I_{i_1 i_2 \dots i_n}|$$

are called the *lower Darboux sum* and the *upper Darboux sum* (of the function f over the interval I corresponding to the partition P of the interval).

Claim 9.3 (Main Properties of Darboux Sums). Let

- I be an interval defined by Equality (1);
- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then:

1. if a partition P' of the interval I is obtained by refining intervals of a partition P of I , then we have

$$\underline{S}(f, P) \leq \underline{S}(f, P') \leq \overline{S}(f, P') \leq \overline{S}(f, P);$$

2. for any two partitions P_1 and P_2 of the interval I we have

$$\underline{S}(f, P_1) \leq \overline{S}(f, P_2).$$

Proof. 1) Follows immediately from Def 9.11 (taking into account the definition of inf and sup).

2) Let P be an auxiliary partition of the interval I obtained by intersecting the intervals of the partitions P_1 and P_2 . For example:

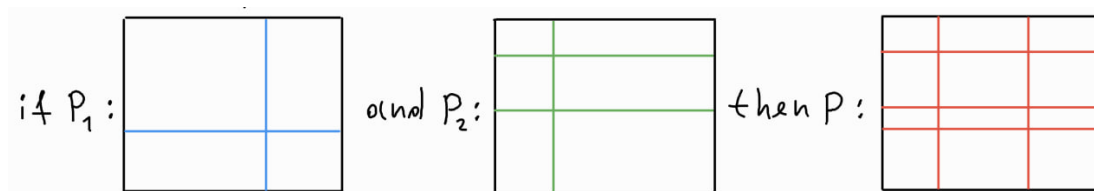


Figure 9.2

Since P is a refinement of each of the partitions P_1 and P_2 , by [Item 1](#)), we have

$$\underline{S}(f, P_1) \leq \underline{S}(f, P) \leq \overline{S}(f, P) \leq \overline{S}(f, P_2).$$

□

Definition 9.12. Let I be an interval defined by [Equality \(1\)](#) and let $f : I \rightarrow \mathbb{R}$ be a function on I . Then, the *lower Darboux integral* and the *upper Darboux integral* (of the function f over the interval I) are respectively defined to be:

$$\int_I f(\bar{x}) d\bar{x} \stackrel{\text{def}}{=} \underline{L}(f) \stackrel{\text{def}}{=} \sup_P \underline{S}(f, P),$$

and

$$\int_I f(\bar{x}) d\bar{x} \stackrel{\text{def}}{=} \overline{U}(f) \stackrel{\text{def}}{=} \inf_P \overline{S}(f, P),$$

where $\underline{S}(f, P)$ and $\overline{S}(f, P)$ are the lower and upper Darboux sums (see [Def 9.11](#)), and the supremum and infimum are taken over all possible partitions P of the interval I .

It should be clear that, due to [Def 9.11](#) and [Def 9.12](#), for any partition P of I , we have

$$\underline{S}(f, P) \leq \underline{L}(f) \leq \overline{U}(f) \leq \overline{S}(f, P).$$

Claim 9.4 (The Darboux Theorem). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see [Def 9.1](#));
- $f : I \rightarrow \mathbb{R}$ be a bounded on I function.

Then:

1.

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f);$$

2.

$$\lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P) = \overline{U}(f).$$

Proof. 1) By the definition of the limit (see Calculus-1) we have:

$$\left(\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) \right) \iff (\forall \varepsilon > 0 \exists \delta > 0 \forall P [\lambda(P) < \delta \Rightarrow |\underline{S}(f, P) - \underline{L}(f)| \leq \varepsilon]). \quad (3)$$

That is, it suffices to prove the right-hand statement. Fix $\varepsilon > 0$. Note that

$$(f \in \mathcal{B}(I)) \Rightarrow (\exists M > 0 \forall \bar{x} \in I [|f(\bar{x})| < M]).$$

Let

$$\widehat{\varepsilon} = \frac{\varepsilon}{2M + 1}.$$

By [Def 9.12](#) (essentially by the definition of the supremum), there exists a partition $\widehat{P}$ of the interval I such that

$$\underline{L}(f) - \widehat{\varepsilon} < \underline{S}(f, \widehat{P}). \quad (4)$$

Let $\widehat{T}$ be the set of all points of the interval I laying on the boundary of the intervals of the partition $\widehat{P}$. Taking into account the simple structure of $\widehat{T}$, it is not that difficult to see that there exists a number $\delta > 0$ such that the sum of the measures of those intervals that intersect $\widehat{T}$ is less than $\widehat{\varepsilon}$ for every partition P of I such that $\lambda(P) < \delta$. Indeed, take a look at the picture:

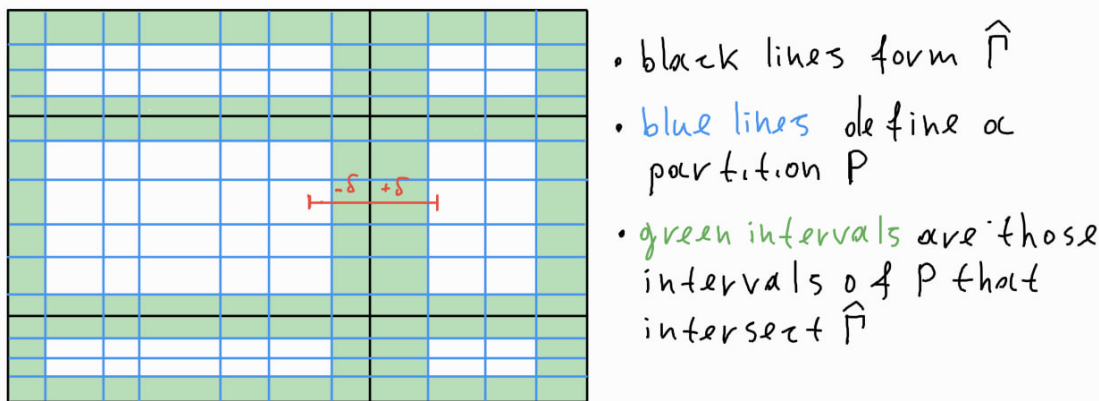


Figure 9.3

Since $\hat{\varepsilon}$ and $\hat{P}$ are fixed, it should be clear that there exists $\delta > 0$ such that the sum of the areas (2-measures) of the green intervals is less than $\hat{\varepsilon}$ for every partition P such that $\lambda(P) < \delta$.

Let P be any partition of I such that $\lambda(P) < \delta$. Due to (4), it remains to prove that

$$|\underline{S}(f, P) - \underline{L}(f)| \leq \varepsilon.$$

Let P' be an auxiliary partition of I obtained by intersecting the intervals of partitions $\hat{P}$ and P . Then, we have

$$\underline{L}(f) - \hat{\varepsilon} \stackrel{(4)}{<} \underline{S}(f, \hat{P}) \leq \underline{S}(f, P') \leq \underline{L}(f). \quad (5)$$

Also note that

$$\begin{aligned} |\underline{S}(f, P') - \underline{S}(f, P)| &\leq \sum_{\substack{I_i \in P \\ I_i \cap \hat{T} \neq \emptyset}} 2M |I_i| \\ &\leq 2M \hat{\varepsilon}. \end{aligned} \quad (6)$$

Indeed, the sums $\underline{S}(f, P')$ and $\underline{S}(f, P)$ both contain all the terms that correspond to intervals of the partition P that do not intersect $\hat{T}$, and if an interval I_i from P is such that $I_i \cap \hat{T} \neq \emptyset$, then, by the construction of a partition P' , there are intervals $I_{i1}, \dots, I_{ik}$ in P' such that their union is I_i . Then

$$\left| \sup_{I_i} f + \dots + \sup_{I_{ik}} f - \inf_{I_i} f \right| \leq 2 \sup_I f \leq 2M.$$

Since $\lambda(P) < \delta$, by the paragraph after Inequality (4), we have

$$\sum_{\substack{I_i \in P \\ I_i \cap \hat{T} \neq \emptyset}} |I_i| < \hat{\varepsilon}.$$

Therefore,

$$\begin{aligned} |\underline{S}(f, P) - \underline{L}(f)| &\leq |\underline{S}(f, P) - \underline{S}(f, P')| + |\underline{S}(f, P') - \underline{L}(f)| \\ &\leq 2M \hat{\varepsilon} + \hat{\varepsilon} = \varepsilon. \end{aligned}$$

Thus, we proved Item 1).

2) Can be proven similarly to Item 1). □

Claim 9.5 (Necessary Condition for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see Def 9.1);

- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then,

$$f \in \mathcal{R}(I) \implies f \in \mathcal{B}(I).$$

That is, every Riemann integrable on I function is bounded on I .

Proof. Let f be unbounded on I . Then, for any partition P of the interval I , there exists an interval I_i of the partition P such that f is unbounded on I_i . If $(P, \bar{\zeta}')$ and $(P, \bar{\zeta}'')$ are two tagged partitions of P (see Def 9.8) with points $\bar{\zeta}'$ and $\bar{\zeta}''$ differing only in the choice of the points $\bar{\zeta}'_i$ and $\bar{\zeta}''_i$, then, by Def 9.8, we have

$$|\sigma(f, P, \bar{\zeta}') - \sigma(f, P, \bar{\zeta}'')| = |f(\bar{\zeta}'_i) - f(\bar{\zeta}''_i)| |I_i|. \quad (7)$$

Since $|I_i| > 0$ (indeed, by Def 9.4 all intervals of any partition of I are nondegenerate, thus, by Def 9.2, their n -measures are positive), by changing the points $\bar{\zeta}'_i, \bar{\zeta}''_i \in I_i$, as a result of unboundedness of f on I_i , we could make the right-hand side of Equality (7) arbitrarily large. Therefore, by the Cauchy criterion (see Calculus-1), this implies that the Riemann sums of f on I do not have a limit as $\lambda(P) \rightarrow 0$. Thus, by Def 9.9, f is not Riemann integrable on I .

It should be clear that $f \in \mathcal{B}(I) \not\Rightarrow f \in \mathcal{R}(I)$. $\square$

Claim 9.6 (The Darboux Criterion for Riemann Integrability). Let

- $I \subset \mathbb{R}^n$ be a nondegenerative closed interval (see Def 9.1);
- $f : I \rightarrow \mathbb{R}$ be a function on I .

Then, f is Riemann integrable on I if and only if it is bounded on I and its lower and upper Darboux integrals are equal. Thus,

$$(f \in \mathcal{R}(I)) \iff (f \in \mathcal{B}(I) \text{ and } \underline{L}(f) = \overline{U}(f)).$$

Proof. Necessity. If $f \in \mathcal{R}(I)$, then, by Claim 9.5, $f \in \mathcal{B}(I)$. Let

$$\int_I f(\bar{x}) d\bar{x} = s \in \mathbb{R}.$$

Then, by Def 9.9, we have

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall (P, \bar{\zeta}) [\lambda(P) < \delta \Rightarrow s - \frac{\varepsilon}{2} \leq \sigma(f, P, \bar{\zeta}) \leq s + \frac{\varepsilon}{2}]. \quad (8)$$

Since in (8) both inequalities hold true for every $\bar{\zeta}$, we have

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow s - \frac{\varepsilon}{2} \leq \inf_{\bar{\zeta}} \sigma(f, P, \bar{\zeta})], \quad (9)$$

and

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow \sup_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) \leq s + \frac{\varepsilon}{2}]. \quad (10)$$

Due to Def 9.11, it should be clear that

$$\inf_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) = \underline{S}(f, P), \quad \sup_{\bar{\zeta}} \sigma(f, P, \bar{\zeta}) = \overline{S}(f, P).$$

Therefore, (9) and (10) imply that

$$\forall \varepsilon > 0 \exists \delta(\varepsilon) > 0 \forall P [\lambda(P) < \delta \Rightarrow \overline{S}(f, P) - \underline{S}(f, P) \leq \varepsilon]. \quad (11)$$

Since, by Def 9.12, for every partition P we have

$$\underline{S}(f, P) \leq \underline{L}(f) \leq \overline{U}(f) \leq \overline{S}(f, P),$$

(11) implies that

$$\forall \varepsilon > 0 \quad 0 \leq \overline{U}(f) - \underline{L}(f) \leq \varepsilon.$$

Since the inequality $\overline{U}(f) - \underline{L}(f) \leq \varepsilon$ does not depend on P , this is obviously equivalent to

$$\underline{L}(f) = \overline{U}(f).$$

Sufficiency. Note that, due to [Def 9.11](#) and [Def 9.12](#), we have

$$\forall(P, \bar{\zeta}) \quad \underline{S}(f, P) \leq \sigma(f, P, \bar{\zeta}) \leq \overline{S}(f, P). \quad (12)$$

Since, by [Claim 9.4](#),

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) \quad \text{and} \quad \lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P) = \overline{U}(f),$$

and it is given that $\underline{L}(f) = \overline{U}(f) = s$ for some $s \in \mathbb{R}$, by the Squeeze theorem (see Calculus-1), from [Inequality \(12\)](#) we have

$$\lim_{\lambda(P) \rightarrow 0} \underline{S}(f, P) = \underline{L}(f) = s \leq \lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \bar{\zeta}) \leq s = \overline{U}(f) = \lim_{\lambda(P) \rightarrow 0} \overline{S}(f, P). \quad (13)$$

That is, due to [Def 9.9](#), $f \in \mathcal{R}(I)$.

Note that, if $f \in \mathcal{R}(I)$, then

$$\underline{L}(f) = \overline{U}(f) = \int_I f(\bar{x}) d\bar{x}.$$

□